

The Sixth International Workshop on Logics and Argumentation for New-Generation Artificial Intelligence (LNGAI 2026)

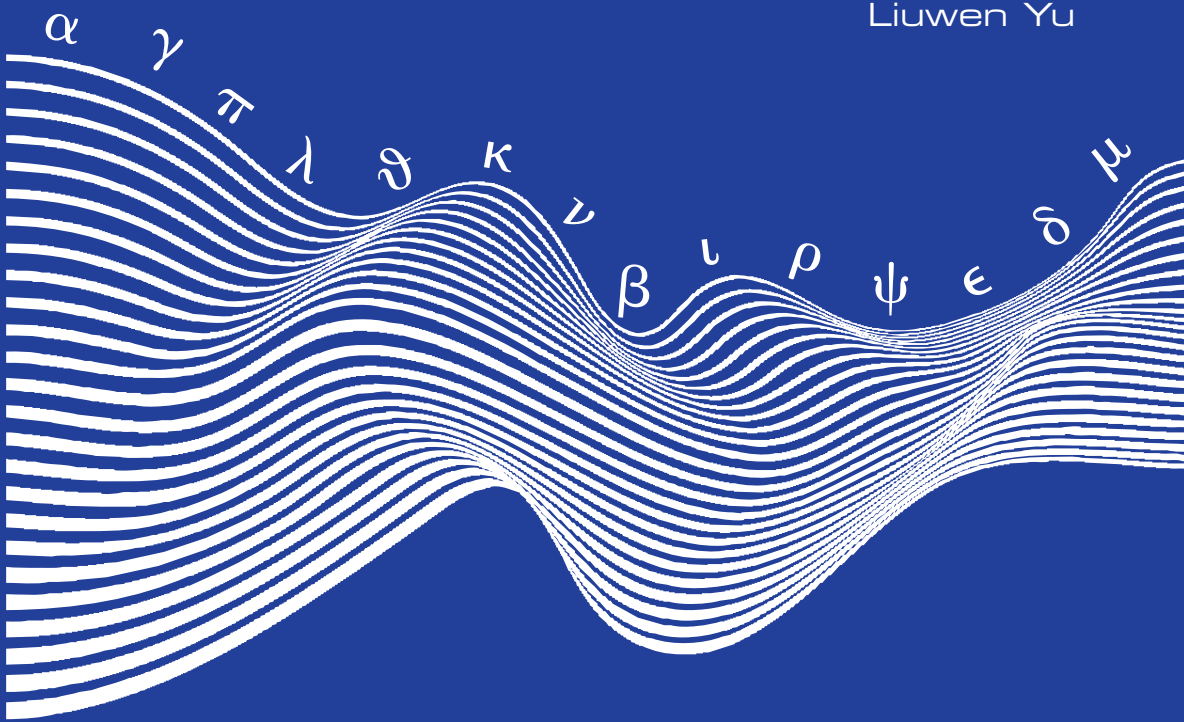
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Beishui Liao

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Logics and Argumentation
for New-Generation AI
Sixth International Workshop
14 September 2026, Barcelona

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Beishui Liao, Jieting Luo and Leendert van der Torre, eds

Volume 2

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Volume 6

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Beishui Liao, Nico Potyka, Leendert van der Torre and Liuwen Yu, eds

Logics and Argumentation
for New-Generation AI
Sixth International Workshop
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Preface

This volume continues the series *Logics for New-Generation Artificial Intelligence* (LNGAI). The series brings together research on non-monotonic logics and formal argumentation, with particular attention to reasoning with incomplete, uncertain and conflicting information and to explaining the grounds of a conclusion. The eight papers collected here address three related areas: logical foundations and norm-governed reasoning; collective and quantitative argumentation under change; and the use of large language models (LLMs) in argument mining, legal rule extraction and explainable classification.

The first group concerns logical systems and the conditions for constructing and admitting arguments. Kees van Berkel, Luca Redondi and Christian Straßer give a sequent-based characterization of ASPIC⁺ that incorporates argument strength and defeat conditions into the derivation process. Their correspondence covers weakest-link and last-link priorities, including combinations of the two, and preserves conclusions under complete, grounded, preferred and stable semantics. Zhiguang Zhao axiomatizes parity modal logic, whose operators express whether an odd or even number of successor worlds satisfy a formula. He proves soundness and completeness with respect to image-finite Kripke frames and also obtains completeness over finite frames. Rob Christiaanse studies which claims are eligible to enter an argumentation framework before attack and defence are considered. His procedural semantics evaluates candidates against a specified normative context and records the grounds of the eligibility judgement and the possibilities for repair. It distinguishes repairable deficiencies from limitations of the normative context itself, while leaving standard Dung-style semantics unchanged for the claims admitted.

A second group investigates how argumentative conclusions behave when individual judgements, graph structures or argument strengths change. Weiwei Chen and Xiaolong Liu study the aggregation of three-valued argument labellings. Besides a static dictatorship result, they show on a seven-argument framework that, under specified support-based axioms, requiring aggregation to commute with regular labelling-preserving revision forces dictatorship even without a uniqueness requirement. They also distinguish componentwise aggregation from the stronger restrictions imposed by global updates. Arunavo Ganguly, Julian Alfredo Mendez and Timotheus Kampik introduce safety, os-

cillation, liveness and fairness for dialogues represented as sequences of quantitative bipolar argumentation graphs. Their threshold-based definitions distinguish arguments that remain credible throughout a dialogue, are credible at its end, or cross the threshold during the exchange. They establish relations between these notions and define discrete and gradual notions of fairness based on the distribution of threshold attainment.

Timotheus Kampik, Arunavo Ganguly and Filip Naudot examine nonmonotonic and pseudo-nonmonotonic effects in quantitative bipolar argumentation. Using DFQuAD semantics, they show that changes in one argument’s initial strength can have nonmonotonic effects on another’s final strength even in attack-only graphs. They also identify pseudo-nonmonotonic effects on the difference between two final strengths in support-only and multitree-like graphs. Such effects concern relative marginal influence and do not require a reversal of the arguments’ ranking.

The remaining three papers connect argumentation with large language models and natural-language sources. Davide Liga, Liuwen Yu and Réka Markovich present the construction methodology and a one-episode pilot for YouCanYouBB, a multilingual and multimodal argument- and value-mining corpus drawn from Chinese television debate. Their annotation scheme combines bilingual claims and premises, argumentation schemes, values and expert interventions, with links to changes in audience votes. They report preliminary automated validation; human evaluation and the full corpus release remain to be completed. Dennis Craandijk, Floris Bex and Henry Prakken propose a neuro-symbolic approach to explainable text classification in which an LLM mines reusable arguments and proposes weighted support and attack relations. A bipolar argumentation framework then performs the final inference under a gradual semantics. The authors motivate the approach through fake-job and hate-speech detection and report preliminary fake-job classification results.

We would like to thank the authors for their contributions to this volume and the reviewers for their careful reading and constructive comments.

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September 1, 2026

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A Characterization of ASPIC⁺ Frameworks in Sequent-Based Formal Argumentation

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Abstract

Structured argumentation frameworks like ASPIC⁺ and sequent-based systems offer contrasting yet complementary approaches to defeasible reasoning. While ASPIC⁺ relies on proof trees to track argument structure—a necessity for applying preference principles like *weakest link* and *last link*—sequent-based approaches represent arguments as flat premise-conclusion pairs, sacrificing this structural detail for brevity. This paper establishes a formal bridge between these paradigms by proposing a sequent-based characterization of ASPIC⁺ that internalizes reasoning with priorities directly into the argument generation process, thus bypassing the need for proof trees. We introduce a generalized sequent calculus where inference rules are indexed with strength metrics, allowing argument strength to be computed and embedded upon rule application. This mechanism ensures that preference relations transform attacks into defeats during derivation, rather than by external evaluation. This work not only clarifies the relationship between tree-based and sequent-based argumentation but also provides a unified framework for handling structured preferences.

Keywords: logical argumentation, nonmonotonic logic, proof theory, ASPIC⁺, sequent-based argumentation, preferences

1 Introduction

Starting with the work of Dung [16] formal argumentation has been shown to serve as a unifying framework for characterizing nonmonotonic logics, that

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is, formal defeasible reasoning (see [1] for an overview of results). Intuitively, defeasibility is captured through an exchange of (counter-)arguments related through argumentative attacks [30]. Over the past decades, various structured argumentation frameworks were developed to uniformly capture different kinds of nonmonotonic logics. Examples of such frameworks are sequent-based argumentation [4], assumption-based argumentation [17], dialectical argumentation [15], and ASPIC⁺ [23] (with partial translation results available [1,18,28,11]). Such frameworks are not logics but formalisms in which logics are represented.

Especially, ASPIC⁺ has shown to be a robust theoretical foundation for modelling defeasible reasoning. At its core, ASPIC⁺ characterizes arguments as proof trees, preserving the sequential order and hierarchical structure of rule applications in constructing arguments. It differentiates strict from defeasible assumptions and rules, allowing for the characterization of a multitude of attack types that integrate preference orders over arguments [22]. The representation of arguments as trees is crucial for handling preferences: the strength of an argument, determined by principles such as *weakest link* or *last link*, depends on the strength and specific order of the rules applied in the argument.

A distinct approach in the study of argumentative characterizations of non-monotonic logics is sequent-based argumentation [4,5]. In sequent-based argumentation, arguments are represented as pairs of premises and conclusions, where the premises constitute the reasons that logically justify the conclusion. Over the last ten years, sequent-based argumentation has shown to be effective in characterizing large classes of nonmonotonic logics, such as default logic [9,29], input/output logics [10,21], and preferred subtheories [7,13], and in providing dynamic proof theories to model iterative inference processes [6,5]. One of the strong sides of sequent-based argumentation is its modularity: rules can be simply added or removed to a sequent calculus [24] to obtain different non-monotonic logics. While its representation is often more concise and easier to integrate with varying base logics [12], one may argue that it suffers from a ‘flat’ nature: the standard sequent represents the construction of an argument by means of a derivation, but the history of this construction is not present in the final argument, as is the case for proof tree arguments in ASPIC⁺.

The above observations raise immediate questions concerning the relation of these two structured argumentation frameworks. This work seeks to address these long due questions and studies the relation between the two frameworks. In particular, with respect to handling priorities, it is not apparent whether the loss of information in the construction of a sequent-style argument complicates the application of priority mechanisms that rely on the depth or path of the argument, such as the aforementioned weakest and last link mechanisms.

Contributions We propose a bridge between these two paradigms and establish a formal translation between ASPIC⁺ and a sequent-based system that effectively internalizes reasoning with prioritized defeasible information into the argument generation process. In fact, we establish a correspondence between a novel instantiation of sequent-based argumentation and a unified generalized representation of ASPIC⁺, which subsumes various frameworks as special cases.

The most distinguishing feature of our sequent-based approach, is that, rather than relying on an external evaluation mechanism to determine the strength of an argument from a flat representation, our approach computes and embeds strength information directly within the calculus' inference rules. By encoding the *weakest link* and *last link* principles directly into the sequent derivation, our system ensures that only successful attacks (defeats) are produced, transforming preference handling into a syntactic feature of the derivation.

The principal contributions of this work are fourfold:

- (i) We show that ASPIC^+ , a leading structured argumentation framework, can be characterized with a sequent-based argumentation framework.
- (ii) Our correspondence is unifying: we prove correspondence for a generalized account of ASPIC^+ with known frameworks as special cases: namely, the flat system; weakest-link systems; and last-link systems.
- (iii) We adopt a novel way to calculate the weakest-link and last-link ordering. That is, we show that reasoning with preferences among defeasible rules can be internalized into the proof-theoretic derivation of arguments, rather than being applied post-hoc by scanning the tree-like structure of an argument (as, e.g., for ASPIC^+).
- (iv) As a consequence of this general approach, we also obtain novel systems. For instance, hybrid systems using both weakest-link and last-link and systems resulting from a generalized notion of strict rules and base logics.

These results allow for a compact representation of arguments, omitting the use of arguments as proof trees while preserving nuanced reasoning with priorities. Furthermore, the above results open the door to further investigations about the relation between these fundamental structured argumentation paradigms, and we will discuss some of these immediate future research steps.

Outline This paper is organized as follows. In Section 2, we define a generalized account of ASPIC^+ . Section 3 introduces our general sequent-based framework and present the metatheory establishing the translation and equivalence between the two frameworks. Section 4 concludes with a discussion of related work and future directions. Proof sketches are found in the appendix.

2 A Unifying Account for ASPIC^+ Frameworks

ASPIC^+ is a general framework for constructing nonmonotonic logics and knows many well-studied variations. We define a generalized version that encompasses several of such variations as specific instances.

2.1 The Generalized Knowledge Base

The ASPIC^+ framework [23] models defeasible reasoning by constructing arguments from a knowledge base using strict and defeasible assumptions, as well as rules. Typically, an ASPIC^+ knowledge base is a tuple $\langle \mathcal{A}, \mathcal{R}_s, \mathcal{R}_d, \overline{} \rangle$, where $\mathcal{A}$ contains strict assumptions, $\mathcal{R}_s$ contains strict rules, $\mathcal{R}_d$ contains defeasible rules, and $\overline{}$ is a contrary function, generalizing the notion of logical negation

by specifying which formulas are contrary to one another. (One may also introduce $\mathcal{A}_d$ for defeasible assumptions but such assumptions can be equivalently expressed as defeasible rules with vacuous antecedents; we take the latter approach.) It is also common to use strict rules to import valid inferences from some base logic bl . We call ASPIC⁺ using the above knowledge base the *flat framework*. The flat framework can be extended with a ranking **strength** over the set of defeasible rules. Then, the strength of each argument is defined as a function of the strengths of the rules applied to construct it. The so-called *weakest-link* principle captures that an argument is as strong as the weakest rule applied in its construction, while the *last-link* principle stipulates that it is rather as strong as its last defeasible rule.

Instead of considering each type of ASPIC⁺ framework separately, we construct a generalized ASPIC⁺ knowledge base that subsumes all the different types as special cases. In this way, we can focus on providing a sequent-based characterization of the general framework, rather than several characterizations for the different variants. First, we partition the set of strict rules into $\mathcal{R}_s^{dom}$ and $\mathcal{R}_s^{bl}$. $\mathcal{R}_s^{dom}$ contains domain-specific rules, i.e., rules designed specifically for the ASPIC⁺ model. $\mathcal{R}_s^{bl}$, if non-empty, imports valid inferences from some Tarskian base logic bl (so a logic with a reflexive, transitive, monotonic and compact consequence relation $\vdash_{bl}$). The ranking **strength** subsumes the flat framework as a special case where the strengths of all defeasible rules are the same. Finally, we partition the set $\mathcal{R}_d$ into $\mathcal{R}_d^w$ and $\mathcal{R}_d^l$, where $\mathcal{R}_d^w$ contains defeasible *weakest-link* rules, while $\mathcal{R}_d^l$ contains defeasible *last-link* rules. In this way we allow for hybrid systems in which some rules adhere to last link, while other to weakest link. We detail argumentation strength calculation below, but already note that ASPIC⁺ purely based on weakest link is captured by letting $\mathcal{R}_d^l = \emptyset$, and similar for purely last-link based frameworks.

Definition 2.1 A *knowledge base* $\mathbb{K}$ is a tuple $\langle \mathcal{A}, \mathcal{R}_s, \mathcal{R}_d, \text{strength}, \bar{\cdot} \rangle$, where:

- $\mathcal{A} \subseteq \text{WFF}$ is a set of sentences in a given logical language (strict assumptions).
- $\mathcal{R} = \mathcal{R}_s \cup \mathcal{R}_d$ is a set of inference rules. It consists of (where $\theta = \theta_1, \dots, \theta_n$ and $\theta, \psi \in \text{WFF}$):
 - the set of *strict rules* $\mathcal{R}_s = \langle \mathcal{R}_s^{bl}, \mathcal{R}_s^{dom} \rangle$ of the form $\theta \rightarrow \psi$, that is divided in *domain-specific rules* $\mathcal{R}_s^{dom}$ and possibly a set of *deductive rules* induced by a given base logic bl : $\mathcal{R}_s^{bl} = \{ \theta \rightarrow \psi \mid \theta \vdash_{bl} \psi \}$. If there is no base logic then $\mathcal{R}_s^{bl} = \emptyset$;
 - the set of *defeasible rules* $\mathcal{R}_d = \langle \mathcal{R}_d^l, \mathcal{R}_d^w \rangle$, containing rules of the form $\theta \rightsquigarrow \psi$, and which is divided into *last-link-rules* $\mathcal{R}_d^l$ and *weakest-link-rules* $\mathcal{R}_d^w$.
- $\text{strength} : \mathcal{R}_d \rightarrow \mathbb{Z}$ associates defeasible information with numerical strength; we write $\text{strength}(r)$ to denote the strength of $r \in \mathcal{R}$.
- $\bar{\cdot} : \text{WFF} \cup \mathcal{R}_d \rightarrow 2^{\text{WFF}}$ associates formulas with contrary formulas.

Rules in $\mathcal{R}$ map sets of formulas to formulas, written as $\theta_1, \dots, \theta_n \rightarrow \psi$ for strict rules and $\theta_1, \dots, \theta_n \rightsquigarrow \psi$ for defeasible rules. For any $r_1, r_2 \in \mathcal{R}_d$, we say

r_1 is strictly *weaker* than r_2 , whenever $\text{strength}(r_1) < \text{strength}(r_2)$.

Remark 2.2 Typically, ASPIC⁺ assumes a naming function in the language that associates defeasible rules with atoms in the language WFF so that undercut attacks can specifically attack defeasible rules. Instead, and without loss of generality, we simply extend the domain of the contrariness function to include contraries to defeasible rules to allow for undercut attacks. Also, here we model defeasible assumptions ϕ as defeasible rules with empty body $\sim \phi$. Finally, we will only study rankings as preferences among defeasible parts of the knowledge base, not more general preorders.

Remark 2.3 We distinguish three types of knowledge bases, yielding different ASPIC⁺ frameworks. $\mathbb{K}$ is

- a *flat* knowledge base if and only if $\text{strength} : r \mapsto 1$.
- a *weakest-link* knowledge base if and only if $\mathcal{R}_d^l = \emptyset$.
- a *last-link* knowledge base if and only if $\mathcal{R}_d^w = \emptyset$.

2.2 Arguments and Attacks

In ASPIC⁺, arguments are defined iteratively, yielding arguments as proof trees. The underlying rationale is the *chaining of rules*: when the conclusion of an argument coincides with the premise of a rule, one can attach the rule to the argument obtaining a new valid argument. Some functions are given to describe the properties of the argument; e.g., given an argument a , $\text{Con}(a)$ denotes the conclusion of a . We adopt this general iterative strategy, and, in addition to the standard functions, we introduce $\text{strength}(a)$, that returns the strength of an argument and depends on whether the last rule applied is strict, weakest-link defeasible or last-link defeasible. Normally, the strength of an argument is computed when the attacks are drawn. Suppose one uses the weakest-link approach: to determine the strength of an argument, the defeasible rules applied in its construction are inspected and its weakest rule is identified. We instead model weakest-link (such as last-link) as a property of the rules and compute its impact on the strength of the arguments immediately when the rule is applied.

Definition 2.4 [Structured Argument] Given a knowledge base $\mathbb{K}$, a *structured argument* a in ASPIC⁺ is a finite proof tree constructed as follows:

- (i) **Leaf:** For any $\phi \in \mathcal{A}$, $a = \langle \phi \rangle$ is an argument with conclusion $\text{Con}(a) = \phi$, premises $\text{Prem}(a) = \{\phi\}$, $\mathcal{R}_s(a) = \mathcal{R}_d(a) = \emptyset$, $\text{Sub}(a) = \{a\}$, $\text{Last}(a) = \emptyset$, and $\text{strength}(a) = \infty$.
- (ii) **Strict Inference:** If $a_1, \dots, a_n$ are arguments such that $r = \text{Con}(a_1), \dots, \text{Con}(a_n) \rightarrow \psi \in \mathcal{R}_s$, then $a = \langle a_1, \dots, a_n \rightarrow \psi \rangle$ is an argument with: $\text{Con}(a) = \psi$, $\text{Last}(a) = \bigcup_{i=1}^n \text{Last}(a_i)$, $\mathcal{R}_s(a) = \bigcup_{i=1}^n \mathcal{R}_s(a_i) \cup \{r\}$, $\mathcal{R}_d(a) = \bigcup_{i=1}^n \mathcal{R}_d(a_i)$, $\text{Sub}(a) = \bigcup_{i=1}^n \text{Sub}(a_i) \cup \{a\}$ and $\text{strength}(a) = \min(\text{strength}(a_i))_{i=1}^n$.
- (iii) **Defeasible Inference:** If $a_1, \dots, a_n$ are arguments such that $r :$

$\text{Conc}(a_1), \dots, \text{Conc}(a_n) \rightsquigarrow \psi \in \mathcal{R}_d$, then $a = \langle a_1, \dots, a_n \rightsquigarrow \psi \rangle$ is an argument with: $\text{Con}(a) = \psi$, $\text{Last}(a) = \{r\}$, $\mathcal{R}_s(a) = \bigcup_{i=1}^n \mathcal{R}_s(a_i)$, $\mathcal{R}_d(a) = \bigcup_{i=1}^n \mathcal{R}_d(a_i) \cup \{r\}$, $\text{Sub}(a) = \bigcup_{i=1}^n \text{Sub}(a_i) \cup \{a\}$ and

$$\text{strength}(a) = \begin{cases} \text{strength}(r) & \text{if } r \in \mathcal{R}_d^l \\ \min(\{\text{strength}(r)\} \cup \bigcup_{i=1}^n \text{strength}(a_i)) & \text{if } r \in \mathcal{R}_d^w \end{cases}$$

We denote the set of all arguments for $\mathbb{K}$ as $\text{Arg}_{\text{as}}(\mathbb{K})$ (with ‘as’ for ASPIC⁺). An argument a is defeasible if $\text{Last}(a) \neq \emptyset$ (which is equivalent to $\mathcal{R}_d(a) = \emptyset$).

Remark 2.5 Definitions 2.1 and 2.4 form our ASPIC⁺-style account for the construction of arguments. As we already mentioned, our framework is general enough to account for flat ASPIC⁺ systems, weighted ASPIC⁺ systems with weakest link, weighted ASPIC⁺ systems with last link, and ASPIC⁺ systems that extends a base logic by imposing domain-specific rules as ad hoc constraints on the knowledge base. However, the account can also express further types of ASPIC⁺ formalisms, that cannot be characterized by any of the standard approaches: namely, knowledge bases that combine weakest-link rules and last-link rules. Although the purpose of this paper is not to investigate how this expressivity is to be used, this result is worth pointing out.

The various argument types generate different attack types.

Definition 2.6 [Attacks and Defeats] Given arguments a and b in $\mathbb{K}$, we say that a *attacks* b if one of the following holds:

- **Rebuttal:** a attacks b via rebuttal if $\text{Conc}(a) \in \overline{\text{Conc}(b')}$, where b' is a subargument of b and $\text{Last}(b') \in \mathcal{R}_d$.²
- **Undercut:** a attacks b via undercut if $\text{Conc}(a) \in \bar{r}$, where $r \in \text{Last}(b')$ is the last applied rule of a subargument b' of b .

A rebuttal is successful if $\text{strength}(a) \geq \text{strength}(b')$. An undercut is always successful. A successful attack is called *defeat*. We collect all pairs of arguments (a, b) where $a \in \text{Arg}_{\text{as}}(\mathbb{K})$ defeats $b \in \text{Arg}_{\text{as}}(\mathbb{K})$ in the set $\text{att}_{\text{as}}(\mathbb{K})$.

The pair $(\text{Arg}_{\text{as}}(\mathbb{K}), \text{att}_{\text{as}}(\mathbb{K}))$ is the *argumentation framework induced by $\mathbb{K}$ and ASPIC⁺*.

Example 2.7 Let $\mathbb{K} = \langle \mathcal{A}, \mathcal{R}_s, \mathcal{R}_d, \text{strength}, \bar{} \rangle$ where: $\mathcal{A} = \{p, q\}$, $\mathcal{R}_s = \{r_0 : r \rightarrow s \wedge u\} \cup \{r_i : \Gamma \rightarrow \phi \mid \Gamma \vdash_{bl} \phi\}$, $\mathcal{R}_d = \mathcal{R}_d^l = \{r_1 : p \rightsquigarrow r, r_2 : q \rightsquigarrow \neg s, r_3 : s \rightsquigarrow t, r_4 : q \rightsquigarrow \neg t, r_5 : q \rightsquigarrow w\}$. Also, let $\text{strength}(r_1) = 1, \text{strength}(r_2) = 2, \text{strength}(r_3) = 3, \text{strength}(r_4) = 2$, and $\text{strength}(r_5) = 2$. Finally, let $\bar{\phi} = \{\psi \mid \psi = \neg\phi \text{ or } \phi = \neg\psi\}$ for all $\phi \in \text{WFF}$ and $\bar{r}_3 = \{w\}$ and $\bar{r}_i = \emptyset$ for all $i \in \{1, 2, 4, 5\}$. We can construct the following arguments:

- | | | |
|---------------------------------------|-------------------------------------|--|
| • $a_1 : \langle p \rangle;$ | • $a_4 : a_3 \rightarrow s;$ | • $a_7 : a_6 \rightsquigarrow_2 \neg s;$ |
| • $a_2 : a_1 \rightsquigarrow_1 r;$ | • $a_5 : a_4 \rightsquigarrow_3 t;$ | • $a_8 : a_6 \rightsquigarrow_2 \neg t;$ |
| • $a_3 : a_2 \rightarrow s \wedge u;$ | • $a_6 : \langle q \rangle;$ | • $a_9 : a_6 \rightsquigarrow_2 w.$ |

² Since we translate defeasible assumptions into defeasible rules, undermining attacks are captured by rebuttals.

For the convenience of the reader, we denote the strength of the (sub-)argument with subscripts. The tree-structure of a_5 is $\langle\langle\langle p \rangle \rightsquigarrow_1 r \rangle \rightarrow s \wedge u \rangle \rightarrow s \rangle \rightsquigarrow_3 t$.

We have three potential attackers of a_5 : a_7, a_8, a_9 . a_7 concludes the negation of the head of r_0 , but since $r_0 \in \mathcal{R}_s$, it cannot rebut a_5 . a_8 attacks the conclusion of the defeasible rule r_3 and, given that $\text{strength}(a_8) = 2 < 3 = \text{strength}(a_5)$, defeats it. Finally, a_9 undercuts a_5 and, as all undercutting attacks, manages to defeat it. Now consider $\mathbb{K}'$ which is identical to $\mathbb{K}$ apart from the fact that all defeasible rules are *weakest-link* rules. We can construct analogous arguments as in $\mathbb{K}$. $a'_1, a'_2, a'_3, a'_4, a'_6, a'_7, a'_8, a'_9$ have the same strength as their *last-link* counterparts, because they involve at most one defeasible rule. a'_5 , instead, is now weaker: $a'_5 : \langle\langle\langle p \rangle \rightsquigarrow_1 r \rangle \rightarrow s \wedge u \rangle \rightarrow s \rangle \rightsquigarrow_1 t$. Notice that $\text{strength}(a'_8) = 2 > 1 = \text{strength}(a'_5)$, therefore a'_8 *defeats* a'_5 .

2.3 Dung-Style Semantics and Consequence Relations

Arguments and attacks constitute so-called argumentation frameworks:

Definition 2.8 A *structured argumentation framework* AF is a tuple $\langle \text{Arg}, \text{Att}, \text{Con} \rangle$ where Arg is a set of arguments, each $a \in \text{Arg}$ associated with a conclusion $\text{Con}(a) \in \text{WFF}$, and Att collects the attacks between the arguments in Arg .

An argumentation framework is a directed graph where each node represents an argument and the arrows capture the attacks. Both ASPIC^+ and sequent-based argumentation induce structured argumentation frameworks relative to knowledge bases. For ASPIC^+ , we let $\text{AF}_{as}(\mathbb{K}) = \langle \text{Arg}_{as}(\mathbb{K}), \text{att}_{as}(\mathbb{K}), \text{Con} \rangle$. ASPIC^+ and sequent-based argumentation rely on semantics [16] to select arguments. The next definitions assume a given $\text{AF} = \langle \text{Arg}, \text{Att}, \text{Con} \rangle$.

Definition 2.9 Let $\mathcal{E} \subseteq \text{Arg}$ and $a \in \text{Arg}$. $\mathcal{E}$ **defends** a if and only if, for all $b \in \text{Arg}$ such that $(b, a) \in \text{Att}$, there is some $c \in \mathcal{E}$ such that $(c, b) \in \text{Att}$. $\mathcal{E}$ is **conflict-free** if and only, for all $a, b \in \mathcal{E}$, $(a, b) \notin \text{Att}$.

Definition 2.10 [Completeness-Based Semantics] Let $\mathcal{E} \subseteq \text{Arg}$, we have:

- $\mathcal{E}$ is **complete** if and it is conflict-free, defends all its arguments, and contains all the arguments that it defends.
- $\mathcal{E}$ is **preferred** if and only if it is maximally complete.
- $\mathcal{E}$ is **stable** if and only if it is preferred and attacks all arguments in $\text{Arg} \setminus \mathcal{E}$.
- $\mathcal{E}$ is **grounded** if and only if it is minimally complete.

We collect the sets $\mathcal{E}$ (often called *extensions*) for each semantics in $\text{Sem}(\text{AF})$. Note that stable, grounded, and preferred sets are complete [16].

Once the relevant extensions are identified, an inference relation is defined.

Definition 2.11 Let AF be a structured argumentation framework and Sem a semantics. We define two skeptical and a credulous entailment relation:

- *shared arguments*: $\text{AF} \vdash_{\cap}^{a, \text{Sem}} \phi$ iff there is an $a \in \cap \text{Sem}(\text{AF})$ with $\text{Con}(a) = \phi$.

- *shared conclusions*: $\text{AF} \vdash_{\cap}^{c, \text{Sem}} \phi$ iff for every $\mathcal{E} \in \text{Sem}(\text{AF})$ there is an $a \in \mathcal{E}$ with $\text{Con}(a) = \phi$.
- *credulous*: $\text{AF} \vdash_{\cup}^{\text{Sem}} \phi$ iff there is a $\mathcal{E} \in \text{Sem}(\text{AF})$ and $a \in \mathcal{E}$ with $\text{Con}(a) = \phi$.

Definition 2.12 Let $\mathcal{W}$ be a class of formulas and Sem an argumentation semantics. We say that two structured argumentation frameworks AF_1 and AF_2 are *Sem-equivalent on $\mathcal{W}$* iff

- For every $\mathcal{E}_1 \in \text{Sem}(\text{AF}_1)$ there is a $\mathcal{E}_2 \in \text{Sem}(\text{AF}_2)$ for which $\text{Con}[\mathcal{E}] \cap \mathcal{W} = \text{Con}[\mathcal{E}_2] \cap \mathcal{W}$, and vice versa.
- For all $\phi \in \mathcal{W}$ and all entailments $\vdash$ from Def. 2.11, $\text{AF}_1 \vdash \phi$ iff $\text{AF}_2 \vdash \phi$.

In this paper, we aim at establishing a correspondence between the argumentation framework generated by chaining rules according to Definition 2.4 and the one generated by applying our sequent calculus on $\mathbb{K}$. Our result will warrant that all the above-mentioned inference relations yield the same outputs through our sequent calculus framework as given by the ASPIC⁺ framework.

3 A Sequent-Based Characterization of ASPIC⁺

3.1 Preliminaries on Sequent-Based Argumentation

In this subsection, we introduce some basic notions of sequent-based argumentation. Our approach belongs to the Argument Calculi paradigm [9,10], where sequent-style calculi are used to derive two types of arguments: arguments drawing conclusions from a knowledge base, and arguments that constitute attackers and, so, capturing defeasibility. Knowledge bases for sequent-based argumentation are often simpler than their counterparts in ASPIC⁺:

Definition 3.1 [Knowledge based for sequent-based argumentation] A knowledge base is a pair $\mathbb{K} = \langle \mathcal{A}_{\text{str}}, \mathcal{A}_{\text{def}} \rangle$ where $\mathcal{A}_{\text{str}} \subseteq \text{WFF}$ collects strict assumptions and $\mathcal{A}_{\text{def}} \subseteq \text{WFF}$ represents defeasible assumptions (with $\mathcal{A}_{\text{str}} \cap \mathcal{A}_{\text{def}} = \emptyset$).

In sequent-based argumentation sequents are arguments. We start from a standard definition of a sequent:

Definition 3.2 [Sequent] Given a language $\mathcal{L}$, a sequent $\Gamma \Rightarrow \Delta$ is a pair of regular, non-ordered, finite sets of propositions $\Gamma = \{\gamma_1, \dots, \gamma_n\}$, $\Delta = \{\delta_1, \dots, \delta_n\}$ in $\mathcal{L}$ (where $\Rightarrow$ is possibly indexed with a natural number to denote its strength). A sequent is interpreted as stating that, all formulas on the right-hand side being true/justified/etc., at least one of the formulas on the left-hand side must be true/justified/etc.

Sequents are used to characterize the inference relation of a logic as a Gentzen-style ‘sequent calculus,’ which is a list of sequent-rules to derive sequents from other sequents. For instance, in many calculi the following rules characterize negation:

$$\frac{\Gamma \Rightarrow \phi, \Delta}{\neg \phi, \Gamma \Rightarrow \Delta} \neg\text{-LEFT} \qquad \frac{\Gamma, \phi \Rightarrow \Delta}{\Gamma \Rightarrow \neg \phi, \Delta} \neg\text{-RIGHT}$$

In sequent-based argumentation, arguments are generated from a sequent calculus applied on a knowledge base:

Definition 3.3 Give a knowledge base $\mathbb{K}$ and a calculus $\mathbf{C}$, $a : \Gamma \Rightarrow \phi$ is an *argument in $\text{Arg}_{\text{sb}}(\mathbb{K})$* if and only if: $\Gamma \Rightarrow \phi$ is derivable in $\mathbf{C}$ and $\Gamma \subseteq \mathcal{A}_{\text{str}} \cup \mathcal{A}_{\text{def}}$.

Some standard attack-relations include (see [3] for a systematic study):

Definition 3.4 Given $a, b \in \text{Arg}_{\text{sb}}(\mathbb{K})$, we have:

- a **rebuts** b iff $\text{Con}(a) \vdash \neg \text{Con}(b)$.
- a **set-undermines** b iff $\text{Con}(a) \vdash \neg \bigwedge \text{Prem}(b) \cap \mathcal{A}$.
- a **pointedly undermines** b iff $\text{Con}(a) \vdash \neg \phi$ for some $\phi \in \text{Prem}(b) \cap \mathcal{A}$.

3.2 The Language

Consider an ASPIC⁺ knowledge base: $\mathbb{K} = \langle \mathcal{A}, \mathcal{R}_s, \mathcal{R}_d, \text{strength}, \overline{} \rangle$. We build on the idea explored in [10] to treat rules as objects in the language. Consider, for instance, a $r_1 \in \mathcal{R}_s$ where $\{\theta_1, \dots, \theta_n\} = \text{Prem}(r_1)$ and $\phi = \text{Con}(r_1)$. We write this rule using the notation (where $\theta = \theta_1, \dots, \theta_n$): (θ/ϕ) . We use an analogous notation for defeasible rules, just switching to square brackets to be able to syntactically distinguish them from the strict ones and reporting their strength as a subscript, e.g.: $[\theta/\phi]_k$. The language of our logic extends a given language WFF, that we import from the ASPIC⁺ model:

$$\text{WFF}_{\text{ext}} : \quad \phi \in \text{WFF} \quad | \quad (\theta_1, \dots, \theta_n/\phi) \quad | \quad [\theta_1, \dots, \theta_n/\phi]_k \quad | \quad \neg[\theta_1, \dots, \theta_n/\phi]_k$$

The language of the logic is given by all the formulas in the language of the ASPIC⁺ model plus the formulas that capture strict and defeasible rules. By expressing rules as formulas, we can treat $\mathbb{K} = \langle \mathcal{A}, \mathcal{R}_s, \mathcal{R}_d, \overline{} \rangle$ as a standard knowledge base for sequent-based argumentation (see Definition 3.1), by taking $\mathcal{A} \cup \mathcal{R}_s$ and $\mathcal{R}_d$ as the set of strict, respectively defeasible assumptions.

3.3 The Sequent Calculus

Recall, ASPIC⁺ arguments are constructed by the so-called *chaining-of-rules* method (see Definition 2.4): one starts from the assumptions in $\mathcal{A}$ and then chain the rules, strict or defeasible, to generate further arguments. We reconstruct this mechanism. First, we provide a zero-premises ID sequent-rule to introduce assumptions, and then we present the DET sequent-rules that capture the detachment—i.e., the application—of a rule. We have several detachment meta-rules, depending on the rule to be applied.

Arguments that state assumptions. The simplest arguments in ASPIC⁺ state some formula in $\mathcal{A}$. We capture these arguments as identity sequents, derivable from no premises by the rule:

$$\frac{}{\phi \Rightarrow_{\infty} \phi} \text{ ID}$$

Remark 3.5 The subscript denotes the strength of the argument. In general, this will be a function of the strengths of the defeasible rules used in its derivation. We approximate strict arguments—i.e., arguments with no defeasible rules—as arguments with infinite strength.

Detachment of domain-specific strict rules. Let $\theta = \theta_1, \dots, \theta_n$, DETS captures the application of a strict domain-specific rule $r : \theta \rightarrow \psi \in \mathcal{R}_s^{dom}$.

$$\frac{\Delta_1 \Rightarrow_{k_1} \theta_1 \quad \dots \quad \Delta_n \Rightarrow_{k_n} \theta_n}{\Delta_1, \dots, \Delta_n, (\theta/\psi) \Rightarrow_{\min\{k_i\}_{i=1}^n} \psi} \text{ DETS}$$

Detachment of strict rules from the base logic. In ASPIC⁺, the application of a base logic rule is the same as the application of any other strict rule. Sequent-based argumentation, however, allows us to import the inferential engine of the base logic, rather than the valid inference pairs one-by-one:

$$\frac{\Delta_1 \Rightarrow_{k_1} \theta_1 \quad \dots \quad \Delta_n \Rightarrow_{k_n} \theta_n \quad \theta \Rightarrow^{bl} \psi}{\Delta_1, \dots, \Delta_n \Rightarrow_{\min\{k_i\}_{i=1}^n} \psi} \text{ DETBL}$$

Remark 3.6 $\theta \Rightarrow^{bl} \psi$ denotes a valid sequent in some base logic bl . One can attach any sequent calculus for the base logic to our system to derive $\theta \Rightarrow^{bl} \psi$. The superscript ensures that the sequent-rules for the base logic do not interact with the sequent-rules in our calculus in any other way but through DETBL.

Remark 3.7 DETBL can be seen as encoding a special version of the well-known CUT-rule, where what is cut is a set of formulas rather than one single formula. Note that DETBL can only be used to cut the conclusions of a set of arguments, substituting them with one of their consequences according to some base logic bl . For instance, suppose have: $(\neg \xi \wedge \psi) \Rightarrow_n \xi \wedge \psi$, then we can apply DETBL to get: $(\neg \xi \wedge \psi) \Rightarrow_n \xi$.

Detachment of defeasible rules. In ASPIC⁺, the application of a defeasible rule may impact the strength of an argument. Its impact depends on whether one uses the weakest-link or last-link principle. In our formalism, we treat weakest-link and last-link as properties of the rules. We introduce a mechanism that tackles the strength of each defeasible rule at the moment of its application, reporting it as a subscript to the rule. Such strength is equal to the strength of the corresponding argument when the rule is applied, and will remain the same as further rules are applied, while the strength of the argument may change. This way, we reproduce, to some extent, the explicit representation of the history of the derivation, which is a typical feature of ASPIC⁺ arguments.³ We have the following meta-rules governing rule application in the context of defeasible argument strength:

$$\frac{\Delta_1 \Rightarrow_{k_1} \theta_1 \quad \dots \quad \Delta_n \Rightarrow_{k_n} \theta_n}{\Delta_1, \dots, \Delta_n, [\theta/\psi]_\pi \Rightarrow_\pi \psi} \text{ DETD where } r = \theta \rightsquigarrow \psi$$

³ Notice that our arguments are more compact than their ASPIC⁺ counterparts: we do not need to preserve the tree-like structure of the construction of the argument, we do not need to structurally keep track of the last rule applied in the argument, and we may in principle also omit the original strength of the rules. Especially, in ASPIC⁺ you have the post-hoc calculation of the strength of the argument by scanning the tree structure of the argument, our approach adopts an integrated proof-theoretic way of tracking the argument's evolving strength throughout construction. Furthermore, our sequent-based arguments do not require the representation of the base logic rules that are applied.

$$\text{and } \pi = \begin{cases} \min(\{k_i\}_{i=1}^n \cup \{\text{strength}(r)\}) & r \in \mathcal{R}_d^w \quad (\text{weakest link}) \\ \text{strength}(r) & r \in \mathcal{R}_d^l \quad (\text{last link}) \end{cases}$$

Remark 3.8 ASPIC⁺ is typically studied for exclusively one strength aggregation principle: either weakest link or last link. In our more general result, we allow for hybrid cases since we encode whether a defeasible rule behaves according to last or weakest link principle. Our result immediately applies to the exclusive case where all rules are considered weakest—resp. last—link.

Attacking arguments. The meta-rules presented above capture the construction of arguments by *chaining-of-rules*. However, our sequent-based model needs additional rules to generate attacks. This relates to its underpinning idea, presented in Section 3.2, to treat rules as objects in the language. Attacking arguments are then arguments that conclude the inapplicability of a rule:⁴

$$\frac{\Delta \Rightarrow_k \psi' \quad \Gamma, [\theta/\psi]_{k'} \Rightarrow_l \sigma}{\Delta \Rightarrow_k \neg[\theta/\psi]_{k'}} \text{ REB} \quad \text{where } k \geq k' \text{ and } \psi' \in \overline{\psi}$$

$$\frac{\Delta \Rightarrow_k \psi' \quad \Gamma, [\theta/\psi]_{k'} \Rightarrow_l \sigma}{\Delta \Rightarrow_k \neg[\theta/\psi]_{k'}} \text{ UCUT} \quad \text{where } \psi' \in \overline{\theta \rightsquigarrow \psi}$$

Remark 3.9 For rebuttals, the strength of the attacker needs to be equal or greater than the strength of the attacked argument at the moment when the attacked defeasible inference step occurred. We track this information in the syntax of the targeted rule, as shown by the definition of DETD. In contrast, undercuts do not need to fulfill any condition regarding the strengths of the arguments involved, as desired (see Definition 2.6).

We now define arguments, attacks and argumentation frameworks:

Definition 3.10 Given a knowledge base $\mathbb{K}$, a sequent-based argumentation framework $\text{AF}_{sb}(\mathbb{K})$ is a triple $\langle \text{Arg}_{sb}(\mathbb{K}), \text{Att}_{sb}(\mathbb{K}), \text{Con} \rangle$ where:

- $\text{Arg}_{sb}(\mathbb{K})$ collects the arguments generated from $\mathbb{K}$ by applying DETS, DETBL, DETD, REB, UCUT, and ID.⁵
- $\text{Att}_{sb}(\mathbb{K})$ collects pairs $(\mathbf{a}, \mathbf{b})$ of arguments such that $\text{Con}(\mathbf{a}) = \neg[\theta, \psi]$ for some $[\theta/\psi] \in \text{Prem}(\mathbf{b})$ and
- $\text{Con}(\Gamma \Rightarrow \gamma) = \gamma$ for each $\Gamma \Rightarrow \gamma \in \text{Arg}_{sb}(\mathbb{K})$.

⁴ We here adopt the way attack rules are defined in works on Argument Calculi [9,10]. In contrast, in many works on sequent-based argumentation so-called elimination rules are utilized that derive objects of the type $\Delta \not\Rightarrow \psi$ signifying the defeat of the argument $\Delta \Rightarrow \psi$. The two forms of presentation can easily be transformed into each other.

⁵ Notice that, when a weakest-link rule is applied using DETD, it might be labeled with some k that differs from its original strength. This requires an adjustment in the standard definition of a $\mathbb{K}$ -based sequent argument. Namely, a derivable sequent $\Gamma \Rightarrow_k \psi$ is an argument iff: 1) for all $\phi \in \text{WFF}$, $\phi \in \mathcal{A}$, 2) for each $(\theta, \phi) \in \Gamma$, $(\theta, \phi) \in \mathcal{R}_s$, and 3) for each $[\theta, \phi]_k \in \Gamma$, there is a $[\theta, \phi]_l \in \mathcal{R}_d$.

In the appendix, we prove that $\text{AF}_{as}(\mathbb{K})$ and $\text{AF}_{sb}(\mathbb{K})$ give equivalent outputs according to $\sim_{\cap}^{a, \text{Sem}}$, $\sim_{\cap}^{c, \text{Sem}}$ and $\sim_{\cup}^{\text{Sem}}$:

Theorem 3.11 *Let $\mathbb{K}$ be a knowledge base. Then, $\text{AF}_{as}(\mathbb{K})$ and $\text{AF}_{sb}(\mathbb{K})$ are Sem-equivalent on WFF for every completeness-based semantics Sem.*

Example 3.12 Consider the knowledge bases $\mathbb{K}$ and $\mathbb{K}'$ in Example 2.7. Recall that they contain the same rules, and the only difference is that the defeasible rules in $\mathbb{K}$ are all *last-link* rules, while those in $\mathbb{K}'$ are all *weakest-link* rules.

We can build arguments in $\text{Arg}(\mathbb{K})$ as follows:

$$\begin{array}{c}
 \frac{\frac{\frac{\overline{\mathbf{a}_1 : p \Rightarrow_{\infty} p}}{\mathbf{a}_2 : p, [p/r]_1 \Rightarrow_1 r} \text{ID}}{\mathbf{a}_3 : p, [p/r]_1, (r/s \wedge u) \Rightarrow_1 s \wedge u} \text{DETD}}{\mathbf{a}_4 : p, [p/r]_1, (r/s \wedge u) \Rightarrow_1 s} \text{DETS} \quad \frac{\vdots}{s \wedge u \Rightarrow^{bl} s} \text{DETBL} \\
 \frac{\mathbf{a}_4 : p, [p/r]_1, (r/s \wedge u) \Rightarrow_1 s}{\mathbf{a}_5 : p, [p/r]_1, (r/s \wedge u), [s/t]_1 \Rightarrow_1 t} \text{DETD} \\
 \\
 \frac{\frac{\frac{\overline{\mathbf{a}_6 : q \Rightarrow_{\infty} q}}{\mathbf{a}_8 : q, [q/\neg t]_2 \Rightarrow_2 \neg t} \text{ID}}{\mathbf{a}_9 : q, [q/w]_2 \Rightarrow_2 w} \text{DETD}}{\mathbf{a}_{*8} : q, [q/\neg t]_2 \Rightarrow_2 \neg[s/t]_1} \text{REB} \quad \frac{\vdots}{\mathbf{a}_5} \text{UCUT} \quad \frac{\frac{\overline{\mathbf{a}_6 : q \Rightarrow_{\infty} q}}{\mathbf{a}_7 : q, [q, \neg s]_2 \Rightarrow_2 \neg s} \text{ID}}{\mathbf{a}_5} \text{DETS}
 \end{array}$$

In $\text{Arg}_{sb}(\mathbb{K}')$, we generate $\mathbf{a}'_1, \mathbf{a}'_2, \mathbf{a}'_3, \mathbf{a}'_4, \mathbf{a}'_6, \mathbf{a}'_7, \mathbf{a}'_8, \mathbf{a}'_9$ with the same premises, conclusion and strength as their counterparts in $\text{Arg}_{sb}(\mathbb{K})$. $\mathbf{a}'_5$ can be derived analogously, but it has a different strength:

$$\frac{\vdots}{\mathbf{a}'_4} \text{DETD} \quad \frac{\mathbf{a}'_4}{\mathbf{a}'_5 : p, [p/r]_1, (r/s \wedge u), [s/t]_3 \Rightarrow_3 t} \text{DETD}$$

As for the attacking arguments, we are not able to generate an analogous of $\mathbf{a}_{*8}$ because the targeted rule is too strong. On the other hand, $\mathbf{a}'_{*9}$ can be generated, because UCUT can be applied independently of the strength of the targeted rule. Note that, in $\text{AF}_{sb}(\mathbb{K})$ and $\text{AF}_{sb}(\mathbb{K}')$, we get equivalent result with respect to the corresponding $\text{AF}_{as}(\mathbb{K})$ and $\text{AF}_{as}(\mathbb{K}')$ (see Example 2.7):

- $(a_7, a_5) \notin \text{Att}_{as}(\mathbb{K})$ and $(a'_7, a'_5) \notin \text{Att}_{as}(\mathbb{K}')$ because $\text{Last}(a_5) \in \mathcal{R}_s$. In the sequent-based model, REB cannot be applied to $\mathbf{a}_7$ (nor to $\mathbf{a}'_7$), thus no attacking argument is constructed.
- $(a_8, a_5) \in \text{Att}_{as}(\mathbb{K})$ but $(a'_8, a'_5) \notin \text{Att}_{as}(\mathbb{K}')$. Similarly, $(\mathbf{a}_{*8}, \mathbf{a}_5) \in \text{Att}_{sb}(\mathbb{K})$, but $\mathbf{a}'_{*8}$ cannot be generated.
- $(a_9, a_5) \in \text{Att}_{as}(\mathbb{K})$ and $(a'_9, a'_5) \in \text{Att}_{as}(\mathbb{K}')$. Similarly, $(\mathbf{a}_{*9}, \mathbf{a}_5) \in \text{Att}_{sb}(\mathbb{K})$, but $\mathbf{a}'_{*9}$ cannot be generated.

4 Related Work and Conclusion

Related and future work. To the best of our knowledge, this is the first work to establish a translation between ASPIC^+ and sequent-based argumentation. The core of our characterization lies in the various detachment sequent-rules, that capture the application of the several types of ASPIC^+ rules. The DET-rules resemble the detachment sequent-rule in [10], which in turns captures the distinctive detachment operation in input/output logics [21]. Indeed, our operators $(\theta_1, \dots, \theta_n/\phi)$ and $[\theta_1, \dots, \theta_n/\phi]$ can be seen as generalized input-output operators, with a set of formulas rather than one single formula as the input. The results in this paper have as an immediate implication the sequent-based characterization of prioritized default logics (weakest-link), and assumption-based argumentation due to known ASPIC^+ translation results (cf. [18,28]). Different attack-types and rules, such as joint attacks in various formulas, have been extensively investigated in sequent-based argumentation [4,3], and future work needs to investigate such rules in ASPIC^+ systems [14,20,19]. Furthermore, the translation can be harnessed to investigate characterizations of non-monotonic logics in the various systems. For instance, prioritized default logic can be expressed in ASPIC^+ [25]. Now we know how a sequent-based characterization looks like. Relatedly, alternative liftings from preference orderings over defeasible rules to strengths of arguments remain to be investigated. For instance, in the context of input/output logics [26,27] –characterized in sequent-based argumentation [10]– and sequent-based argumentation [2] prioritized extensions have been proposed that go beyond weakest-link and last-link liftings. The modularity of sequent-based approaches allows for the introduction of additional sequent-rules, such as a rule for disjunctive reasoning (by cases) as introduced in [21] and characterized in [10,9]. Last, our generalized approach to ASPIC^+ allows for hybrid systems, such as those mixing weakest-link and last-link reasoning. The logical properties and application potential of such hybrid methods remain to be studied.

Last, recent results on Argument Calculi [8] establish finite, terminating proof-search procedures for sequent-style arguments (including attackers), such as those generated by our approach. Those procedures decompose the (strict and defeasible) rules of an argument to formulas of the base logic, reducing derivability to derivability in the base logic. These results apply to our flat approach but require future work to incorporate reasoning with preferences.

Conclusion. In this work, we investigated the relations between ASPIC^+ and sequent-based argumentation as two central paradigms of logical argumentation. We showed that, despite the more “flat” nature of sequent-based arguments, compared to tree-like arguments in ASPIC^+ , reasoning with priorities can be integrated into sequent-style argumentation, thus resulting in a first correspondence result between these frameworks. The immediate consequences of this result are as follows: i) sequent-based argumentation allows for flexibility in choosing a base logic, as well as distinguishing between base logic reasoning and additional strict reasoning assumptions (avoiding the need for infinite

sets of base logic axioms); ii) different attack rules can be expressed through calculus rules expressing the conditions of such attacks logically, instead of defining attacks on a meta-level, which conceptually means that we integrate meta-argumentative reasoning about attacks, inapplicability of rules, priorities, and argument strength within the calculus; iii) moreover, sequent-based argumentation allows for a compact representation of arguments, where one does not need to specify whole derivation trees while preserving the possibility of representing nuanced reasoning with priorities.

Appendix

We now outline the proof strategy of our main result (Thm. 3.11).

Definition .1 We define $\mathcal{R}_d^+$ iteratively as follows:

- If $a = \langle \phi \rangle$ with $\phi \in \mathcal{A}$ we let $\mathcal{R}_d^+(a) = \emptyset$
- If $a = \langle a_1, \dots, a_n \rightarrow \phi \rangle$ we let $\mathcal{R}_d^+(a) = \bigcup_{i=1}^n \mathcal{R}_d^+(a_i)$.
- If $a = \langle a_1, \dots, a_n \rightsquigarrow \phi \rangle$ then let $\mathcal{R}_d^+(a) =$

$$\bigcup_{i=1}^n \mathcal{R}_d^+(a_i) \cup \begin{cases} \{(r, \min(\{\text{strength}(r)\} \cup \bigcup_{(r',x) \in \bigcup_{i=1}^n \mathcal{R}_d^+(a_i)} \{x\}))\} & \text{if } r \in \mathcal{R}_d^w \\ \{(r, \text{strength}(r))\} & \text{if } r \in \mathcal{R}_d^l \end{cases}$$

That is: $\mathcal{R}^+(a)$ stores the information about the strength of the rules in $\mathcal{R}_d(a)$ at the moment of their application. The following translation functions are at the core of the proof:

Definition .2 We define two translation functions $\tau_{as2sb} : \text{Arg}_{\text{as}}(\mathbb{K}) \rightarrow \text{Arg}_{\text{sb}}(\mathbb{K})$ and $\tau_{as2sb}^+ : \text{Arg}_{\text{as}}(\mathbb{K}) \rightarrow \wp(\text{Arg}_{\text{sb}}(\mathbb{K}) \setminus \{\emptyset\})$ iteratively as follows:

- Let $a = \langle \phi \rangle$ where $\phi \in \mathcal{A}$. Then let $\tau_{as2sb}(a) = a' = \phi \Rightarrow_{\infty} \phi$ and $a' \in \tau_{as2sb}^+(a)$.
- Let $a = \langle a_1, \dots, a_n \rightarrow \psi \rangle$ where $\text{Con}(a_1), \dots, \text{Con}(a_n) \rightarrow \psi \in \mathcal{R}_s^{\text{dom}}$. Consider, where $\text{strength}(a_i) = k_i$,

$$\frac{\frac{\vdots}{\tau_{as2sb}(a_1) : \Delta_1 \Rightarrow_{k_1} \phi_1} \quad \dots \quad \frac{\vdots}{\tau_{as2sb}(a_n) : \Delta_n \Rightarrow_{k_n} \phi_n}}{a' : \Delta_1, \dots, \Delta_n, (\phi/\psi) \Rightarrow_{\min\{k_1, \dots, k_n\}} \psi} \text{DETS}$$

Then let $a' = \tau_{as2sb}(a)$ and $a' \in \tau_{as2sb}^+(a)$.

- Let $a = \langle a_1, \dots, a_n \rightarrow \psi \rangle$ where $\text{Con}(a_1), \dots, \text{Con}(a_n) \rightarrow \psi \in \mathcal{R}_s^{\text{bl}}$ with $\text{strength}(a_i) = k_i$. Consider

$$\frac{\frac{\vdots}{\tau_{as2sb}(a_1) : \Delta_1 \Rightarrow_{k_1} \phi_1} \quad \dots \quad \frac{\vdots}{\tau_{as2sb}(a_n) : \Delta_n \Rightarrow_{k_n} \phi_n} \quad \frac{\vdots}{\phi \Rightarrow_{\text{bl}} \psi}}{a' : \Delta_1, \dots, \Delta_n \Rightarrow_{\min\{k_1, \dots, k_n\}} \psi} \text{DETBL}$$

Then let $a' = \tau_{as2sb}(a)$ and $a' \in \tau_{as2sb}^+(a)$.

- Let $a = \langle a_1, \dots, a_n \rightsquigarrow \psi \rangle$ based on the defeasible rule $r : \phi_1, \dots, \phi_n \rightsquigarrow \psi \in \mathcal{R}_d^w$ with $\text{strength}(r) = k$ and $\text{strength}(a_i) = k_i$. Consider

$$\frac{\frac{\vdots}{\tau_{as2sb}(a_1) : \Delta_1 \Rightarrow_{k_1} \phi_1} \quad \dots \quad \frac{\vdots}{\tau_{as2sb}(a_n) : \Delta_n \Rightarrow_{k_n} \phi_n}}{a' : \Delta_1, \dots, \Delta_n, [(\phi/\psi)]_{\min\{k, k_1, \dots, k_n\}} \Rightarrow_{\min\{k, k_1, \dots, k_n\}} \psi} \text{DET D}$$

Then let $a' = \tau_{as2sb}(a)$ and $a' \in \tau_{as2sb}^+(a)$.

- Let $a = \langle a_1, \dots, a_n \rightsquigarrow \psi \rangle$ based on the defeasible rule $r : \phi_1, \dots, \phi_n \rightsquigarrow \psi \in \mathcal{R}_d^l$ with $\text{strength}(r) = k$ and $\text{strength}(a_i) = k_i$. Consider

$$\frac{\frac{\vdots}{\tau_{as2sb}(a_1) : \Delta_1 \Rightarrow_{k_1} \phi_1} \quad \dots \quad \frac{\vdots}{\tau_{as2sb}(a_n) : \Delta_n \Rightarrow_{k_n} \phi_n}}{a' : \Delta_1, \dots, \Delta_n, [(\phi/\psi)]_k \Rightarrow_k \psi} \text{DET D}$$

Then let $a' = \tau_{as2sb}(a)$ and $a' \in \tau_{as2sb}^+(a)$.

- If a with $\text{strength}(a) = k$ directly attacks (i.e., an attack in the top defeasible rule of b) $b = \langle b_1, \dots, b_n \rightsquigarrow \psi \rangle$ with $\text{strength}(b) = k'$ and $\text{Con}(b_i) = \phi_i$, then consider

$$\frac{\frac{\vdots}{\tau_{as2sb}(a) : \Delta \Rightarrow_k \sigma} \quad \frac{\vdots}{\tau_{as2sb}(b) : \Gamma, [\phi/\psi]_{k'} \Rightarrow_l \psi}}{a' : \Delta \Rightarrow_k \neg[\phi/\psi]_{k'}} \text{REB/UCUT}$$

Then $a' \in \tau_{as2sb}^+(a)$.

Definition .3 $a' \in \text{Arg}_{bs}(\mathbb{K})$ is a sub-argument of $b' \in \text{Arg}_{bs}(\mathbb{K})$ —written $a' \in \text{Sub}(b')$ —if and only if, $\text{Pre}(a') \subseteq \text{Pre}(b')$.

Lemma .4 (i) For every $a' \in \text{Arg}_{sb}(\mathbb{K})$ there is an $a \in \text{Arg}_{as}(\mathbb{K})$ for which $a' \in \tau_{as2sb}^+(\mathbb{K})$. (ii) If $\text{Con}(a') \in \text{WFF}$, then $a' = \tau_{as2sb}(a)$. (iii) Let $b \in \text{Sub}(a)$. Then, for all $b' \in \tau_{as2sb}^+(b)$ and all $a' \in \tau_{as2sb}^+(a)$, $b' \in \text{Sub}(a')$.

Lemma .5 Let $a \in \text{Arg}_{as}(\mathbb{K})$ and $a' \in \tau_{as2sb}^+(a)$. Then,

- (i) for each $(\phi \rightsquigarrow \psi, x) \in \mathcal{R}_d^+(a)$, $[\phi/\psi]_x \in \text{Pre}(a')$,
- (ii) for each $[\phi/\psi]_x \in \text{Pre}(a')$, $(\phi \rightsquigarrow \psi, x) \in \mathcal{R}_d(a)$,
- (iii) $\text{strength}(a) = \text{strength}(a')$, and
- (iv) $\text{Con}(a) = \text{Con}(a')$ if $a' = \tau_{as2sb}(a)$.

Lemma .6 (i) Let $a' \in \tau_{as2sb}^+(a)$ and $b' \in \tau_{as2sb}^+(b)$. If a' attacks b' in $\text{AF}_{sb}(\mathbb{K})$ then a attacks b in $\text{AF}_{as}(\mathbb{K})$. (ii) If a attacks b in $\text{AF}_{as}(\mathbb{K})$, then there is an $a' \in \tau_{as2sb}^+(a)$ that attacks every $b' \in \tau_{as2sb}^+(b)$. (iii) If a' defends b' in $\text{AF}_{sb}(\mathbb{K})$ then it defends every $b'' \in \tau_{as2sb}^+(b)$.

Lemma .7 *Let $\text{Sem} \in \{\text{Admissible}, \text{Complete}, \text{Stable}\}$. (i) If $\mathcal{E} \in \text{Sem}(\text{AF}_{as}(\mathbb{K}))$ then $\bigcup \tau_{as2sb}^+[\mathcal{E}] \in \text{Sem}(\text{AF}_{sb}(\mathbb{K}))$. (ii) If $\mathcal{E}' \in \text{Sem}(\text{AF}_{sb}(\mathbb{K}))$ then $\bigcup \tau_{as2sb}^{+,-1}[\mathcal{E}'] \in \text{Sem}(\text{AF}_{as}(\mathbb{K}))$ (with $\tau_{as2sb}^{+,-1}[\mathcal{E}'] =_{\text{df}} \{a \in \text{Arg}_{as}(\mathbb{K}) \mid \tau_{as2sb}^+(a) \subseteq \mathcal{E}'\}$).*

Proof. We prove (i) paradigmatically for conflict-freeness and admissibility. Let $\mathcal{E}' = \bigcup \tau_{as2sb}^+[\mathcal{E}]$.

Conflict-freeness. Suppose there are $a', b' \in \mathcal{E}'$ such that a' attacks b' . Thus, there is a $a \in \mathcal{E}$ for which $a' \in \tau_{as2sb}^+(a)$ and there is a $b \in \mathcal{E}$ for which $b' \in \tau_{as2sb}^+(b)$. Moreover, by Lemma .6, a attacks b in $\text{AF}_{as}(\mathbb{K})$, in contradiction to the conflict-freeness of $\mathcal{E}$.

Admissibility. Suppose some $b' \in \text{Arg}_{bs}(\mathbb{K})$ attacks $a' \in \mathcal{E}'$. By Lemma .4, there is a $b \in \text{Arg}_{as}(\mathbb{K})$ such that $b' \in \tau_{as2sb}^+(b)$ and $a \in \mathcal{E}$ such that $a' \in \tau_{as2sb}^+(a)$. By Lemma .6, b attacks a in $\text{AF}_{as}(\mathbb{K})$ and by the admissibility of $\mathcal{E}$ there is a $c \in \mathcal{E}$ that attacks b . By Lemma .6, there is $c' \in \tau_{as2sb}(c)$ such that c' attacks b' . Since $c' \in \mathcal{E}'$, $\mathcal{E}'$ defends a' . $\square$

Lemma .8 *Let $\text{Sem} \in \{\text{Preferred}, \text{Grounded}\}$. Then,*

- (i) *If $\mathcal{E} \in \text{Sem}(\text{AF}_{as}(\mathbb{K}))$ then $\bigcup \tau_{as2sb}^+[\mathcal{E}] \in \text{Sem}(\text{AF}_{sb}(\mathbb{K}))$.*
- (ii) *If $\mathcal{E}' \in \text{Sem}(\text{AF}_{sb}(\mathbb{K}))$ then $\bigcup \tau_{as2sb}^{+,-1}[\mathcal{E}'] \in \text{Sem}(\text{AF}_{as}(\mathbb{K}))$.*

Proof. Ad 1 (item 2 is analogous). Let $\text{Sem} = \text{Preferred}$, $\mathcal{E} \in \text{Sem}(\text{AF}_{as}(\mathbb{K}))$ and $\mathcal{E}' = \bigcup \tau_{as2sb}^+[\mathcal{E}]$. By Lemma .7, we know that $\mathcal{E}'$ is complete in $\text{AF}_{sb}(\mathbb{K})$. Assume there is a $\mathcal{E}_2' \in \text{Sem}(\text{AF}_{sb}(\mathbb{K}))$ for which $\mathcal{E}_2' \supset \mathcal{E}'$. By Lemma .7, $\mathcal{E}_2 = \tau_{as2sb}^{+,-1}(\mathcal{E}_2') \in \text{Complete}(\text{AF}_{as}(\mathbb{K}))$. Let $a' \in \mathcal{E}_2' \setminus \mathcal{E}'$. Thus, $\tau_{as2sb}^{+,-1}(a') \subseteq \mathcal{E}_2 \setminus \mathcal{E}$. This immediately implies that $\mathcal{E}_2 \supset \mathcal{E}$ which contradicts the $\subseteq$ -maximality of $\mathcal{E}$ in $\text{Complete}(\text{AF}_{as}(\mathbb{K}))$. The case for $\text{Sem} = \text{Grounded}$ is analogous. $\square$

Corollary .9 *Let Sem be completeness-based. Then,*

- (i) *For every $\mathcal{E} \in \text{Sem}(\text{AF}_{as}(\mathbb{K}))$ there is a $\mathcal{E}' \in \text{Sem}(\text{AF}_{sb}(\mathbb{K}))$ such that $\text{Con}[\mathcal{E}] = \text{Con}[\mathcal{E}'] \cap \text{WFF}$.*
- (ii) *For every $\mathcal{E}' \in \text{Sem}(\text{AF}_{sb}(\mathbb{K}))$ there is a $\mathcal{E} \in \text{Sem}(\text{AF}_{as}(\mathbb{K}))$ such that $\text{Con}[\mathcal{E}] = \text{Con}[\mathcal{E}'] \cap \text{WFF}$.*

Theorem .10 *Let Sem be completeness-based and $\psi \in \text{WFF}$. Then,*

- (i) $\text{AF}_{as}(\mathbb{K}) \vdash_{\cap}^{\star, \text{Sem}} \psi$ iff $\text{AF}_{sb}(\mathbb{K}) \vdash_{\cap}^{\star, \text{Sem}} \psi$, where $\star \in \{a, c\}$.
- (ii) $\text{AF}_{as}(\mathbb{K}) \vdash_{\cup}^{\star, \text{Sem}} \psi$ iff $\text{AF}_{sb}(\mathbb{K}) \vdash_{\cup}^{\star, \text{Sem}} \psi$.

Proof. We paradigmatically show Item 1 for $\star = a$. Suppose $\text{AF}_{sb}(\mathbb{K}) \vdash_{\cap}^{a, \text{Sem}} \psi$ (the other direction is similar). Thus, there is an $a' \in \bigcap \text{Sem}(\text{AF}_{sb}(\mathbb{K}))$ with $\text{Con}(a') = \psi$. Consider $\mathcal{E} \in \text{Sem}(\text{AF}_{as}(\mathbb{K}))$. As previously shown, $\mathcal{E}' = \tau_{as2sb}^+[\mathcal{E}] \in \text{Sem}(\text{AF}_{sb}(\mathbb{K}))$. So, $a' \in \mathcal{E}'$ and so $a = \tau_{as2sb}^{-1}(a') \in \mathcal{E}$. Thus, $a \in \bigcap \text{Sem}(\text{AF}_{as}(\mathbb{K}))$ and therefore $\text{AF}_{as}(\mathbb{K}) \vdash_{\cap}^{a, \text{Sem}} \psi$. $\square$

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Admissibility Before Argumentation: Procedural Semantics for Norm-Governed Reasoning

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Abstract

Abstract argumentation typically presupposes that candidate arguments are already legitimate participants in a framework. In institutional and AI-supported contexts, however, a logically prior question arises: which claims are eligible to participate as arguments in the first place? This paper develops a procedural semantics of admissibility for norm-governed reasoning. Admissibility is treated not as a relational property derived from attacks, but as preservation of an externally specified normative invariant. A norm-indexed reasoning task computes a typed judgment object containing an admissibility value, clause-level diagnostics, and a repairability classification. This makes the proposal more than a pre-filter: it preserves the grounds of eligibility, distinguishes repairable deviation from structural expressiveness limitation, and supports auditable transitions from candidate claims to arguments. We state the assumptions on which the construction depends, refine the scope of the non-relationality result, and illustrate the architecture through an AI-supported regulatory-assurance case. The framework remains conservative: after restriction to the admissible domain, standard Dung-style semantics apply unchanged. Procedural legitimacy and argumentative success are thereby separated without modifying classical argumentation frameworks.

Keywords: abstract argumentation, admissibility, procedural semantics, norm-governed reasoning, evaluation procedures

1 Introduction

Abstract argumentation provides precise semantics for conflict, defence, and justification once a set of arguments and an attack relation are fixed [3]. In many institutional and AI-supported contexts, however, an antecedent question dominates: which claims qualify as admissible arguments *before* attack and defence are even defined? In governance, auditing, compliance checking, and evidence-based evaluation, legitimacy frequently depends on authority, provenance, procedural correctness, and norm alignment rather than on argumentative interaction.

Standard frameworks typically treat these entry conditions informally (e.g. by pre-filtering candidate arguments) or indirectly (e.g. by encoding illegitimacy as attacks). Both moves conflate *procedural illegitimacy* with *argumentative defeat* and erase diagnostic information needed for explanation and repair.

Aim and contribution. This paper introduces a norm-indexed, pre-argumentative layer: a *procedural semantics of admissibility*. The central thesis is that admissibility is best understood as preservation of an externally specified invariant induced by the normative context. The contribution is fourfold:

- (i) an invariant-based reconstruction of admissibility grounded in bilateral (two-agent) role coherence;
- (ii) a typed formulation of the norm-indexed reasoning task that preserves admissibility values, clause-level diagnostics, and repairability information;
- (iii) a scoped non-relationality result: extension outputs over a bare attack relation cannot, in general, recover distinctions that have been discarded before argumentation begins;
- (iv) a conservative interface through which the procedurally admissible domain is handed to standard Dung-style semantics unchanged.

The architectural claim is therefore stronger than “pre-filtering.” A filter records only passage or exclusion. The proposed task also records the normative grounds of that outcome, the reachable repairs, and whether failure belongs to the candidate, the procedure, or the expressive resources of the normative context.

2 Related Work and Positioning

2.1 Abstract argumentation and admissibility

Abstract argumentation frameworks, as introduced by Dung, provide semantics for acceptance based on attack and defence relations over a fixed set of arguments [3]. Within this setting, notions such as admissibility, groundedness, or stability are defined relationally and presuppose that candidate arguments are already legitimate participants in the framework. While structured argumentation frameworks, such as ASPIC+ [5,8,7], allow for premises to be attacked to model invalidity, this approach implies that procedural failures must be resolved through adversarial conflict. By contrast, our work isolates procedural eligibility as a pre-relational condition. The present work therefore does not revise or compete with these semantics. Instead, it addresses a logically prior question: under which conditions do candidate claims qualify to enter an argumentation framework? This entry problem is largely abstracted away in standard frameworks, where illegitimate or procedurally flawed claims are either excluded informally or encoded indirectly as defeated arguments.

2.2 Three different uses of admissibility

The term *admissibility* carries different burdens at different levels of a reasoning architecture. To avoid terminological collision, let

$$\text{Elig}_{\mathcal{N},\tau}(c) \text{ mean } \mathcal{P}_{\mathcal{N}}(c) \succeq \tau,$$

where c is a candidate claim. This is a unary, norm-indexed eligibility judgment. By contrast, Dung admissibility is a property of a set $S \subseteq A$ in an already constituted framework $AF = (A, R)$: S is conflict-free and defends each of its members. A third notion occurs in structured argumentation, where the internal construction of an argument may depend on premises, rules, and preferences.

These notions answer different questions. Procedural eligibility asks whether a candidate may enter the argumentative domain under $\mathcal{N}$. Structured construction asks how an argument is built. Dung admissibility asks whether a set of arguments can withstand attack. The present proposal concerns the first question and remains orthogonal to the latter two. In particular,

$$\text{Elig}_{\mathcal{N},\tau}(c) \neq \{c\} \in \text{Adm}(AF).$$

The left-hand side can be evaluated before R exists; the right-hand side is meaningful only after an argumentation framework has been constructed.

2.3 Norm-governed and institutional reasoning

In institutional, legal, and governance contexts, admissibility is governed by authority, provenance, procedural correctness, and compliance with explicit norms. Whether a report, claim, or assertion may participate in reasoning is often decided independently of its argumentative relations to other claims. Research on normative multi-agent systems has extensively analysed how norms regulate agent behaviour and define obligations [1]. Our proposal complements this by focusing specifically on the constitutive role of norms for argumentation: defining the procedural invariants that determine whether a claim counts as an argument in the first place.

The framework proposed here is motivated by such settings. Admissibility is treated as a norm-indexed judgment grounded in externally specified invariants, rather than as a derivative notion defined by relational semantics.

2.4 Positioning of the contribution

The contribution of this paper is architectural and can be framed within a “logic toolbox” [4] or “agreement technologies” [2,6] perspective. Within such modular architectures, argumentation is typically one downstream component alongside others such as norms, trust, and negotiation. Our proposal functions as an upstream “entry discipline” or evaluation module for this pipeline. It introduces a pre-argumentative layer that determines which claims are eligible to participate as arguments in the first place. By treating admissibility as a norm-indexed check ($\mathcal{P}_{\mathcal{N}}$) with its own diagnostics and repair mechanisms, we isolate the procedural preconditions of reasoning from the adversarial dynamics

of attack and defense. Once admissibility has been procedurally established, standard Dung-style semantics apply unchanged as a downstream engine.

3 Admissibility as an Invariant-Based Concept

Let $\mathcal{C}$ be a set of candidate claims and let $\mathcal{N}$ denote a normative context (clauses, authorities, procedures) governing legitimate participation. We write $(\mathcal{C}, \mathcal{N})$ for a norm-indexed reasoning setting.

Definition 3.1 Invariant. An invariant $I_{\mathcal{N}}$ induced by $\mathcal{N}$ is a condition on claims such that preservation of $I_{\mathcal{N}}$ guarantees coherence under role reversal, perspective switching, or bilateral interpretation within the given reasoning context.

Definition 3.2 Invariant-Based Admissibility. A claim $c \in \mathcal{C}$ is *admissible relative to $\mathcal{N}$* iff c preserves the invariant $I_{\mathcal{N}}$.

3.1 Assumptions and proof dependencies

The results below depend on a small set of explicit assumptions. They are separated here so that the theorem-level claims do not hide their proof obligations.

A1 (Norm determinacy). The normative context $\mathcal{N}$ identifies the clauses, authorities, procedures, and metadata relevant to eligibility.

A2 (Indexed evaluation). Local evaluations remain associated with the clauses that generated them; aggregation does not erase their indices.

A3 (Threshold determinacy). The ordered outcome space $\mathcal{D}$ and threshold τ determine whether a candidate enters $\mathcal{C}_{\text{adm}}$.

A4 (Repair discipline). The family $\text{Rep}_{\mathcal{N}}$ contains exactly the transformations permitted without revising $\mathcal{N}$.

A5 (Extension-only observability). The downstream comparison observes the family of σ -extensions, but no additional clause labels, provenance annotations, or repair traces.

A6 (Conservative hand-off). Attack and defence are evaluated only after the domain $\mathcal{C}_{\text{adm}}$ has been fixed.

The proof architecture concerns the internal structure of the procedural admissibility judgment. It proceeds in three stages:

$$\begin{array}{ccc}
 \text{clause-indexed evaluation} & & \\
 \Downarrow & & \\
 \text{diagnostic traceability} & & (1) \\
 \Downarrow & & \\
 \text{classification by threshold and repair reachability.} & &
 \end{array}$$

First, each candidate claim is evaluated against the individually indexed clauses of the normative context. Second, the clause indices are preserved in the resulting diagnostic object, so that the grounds of the judgment remain recoverable.

Third, the admissibility threshold and the repair closure determine whether the claim is directly admissible, inadmissible but repairable, or subject to an expressiveness limitation.

Assumptions A1–A4 govern these three stages. Assumption A5 is used only for the non-relationality result: it specifies that the observable output of the downstream argumentation semantics consists solely of its extensions. Assumption A6 concerns the subsequent architectural interface. Once the procedurally admissible domain has been determined, it can be passed unchanged to a classical argumentation framework, thereby establishing semantic conservativity.

3.2 Two-agent role coherence (minimal motivation)

Consider two agents A and B and a role-inversion operation $\theta : \mathcal{C} \rightarrow \mathcal{C}$ exchanging their roles.

Definition 3.3 Role-invariance. A claim $c \in \mathcal{C}$ is *role-invariant* if c and $\theta(c)$ remain coherent under the normative interpretation induced by $\mathcal{N}$.

Proposition 3.4 Role coherence as a necessary admissibility condition. *In a two-agent reasoning setting, admissibility (as legitimate participation) entails preservation of role-invariance.*

Remark 3.5 Role-invariance is a structural precondition for admissible participation; it does not imply agreement. Disagreement concerns acceptance outcomes; role coherence concerns the stability of the reasoning situation under role reversal. Role coherence provides a minimal motivating instance of invariant preservation; it is not intended as an exhaustive characterisation of all normative invariants. In institutional settings, invariants may additionally concern authority, provenance, evidential form, temporal validity, or procedural competence.

Two-agent role coherence should be understood as a minimal instance of a more general requirement: participants must share sufficient normative structure for their claims to be mutually interpretable and evaluable. Role reversal tests whether a claim remains coherent when the positions of the participating agents are exchanged. Institutional settings formalise and extend this reciprocal structure by specifying the common ground against which claims are assessed. Such common ground may include recognised authorities, shared evidential categories, provenance requirements, procedural competences, and rules governing the interpretation of claims. Institutional invariants therefore do not replace bilateral role coherence; they make its underlying demand for reciprocal intelligibility operational within a norm-governed domain.

4 Limits of Relational Admissibility Semantics

In Dung-style frameworks, admissibility is defined relationally over (A, R) where A is a set of arguments and $R \subseteq A \times A$ is attack [3]. This presupposes that all elements of A are already legitimate arguments. In norm-governed settings, that presupposition fails: a claim may be disqualified by authority/provenance/procedure independently of attack and defence.

The key mismatch is diagnostic: institutional admissibility requires *typed* or clause-indexed failure modes, while Dung-style semantics deliver extension sets (accepted/rejected) without guaranteeing recovery of distinct normative violation types.

Proposition 4.1 No-Go: Relational encoding of norm-indexed diagnostics. *There exist norm-governed reasoning settings $(\mathcal{C}, \mathcal{N})$ and an admissibility procedure $\mathcal{P}_{\mathcal{N}} : \mathcal{C} \rightarrow \mathcal{D}$ with diagnostic traceability such that for no attack relation $R \subseteq \mathcal{C} \times \mathcal{C}$ and no Dung-style semantics σ whose **only observables** are the σ -extensions, can the following be simultaneously ensured:*

- (i) **Extensional faithfulness:** *for all $c \in \mathcal{C}$, $\mathcal{P}_{\mathcal{N}}(c) \succeq \tau$ iff c is accepted according to σ in $(\mathcal{C}, R)$ (under a fixed acceptance convention);*
- (ii) **Diagnostic recoverability:** *if $\mathcal{P}_{\mathcal{N}}(c_1) \neq \mathcal{P}_{\mathcal{N}}(c_2)$ due to different clause-violations (or different violation types), then this distinction is recoverable from the observable σ -extensions alone.*

Argument. Choose c_1, c_2 that are both procedurally inadmissible but for different clause-indexed reasons. Any semantics whose output is restricted to extension sets can make both unavailable or rejected to match extensional faithfulness. Under A5, however, the observable contains no clause index or repair trace from which the different reasons can be reconstructed. The distinction has been forgotten before or at the relational interface and is therefore not recoverable from the extensions alone. $\square$

Scope of the result. The proposition does not claim that argumentation systems are incapable of representing diagnostics. Auxiliary arguments, labels, meta-arguments, provenance structures, or richer semantics may encode them. The point is more precise: once typed normative and repair information is added, diagnostic recoverability is supplied by that additional structure rather than by extension-based relational semantics alone. The procedural layer makes this dependency explicit instead of hiding it inside an ad hoc relational encoding.

4.0.1 Illustrative example

Consider a normative context $\mathcal{N}$ governing admissibility of analytical reports in an institutional setting. Suppose $\mathcal{N}$ requires that admissible claims (i) be issued by an authorised role, and (ii) be supported by documented evidence of a prescribed form.

Let φ be a claim produced by an authorised agent and supported by extensive qualitative reasoning, but not by evidence conforming to the prescribed documentation standard. Assume further that no repair operation permitted by $\mathcal{N}$ allows the evidential form itself to be transformed into the required type; repairs may correct omissions or formatting errors, but may not introduce new evidential categories.

In this case, φ is not defeated by counterargument, nor does it violate authority constraints. Nevertheless, for all $\psi \in \text{Cl}_{\mathcal{N}}(\varphi)$, the admissibility procedure yields $\mathcal{P}_{\mathcal{N}}(\psi) \prec \tau$. The failure is therefore not attributable to incorrect

content or procedural error, but to the limited expressive resources of $\mathcal{N}$ itself.

This example illustrates an expressiveness limitation: the normative context cannot license the claim as a legitimate reasoning act without extending its evaluative vocabulary or admissible evidence types. Norm updates in such cases are best understood as expressiveness extensions rather than as relaxations of standards.

5 Procedural Semantics for Admissibility

The no-go result suggests a disciplined response: keep classical argumentation untouched, but make admissibility explicit as a pre-argumentative, norm-indexed procedure. To frame this precisely in Knowledge Representation (KR) terms, we formalise the reasoning task as a modular input-output transformation. The upstream module takes the raw claims and metadata as input, produces a diagnostic object $\Delta_{\mathcal{N}}$ and the admissible domain $\mathcal{C}_{adm}$ as output, and passes only the latter downstream for extensional evaluation under a chosen argumentation semantics σ .

Definition 5.1 Admissibility Reasoning Task. Given a set of candidate claims $\mathcal{C}$ and a normative context $\mathcal{N}$ (comprising clauses, procedures, and metadata), the admissibility reasoning task is to compute for each $c \in \mathcal{C}$ a tuple $\langle v, \Delta_{\mathcal{N}}, \kappa_{\mathcal{N}} \rangle$, where:

- (i) $v = \mathcal{P}_{\mathcal{N}}(c) \in \mathcal{D}$ is the admissibility value, determining whether $c \in \mathcal{C}_{adm}$ relative to a threshold τ ;
- (ii) $\Delta_{\mathcal{N}}(c)$ is a diagnostic vector recording the evaluation status of specific clauses in $\mathcal{N}$ (e.g., {authority: OK, evidence: fail});
- (iii) $\kappa_{\mathcal{N}}(c)$ is a repairability classification, distinguishing between *repairable deviation* (where $\exists c' \in \text{Cl}_{\mathcal{N}}(c) : \mathcal{P}_{\mathcal{N}}(c') \succeq \tau$) and *expressiveness limitation* (where no such c' exists).

This explicit output structure ensures that inadmissibility is not merely a binary rejection, but a transparent state that guides subsequent repair or system evaluation.

Definition 5.2 Admissibility procedure. An admissibility procedure is a function

$$\mathcal{P}_{\mathcal{N}} : \mathcal{C} \rightarrow \mathcal{D},$$

where $\mathcal{D}$ is an outcome space (binary or graded) and $\mathcal{P}_{\mathcal{N}}(c)$ reflects the degree to which c preserves $I_{\mathcal{N}}$.

Definition 5.3 Procedural admissibility. Fix a threshold $\tau \in \mathcal{D}$. A claim c is procedurally admissible iff $\mathcal{P}_{\mathcal{N}}(c) \succeq \tau$.

Procedural admissibility separates (i) eligibility for participation from (ii) argumentative success under attack and defence. It also makes admissibility auditable: outcomes are indexed to $\mathcal{N}$ and can be traced to clause-level criteria.

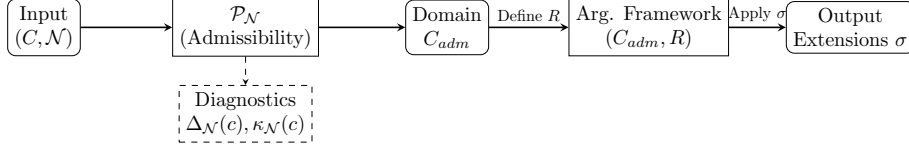


Fig. 1. The Architectural Pipeline: Admissibility as an upstream entry discipline. The procedure $\mathcal{P}_N$ filters candidates and produces diagnostics, while standard semantics σ operate conservatively on the admissible domain C_{adm} .

Table 1. Comparison of Relational and Procedural Admissibility

Aspect	Relational admissibility	Procedural admissibility
Primary object	Arguments	Candidate claims
Failure mode	Defeat by attack	Normative deviation
Norm sensitivity	Implicit or indirect	Explicit and indexed
Outcome type	Binary acceptance	Graded/diagnostic
Repair support	Not represented	First-class operation
Role of norms	Encoded in attacks	Governing invariants

6 Deviation, Diagnosis, and Repair

Procedural failure is not defeat: it is deviation from the invariant induced by $\mathcal{N}$.

Definition 6.1 Deviation. A claim c exhibits a *deviation* if $\mathcal{P}_N(c) \prec \tau$.

Definition 6.2 Diagnostic outcome. A diagnostic outcome is an admissibility result that records whether I_N is preserved and, if not, which clause-components (or violation types) fail.

Definition 6.3 Repair. A repair operation is a transformation $\rho : \mathcal{C} \rightarrow \mathcal{C}$ such that for deviating c ,

$$\mathcal{P}_N(\rho(c)) \succ \mathcal{P}_N(c).$$

Proposition 6.4 Admissibility vs. acceptance. *Repair concerns eligibility for participation and may restore admissibility without implying acceptance under any argumentation semantics.*

6.1 Expressiveness limits of repair

Not all inadmissibility is repairable under a fixed normative context. Some claims remain inadmissible because $\mathcal{N}$ lacks the expressive resources to license them as reasoning acts.

Definition 6.5 Repair closure. Let Rep_N be the set of repair operations allowed under $\mathcal{N}$. Define the repair closure

$$\text{Cl}_N(\varphi) := \{\rho(\varphi) \mid \rho \in \text{Rep}_N^*\},$$

where Rep_N^* is the closure under composition and identity.

Remark 6.6 Algorithmic bounds on repair. In practical KR implementations, computing the full abstract closure $\text{Cl}_{\mathcal{N}}(\varphi)$ may be computationally intractable. Realistic systems operationalise this by imposing bounded repair depths or by introducing a cost model for the operations in $\text{Rep}_{\mathcal{N}}$. Furthermore, if the admissibility aggregation is monotone, heuristic search strategies can efficiently guide the repair process toward the threshold τ without computing the entire closure.

Definition 6.7 Expressive adequacy. $\mathcal{N}$ is *expressively adequate* for φ if $\exists \psi \in \text{Cl}_{\mathcal{N}}(\varphi)$ such that $\mathcal{P}_{\mathcal{N}}(\psi) \succeq \tau$.

Definition 6.8 Expressiveness limitation. φ exhibits an *expressiveness limitation* relative to $\mathcal{N}$ if

$$\forall \psi \in \text{Cl}_{\mathcal{N}}(\varphi), \quad \mathcal{P}_{\mathcal{N}}(\psi) \prec \tau.$$

This notion isolates a precise failure mode: the claim is not defeated by counterargument and not correctable by internal repair; rather, it lies beyond what $\mathcal{N}$ can licence as an admissible reasoning act. Norm updates may therefore be understood as *expressiveness extensions* (richer clause vocabularies, additional evaluative dimensions, or new admissible repair moves) rather than as “lowering standards.”

6.2 Admissibility, expressiveness, and non-reductive rationality

An expressiveness limitation is not a deficiency of rationality or evidence of inconsistency. It marks a boundary of what a fixed normative context can recognise as a legitimate reasoning act. A claim may be factually correct yet procedurally inadmissible, or procedurally admissible yet rejected downstream; admissibility therefore concerns participation conditions rather than truth or argumentative success. This distinction also clarifies institutional responses. Repairable deviations call for correction within $\mathcal{N}$, whereas persistent limitations call for reflection on the expressive adequacy of $\mathcal{N}$. A norm update may then be understood as an expressive extension—adding admissible vocabularies, evidence types, evaluative dimensions, or repair moves—rather than simply lowering a standard. Conversely, retaining a restrictive $\mathcal{N}$ may be rational where stability, comparability, or auditability outweigh coverage.

6.3 Illustrative example: inadmissibility without defeat

Let $\mathcal{N} = \{n_{\text{auth}}, n_{\text{source}}, n_{\text{proc}}\}$ require identification of a responsible authority, traceability to an admissible source, and evaluability under a shared procedural standard. Consider three claims about the same subject: c_1 is complete, sourced, and well formed; c_2 omits source attribution; and c_3 rejects the applicability of the procedural standard itself. Let $\mathcal{P}_{\mathcal{N}}(c)$ return a vector of clause satisfactions.

For c_1 , $\mathcal{P}_{\mathcal{N}}(c_1) = \langle \text{ok}, \text{ok}, \text{ok} \rangle$, so $c_1 \in C_{\text{adm}}$. For c_2 , the vector is $\langle \text{ok}, \text{fail}_{\text{source}}, \text{ok} \rangle$. Supplying the missing source yields a compliant vector, so the deviation is internal to $\mathcal{N}$ and repairable. For c_3 , the vector is

$\langle \text{ok}, \text{ok}, \text{fail}_{proc} \rangle$. No repair permitted by $\mathcal{N}$ can make c_3 accept n_{proc} without changing the normative context; this is therefore an expressiveness limitation.

A standard extension-based semantics may map both c_2 and c_3 to “rejected”, whereas the procedural layer preserves the diagnostic difference: c_2 creates a repair task for the agent, while c_3 creates a norm-design question for the institution.

Proposition 6.9 Procedural trichotomy and non-relationality. *Fix a norm-indexed setting $(\mathcal{C}, \mathcal{N})$ with admissibility procedure $\mathcal{P}_{\mathcal{N}} : \mathcal{C} \rightarrow \mathcal{D}$, threshold $\tau \in \mathcal{D}$, and repair family $\text{Rep}_{\mathcal{N}} \subseteq \mathcal{C}^{\mathcal{C}}$. Let $\text{Rep}_{\mathcal{N}}^*$ be the closure of $\text{Rep}_{\mathcal{N}}$ under composition and identities, and define the repair closure*

$$\text{Cl}_{\mathcal{N}}(c) := \{\rho(c) \mid \rho \in \text{Rep}_{\mathcal{N}}^*\}.$$

For each $c \in \mathcal{C}$ exactly one of the following holds:

- (i) $\mathcal{P}_{\mathcal{N}}(c) \succeq \tau$.
- (ii) $\mathcal{P}_{\mathcal{N}}(c) \prec \tau$ and $\exists \psi \in \text{Cl}_{\mathcal{N}}(c)$ with $\mathcal{P}_{\mathcal{N}}(\psi) \succeq \tau$.
- (iii) $\mathcal{P}_{\mathcal{N}}(c) \prec \tau$ and $\forall \psi \in \text{Cl}_{\mathcal{N}}(c), \mathcal{P}_{\mathcal{N}}(\psi) \prec \tau$.

Moreover, for any attack relation $R \subseteq \mathcal{C} \times \mathcal{C}$ and any Dung-style semantics σ whose observable output is the family of σ -extensions, there exist choices of $(\mathcal{N}, \mathcal{P}_{\mathcal{N}}, \tau, \text{Rep}_{\mathcal{N}})$ and claims $c_2, c_3 \in \mathcal{C}$ such that c_2 satisfies (ii), c_3 satisfies (iii), and the σ -extensions of $(\mathcal{C}, R)$ do not determine which of (ii) or (iii) holds.

Proof sketch. The trichotomy is immediate from the definitions: either $\mathcal{P}_{\mathcal{N}}(c) \succeq \tau$ (case (i)), or $\mathcal{P}_{\mathcal{N}}(c) \prec \tau$, and then either admissibility is reachable within the repair closure (case (ii)) or it is not (case (iii)); mutual exclusivity follows from the $\exists/\forall$ dichotomy over the same set $\text{Cl}_{\mathcal{N}}(c)$.

For the non-relationality statement, pick $c_2, c_3 \in \mathcal{C}$ such that both are extensionally rejected under the chosen acceptance convention, but where c_2 satisfies (ii) (it can be repaired into the threshold) and c_3 satisfies (iii) (it cannot). Since σ -outputs are sets of accepted arguments (extensions), they need not encode the clause-indexed or repair-reachability distinction between c_2 and c_3 without additional typed structure beyond extensions. $\square$

7 An Exemplar Evaluation Procedure

We present a compact exemplar showing how $\mathcal{P}_{\mathcal{N}}$ can be implemented as clause-indexed evaluation plus aggregation, while retaining diagnostic traceability.

Let $\mathcal{N} = \{n_1, \dots, n_k\}$ be a finite set of normative clauses. For each clause define a local evaluation function

$$e_i : \mathcal{C} \rightarrow \mathcal{V},$$

where $\mathcal{V}$ is a partially ordered valuation domain (e.g. satisfied / partial / violated, or a graded scale).

Definition 7.1 Aggregation. An aggregation function is

$$A : \mathcal{V}^k \rightarrow \mathcal{D}.$$

Define the admissibility procedure by

$$\mathcal{P}_{\mathcal{N}}(c) = A(e_1(c), \dots, e_k(c)).$$

Proposition 7.2 Traceability. *If A is computed from the tuple $(e_1(c), \dots, e_k(c))$ without discarding indices, then $\mathcal{P}_{\mathcal{N}}(c)$ admits a diagnostic trace back to the clause-level evaluations.*

Proposition 7.3 Repair guidance. *If $\mathcal{P}_{\mathcal{N}}(c) \prec \tau$, then improving at least one violated clause-evaluation $e_i(c)$ yields a non-decreasing admissibility outcome under monotone A .*

Finally define the admissible domain

$$\mathcal{C}_{\text{adm}} = \{c \in \mathcal{C} \mid \mathcal{P}_{\mathcal{N}}(c) \succeq \tau\},$$

which is the input set of arguments for subsequent relational evaluation.

8 Domain Case: AI-Supported Regulatory Assurance

Consider an AI system assisting an assurance professional in evaluating whether a corporate report satisfies a disclosure requirement. The system is not yet deciding whether the company’s substantive claim is ultimately justified. Its first task is to determine whether the extracted claim is a legitimate input to that decision.

Let the normative context be

$$\mathcal{N}_A = \{n_{\text{scope}}, n_{\text{authority}}, n_{\text{source}}, n_{\text{form}}, n_{\text{trace}}\},$$

where the clauses require that the disclosure falls within scope, is attributable to a responsible governance role, is linked to an admissible source, instantiates the required evidence form, and is traceable to a report location. For a candidate c , the AI produces

$$J_{\mathcal{N}_A}(c) = \langle \mathcal{P}_{\mathcal{N}_A}(c), \Delta_{\mathcal{N}_A}(c), \kappa_{\mathcal{N}_A}(c) \rangle,$$

where κ classifies the candidate as admissible, repairable, or expressively unsupported.

Suppose the system extracts three candidates concerning the same governance policy. Candidate c_1 identifies the policy, the responsible body, a report location, and the prescribed evidence. Candidate c_2 expresses substantially the same proposition but lacks a traceable page or source reference. Candidate c_3 contains a detailed and plausible narrative, yet the reporting framework recognises only a different evidence category and offers no permitted transformation from the supplied narrative to that category.

The resulting judgments may be represented as follows:

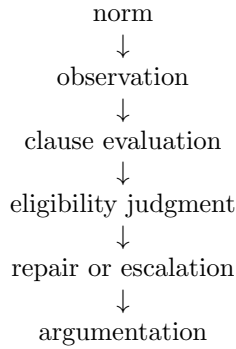
$$\begin{aligned} J_{\mathcal{N}_A}(c_1) &= \langle 1, \langle ok, ok, ok, ok, ok \rangle, \text{admissible} \rangle, \\ J_{\mathcal{N}_A}(c_2) &= \langle 0, \langle ok, ok, fail, ok, fail \rangle, \text{repairable} \rangle, \\ J_{\mathcal{N}_A}(c_3) &= \langle 0, \langle ok, ok, ok, fail, ok \rangle, \text{expressiveness limitation} \rangle. \end{aligned}$$

For c_2 , the system can request a source location or documentary link. This is a workload for the reporting agent or assessor. For c_3 , no repair allowed under $\mathcal{N}_A$ changes the evidence type; the issue must be escalated to the designer or owner of the normative framework. This is a workload for the institution. Neither case is yet an argumentative defeat.

Only c_1 , and a repaired version of c_2 , enter the downstream framework. They may subsequently attack or support other arguments and may still fail under the selected semantics. The architecture therefore preserves four operationally different outcomes: eligibility, repairable ineligibility, structural ineligibility, and downstream defeat. Collapsing them into a single “rejected” status would obscure who must act, which evidence is missing, and whether the norm itself is inadequate.

8.1 Why this is an assurance task rather than a filter

A conventional filter computes a characteristic function $\chi : C \rightarrow \{0, 1\}$. The assurance task computes a richer judgment object $J_{\mathcal{N}}$ and retains the evidence path by which the characteristic decision was reached. That path supports review, challenge, and reproducibility:



The binary domain restriction is only one projection of this object. In regulated AI, the preserved projections are often as important as the final eligibility decision because they provide the basis for explanation and accountability.

9 Compatibility with Abstract Argumentation

Let $A := \mathcal{C}_{\text{adm}}$ and let $R \subseteq A \times A$ be any attack relation defined *only on admissible claims*. Let σ be any Dung-style semantics.

Restriction view. For each normative context $\mathcal{N}$ and threshold τ , the admissibility procedure induces a domain selection operation

$$F_{\mathcal{N},\tau} : \mathcal{C} \rightharpoonup \mathcal{C}_{\text{adm}},$$

where $F_{\mathcal{N},\tau}$ may be read as the canonical “restriction to admissible candidates” given by

$$F_{\mathcal{N},\tau}(c) = \begin{cases} c & \text{if } \mathcal{P}_{\mathcal{N}}(c) \succeq \tau, \\ \text{undefined} & \text{otherwise.} \end{cases}$$

Accordingly, any abstract argumentation framework considered in this paper is obtained by first restricting the candidate universe and then defining attacks on the restricted domain:

$$\mathcal{C} \xrightarrow{F_{\mathcal{N},\tau}} \mathcal{C}_{\text{adm}} \rightsquigarrow (A, R) \text{ with } A = \mathcal{C}_{\text{adm}}.$$

In this sense the procedural layer is an *input discipline* for abstract argumentation, not a competing acceptance semantics.

Proposition 9.1 Layer separation. *Procedural admissibility determines A but does not constrain the definition of R beyond its domain restriction to A .*

Proposition 9.2 Semantic conservativity. *For any semantics σ , the σ -extensions of (A, R) are computed exactly as in classical abstract argumentation, since neither R nor σ is modified.*

Proposition 9.3 No conflation. *Procedurally inadmissible claims are not treated as defeated arguments; they are excluded as non-arguments relative to $\mathcal{N}$.*

9.1 Categorical factorisation via a restriction (partial) map

We isolate the procedural layer as a restriction operation expressed in the category of partial maps. Let $\mathbf{Set}_{\text{par}}$ denote the category whose objects are sets and whose morphisms $X \rightharpoonup Y$ are partial functions (equivalently, spans $X \xleftarrow{m} D \xrightarrow{f} Y$ with m a monomorphism).

Fix $(\mathcal{C}, \mathcal{N})$ with $\mathcal{P}_{\mathcal{N}} : \mathcal{C} \rightarrow \mathcal{D}$ and threshold τ . Define the admissible domain

$$\mathcal{C}_{\text{adm}} := \{c \in \mathcal{C} \mid \mathcal{P}_{\mathcal{N}}(c) \succeq \tau\},$$

and let $m_{\mathcal{N},\tau} : \mathcal{C}_{\text{adm}} \hookrightarrow \mathcal{C}$ be the inclusion. Define the partial identity (restriction) map

$$F_{\mathcal{N},\tau} : \mathcal{C} \rightharpoonup \mathcal{C}, \quad F_{\mathcal{N},\tau}(c) = c \text{ iff } c \in \mathcal{C}_{\text{adm}}.$$

Equivalently, $F_{\mathcal{N},\tau}$ is represented by the span $\mathcal{C} \xleftarrow{m_{\mathcal{N},\tau}} \mathcal{C}_{\text{adm}} \xrightarrow{m_{\mathcal{N},\tau}} \mathcal{C}$.

Proposition 9.4 Restriction factorisation of argumentation. *Let $\mathbf{AF}$ be the class of abstract argumentation frameworks. For each $(\mathcal{N}, \tau)$, any choice of attack relation $R \subseteq \mathcal{C}_{\text{adm}} \times \mathcal{C}_{\text{adm}}$ determines a framework $(A, R) \in \mathbf{AF}$ with $A = \mathcal{C}_{\text{adm}}$. Moreover, the construction factorises as*

$$\mathcal{C} \xrightarrow{F_{\mathcal{N},\tau}} \mathcal{C}_{\text{adm}} \xrightarrow{\iota} (\mathcal{C}_{\text{adm}}, R),$$

where $F_{\mathcal{N},\tau}$ is a morphism in $\mathbf{Set}_{\text{par}}$ (a partial restriction) and ι is the inclusion of the admissible domain as the argument set of an abstract framework. In particular, for any Dung-style semantics σ , the computation of σ -extensions is performed entirely on $(\mathcal{C}_{\text{adm}}, R)$ and is independent of the behaviour of $\mathcal{P}_{\mathcal{N}}$ on $\mathcal{C} \setminus \mathcal{C}_{\text{adm}}$.

Proof sketch. The partial map $F_{\mathcal{N},\tau}$ is the identity on its domain of definition $\mathcal{C}_{\text{adm}}$ and undefined elsewhere, hence is a morphism in $\mathbf{Set}_{\text{par}}$ represented by the indicated span. Given any R on $\mathcal{C}_{\text{adm}}$, the resulting framework has argument set $A = \mathcal{C}_{\text{adm}}$. Dung-style semantics are defined on frameworks (A, R) , so their evaluation depends only on A and R and not on inadmissible candidates. $\square$

Note that while we treat arguments as abstract entities in $\mathcal{C}_{\text{adm}}$, their internal structure can essentially be anything. Therefore, our procedural layer is orthogonal to structured argumentation frameworks like ASPIC+ [5,8,7], which can be applied downstream to model the internal logic of the procedurally admissible claims.

9.2 Restriction and information loss

The categorical factorisation also clarifies what is lost at the interface. Let the full procedural judgment be

$$J_{\mathcal{N}}(c) = \langle \mathcal{P}_{\mathcal{N}}(c), \Delta_{\mathcal{N}}(c), \kappa_{\mathcal{N}}(c) \rangle,$$

where κ records repairability. The restriction map depends only on the first component relative to τ . There is therefore a projection

$$U_{\tau} : J_{\mathcal{N}}(\mathcal{C}) \longrightarrow \{\text{in}, \text{out}\},$$

which preserves eligibility but forgets the clause-level grounds and repair classification. This is the precise sense in which a purely extensional view is myopic: it may preserve the correct restricted domain while losing the reasons by which that domain was constituted.

The procedural architecture does not prohibit this forgetting; it controls it. Diagnostics remain available upstream for audit and repair, whereas the downstream argumentation engine receives exactly the structure it requires. The separation is therefore both informational and conservative: richer information is retained without burdening R or σ with responsibilities that do not belong to attack and defence.

10 Epistemic Status, Assurance, and Limitations

Procedural eligibility is neither truth nor knowledge. A true claim may be ineligible because its source or form is not licensed; an eligible claim may be false; an eligible claim may be defeated downstream; and an accepted argument may still fail to justify knowledge. The relevant distinctions are therefore

$$\text{eligibility} \neq \text{acceptance} \neq \text{truth} \neq \text{knowledge}.$$

The procedural judgment supports a narrower but important assurance claim: given the stated normative context, evaluative functions, and evidence trace, the candidate was admitted or excluded for reproducible reasons. It warrants the integrity of participation, not the truth of the proposition.

Several limitations follow. First, norm determinacy may fail where clauses are contested or open-textured. In such cases $\mathcal{P}_N$ may itself require interpretation or argumentation. This does not collapse the architecture, but it makes the boundary recursive: argumentation may be needed to settle the governing interpretation before the candidate-level procedure can run. Second, the repair closure may be computationally large, requiring bounded search or cost-sensitive approximation. Third, threshold and aggregation choices can introduce institutional bias; their governance must therefore be visible rather than treated as neutral implementation details. Finally, the present account specifies the interface but not a complete implementation language for norms, evidence types, or repairs. These are natural subjects for domain-specific extensions.

11 Conclusion

This paper developed procedural admissibility as a distinct norm-indexed reasoning task rather than an informal pre-filter. Its output is a typed judgment object that combines an eligibility value, indexed diagnostics, and a repairability classification. On that basis, the framework distinguishes direct admissibility, repairable deviation, and expressiveness limitation before any attack relation is defined. The scoped non-relationality result shows why extension outputs alone cannot reconstruct distinctions that have already been forgotten at the interface, while the categorical factorisation identifies the downstream hand-off as a partial restriction. The regulatory-assurance case demonstrated the operational consequence: candidate repair, institutional escalation, and argumentative defeat allocate different responsibilities and should not be collapsed. The proposal remains conservative with respect to abstract argumentation, because attacks and Dung-style semantics operate unchanged after restriction to C_{adm} . It thereby preserves diagnostic and assurance information upstream while providing a clean interface to classical argumentation downstream.

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A Multilingual Argument and Value Mining Corpus from Public Debate

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Abstract

We introduce YOU CAN YOU BB (YCYBB), a multimodal and multilingual resource for argument mining and value mining, based on the Chinese debate television show *I Can I BB* (*qípā shuō*, 奇葩说). The show combines structured pro/con argumentation, expert panellists who challenge implicit assumptions, and audience votes collected before, during, and after each debate. To our knowledge, YCYBB is the first planned corpus to integrate multimodal Chinese debate, argument and value mining, expert normative interventions, and observable persuasion outcomes.

This paper presents the dataset construction methodology and a proof-of-concept pilot. We release: (i) an end-to-end open-source annotation pipeline for downloading episodes, WhisperX-based diarised transcription, batched LLM annotation with local or proprietary backends, and structured export; (ii) an annotation schema combining Walton-style argumentation schemes, an open value vocabulary, expert-intervention types such as hidden premise, value reframe, fallacy identification, and perspective shift, plus bilingual Chinese-English claim and premise fields; and (iii) a pilot annotation of one full episode, with 156 utterances and 140–156 LLM-generated annotations depending on backend.

We also propose a six-step LLM-as-judge validation protocol, discuss pilot limitations (fragmented value labels and overuse of **other**), and report results for its first four steps on the pilot episode.

Keywords: television debate, argument mining, value mining, LLM-as-judge, argumentation schemes, multimodal annotation, multilingual corpus

1 Introduction

Argument mining (AM) and value mining have matured rapidly over the past decade [12,11], but the resources that drive the field remain dominated by English text, by static genres (essays, online comments, parliamentary records), and by a separation between the *content* of argumentation (claims, premises, schemes) and the *values* that underwrite it. Multimodal settings, in which arguments unfold in real time before an audience whose preferences shift visibly, are underrepresented; non-Western argumentative cultures even more so.

This paper takes a first step toward filling that gap. We are building the YOU CAN YOU BB corpus, a multimodal and multilingual Chinese argumentation-and-value-mining resource constructed from *I Can I BB* (奇葩说), a Chinese debate show that has run since 2014 and that combines (i) teams of debaters arguing for and against a motion; (ii) expert panellists from law, philosophy, comedy and the social sciences who intervene to expose hidden premises, reframe values, or flag fallacies; and (iii) a studio audience whose live votes are recorded across the debate, making persuasion an observable outcome [14]. The conjunction of these three features makes the show an unusually rich *laboratory* for computational argumentation [21], and the corpus we are constructing is, to our knowledge, the first to combine multimodal debate, joint argument and value mining, expert normative interventions, and observable persuasion outcomes in a single multilingual resource.

The present submission is a *proof-of-concept toward that corpus*, not its release. We deliver three concrete artefacts that together establish the construction methodology and demonstrate it on one fully-annotated episode.

(C1) An open-source annotation pipeline (download, WhisperX + pyannotate diarised transcription, batched LLM annotation via Ollama-served Gemma4 31B or the Anthropic API, and JSON + CSV export). The annotator LLM is a configuration choice, so the corpus can be re-annotated with successive open-weights models at zero marginal cost — long-lived infrastructure combining human supervision with LLM-boosted automation.

(C2) An annotation schema that captures speaker role, speech act, claim, premises, a Walton-style scheme [19], an open value list, and a relation to other utterances, plus a distinctive *expert intervention* layer with normative types. Claim and premise fields are emitted bilingually (Chinese + English).

(C3) A pilot annotation and validation protocol. We annotate one full episode end to end and specify a six-step LLM-as-judge validation protocol with judges (Qwen3 32B, GPT-5 mini) disjoint from the annotator backends, followed by human evaluation. Steps 1–4 are executed and reported here; the human tier is in progress.

This is a *preliminary work* documenting the construction infrastructure of a corpus whose full release is forthcoming. This work sits at the intersection of formal argumentation frameworks [8,4,2,13] and normative reasoning, and the expert-intervention layer in particular is intended to make normative judgements about arguments computationally tractable.

2 Related Work

Argument-mining corpora. The main existing argument mining (AM) resources cluster around three written genres: persuasive essays [17], user-generated web discourse [9], and public debates exemplified by the US2016 presidential corpus [18]. End-to-end systems such as IBM DEBATER [16] and IAM [6] extend the toolchain but their argumentation units remain primarily written. None of these resources jointly capture (a) live spoken argument, (b) audience vote shifts, and (c) explicit normative interventions by experts in the debate. Moreover, Chinese AM is an even less explored area [20]. Inference Anchoring Theory [3] (IAT) is a standard framework for relating dialogical and argumentative structure in spoken corpora. Our schema is designed to be migratable to it in the future.

Value mining. Following the human-value identification task of Kiesel et al. [11], value mining typically uses a fixed Schwartz-style taxonomy [15] or a moral-foundations inventory [10]. These taxonomies were validated predominantly on Western respondents and may miss culturally specific values present in Chinese debate (e.g., *filial piety*, *face/dignity*, *harmony*). Our schema (Section 5) therefore leaves the value list *open* and defers consolidation to downstream analysis.

LLM-based annotation and LLM-as-judge. Annotation by large language models is now widely used as a low-cost seed for human-corrected corpora. The methodology of using LLMs as *judges* of model output, popularised by *MT-Bench* [22], brings well-known biases (self-preference, position, length) documented in subsequent work [5]; our protocol (Section 7) is designed with these biases in mind.

3 The Debate Show as an Empirical Setting

I Can I BB (奇葩说) is an iQIYI-produced Chinese-language debate programme. Three features make it attractive as a corpus source. (i) *Pro/con structure*: each episode debates a motion drawn from ethics, social policy, relationships, technology, or popular culture (e.g., S6E4: *If a museum is on fire, save the painting or save the cat?*); two teams alternate opening statements, free-debate rounds, and closing statements, with stable speaker roles that make diarisation followed by speaker-to-role mapping tractable. (ii) *Expert interventions*: a standing panel of mentors intervenes between rounds, typically identifying hidden premises, reframing the dilemma in terms of a different value, flagging a fallacy, or introducing an unrepresented perspective. The expert turn is a natural locus for studying normative judgement *about* an argument [14], not just argumentation itself. (iii) *Audience vote dynamics*: the studio audience votes pro/con before, during, and after the debate; the resulting vote-shift trajectory gives a behavioural measure of persuasion that can be matched to specific argumentative moves (Figure 1). Pro/con structure alone exists in parliamentary corpora; expert moderation in court transcripts; vote outcomes in legislative records. We are not aware of an existing resource

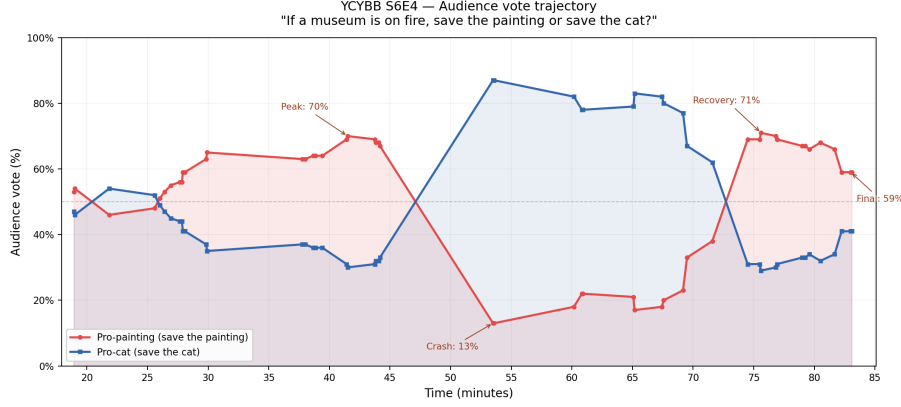


Fig. 1. Audience vote trajectory for S6E4 (49 frame-based snapshots). Pro-painting (red) and pro-cat (blue) shares move sharply: a 54-point fall between minutes 44 and 54 (67% to 13% pro-painting), a recovery to a peak of 71% by minute 76, and a final reading of 59%. Vote percentages are extracted automatically: the pipeline samples video frames and reads the on-screen vote bar with a vision-capable LLM (Gemma 4), yielding an utterance-level `vote_delta` alignable to specific argumentative moves (Section 4).

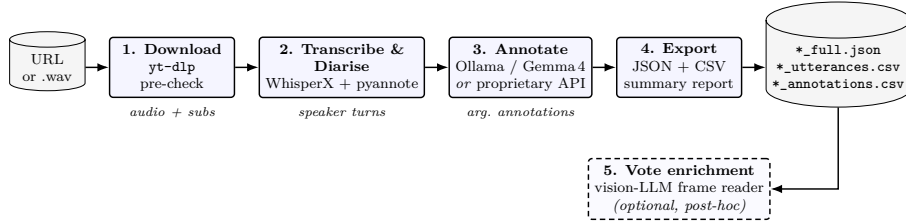


Fig. 2. The YCYBB pipeline. Cylinders denote on-disk artefacts; rectangles denote processing stages. Each stage can be re-run independently from the previous stage’s output.

that combines the three (i.e., pros/cons structures, expert moderation, opinion votes), in any language.

4 Annotation Pipeline Architecture

The pipeline (Figure 2) is implemented in a single Python script. Its four core stages are deliberately decoupled: each can be re-run in isolation, and the output of any stage is a stable, inspectable file. An optional fifth stage (vote enrichment) runs post-hoc on the completed episode.

Stage 1 (Download). We use `yt-dlp`¹ for URL-based ingestion (YouTube, Bilibili) and accept a local `.wav` for sources that `yt-dlp` does not handle (notably iQIYI, the show’s original distributor). A preliminary check

¹ <https://github.com/yt-dlp/yt-dlp>

extracts metadata without downloading and reports potential error messages and remediations (e.g., cookie errors). The stage outputs a 16 kHz mono WAV suitable for Whisper.

Stage 2 (Transcribe + diarise). We use WhisperX [1] (Whisper `large-v3`, with forced-alignment of word-level timestamps). Speaker diarisation is performed with `pyannote`². After diarisation we merge consecutive turns from the same speaker that are separated by less than two seconds, which produces discourse-level utterances rather than ASR-level segments.

Stage 3 (LLM annotation). Two backends: *Gemma 4 31B*³ (Ollama tag `gemma4:31b`) via Ollama⁴ (default; zero-cost) and a proprietary API (Anthropic, model `claude-sonnet-4-20250514`). Utterances are batched with a system prompt specifying the schema, strict JSON output, bilingual claim/premise fields (Section 5.4), temperature 0.1, and a 32,768-token context window.

Stage 4 (Export). Output: a canonical `*_full.json`, two CSVs, and a human-readable summary report.

Stage 5 (Vote enrichment, optional). Audience vote shares are not present in the transcript and are recovered post-hoc from the video track by a separate enrichment on the completed episode. The pipeline samples video frames at a configurable interval (5 s in the pilot; 30 s default) and sends each frame to a vision-capable LLM (Gemma 4) which reads the on-screen vote bar and returns pro/con percentages as JSON. Frames in which the vote bar is not visible are discarded. The result is a step function of vote share over episode time. For an annotation spanning $[t_{\text{start}}, t_{\text{end}}]$ in the WhisperX word-level alignment, `vote_delta` is $v(t_{\text{end}}) - v(t_{\text{start}})$ with v linearly interpolated between adjacent snapshots.

Reproducibility and cost. A 101-minute episode takes 30–60 min on the local backend (M2 Max); no paid services in the default configuration. The cloud backend costs $\sim$ USD 0.80 per episode under Claude Sonnet 4.

5 Annotation Schema

The JSON schema contains *utterance annotations* and a *debate-context* block. We describe each field below; Table 1 summarises the core annotation and Table 3 shows a complete instance from the pilot.

5.1 Argumentation schemes

Following Walton, Reed and Macagno [19], we expose an initial inventory of seven schemes (argument from consequences, from analogy, from authority, from example, and from values; appeal to emotion; and reductio ad absurdum) plus an `other` category. These patterns cover some of the most frequent Walton schemes in argumentation theory, and were chosen for tractable reuse in

² <https://github.com/pyannote/pyannote-audio>

³ <https://ai.google.dev/gemma>

⁴ <https://ollama.com>

Table 1
Core fields of an **ArgumentAnnotation**. The schema is a flat JSON object per utterance (or per group of utterances).

Field	Description
utterance_ids	Utterance(s) the annotation covers.
role	debater-pro / -con / expert / host / audience.
speech_act	assert / rebut / concede / question / meta-comment / narrative.
current_phase	Debate phase: opening / round_1 / round_2 / ... / free_debate / expert_round / closing.
claim	Single-sentence claim (English).
claim_zh	Chinese rendering of claim .
premises	List of supporting reasons (English).
premises_zh	Chinese rendering of premises .
scheme	Walton-style scheme (Sec. 5.1).
relation_type	attack / support / neutral.
related_to	Utterance ids responded to.
values_invoked	Open list of values (Sec. 5.2).
is_expert_intervention	Boolean.
expert_norm_type	hidden premise / value reframe / fallacy id / perspective shift / none.
vote_delta	Change in audience support (%) during the utterance, interpolated from vision-extracted vote snapshots.
confidence	LLM self-reported [0,1].

Note: **speech_act** uses purely communicative categories intended to map onto IAT/AIF illocutionary forces, while structural phase information is captured separately in **current_phase**; **narrative** marks a turn that recounts a case or personal story rather than asserting a proposition directly, and **meta-comment** a turn about the conduct of the debate rather than about the motion. **vote_delta** is not produced by the annotation LLM but derived from the vote-enrichment stage (Section 4).

downstream formal frameworks (ASPIC+ [13] treats each scheme as a defeasible inference rule with named critical questions). Section 6 reports the actual scheme distribution on the pilot and motivates extending the typology in the next iteration: 49% of Gemma’s annotations and 40% of Claude’s were tagged as **other**; Section 9 decomposes this bucket, part of which reflects the show’s humour, anecdote and idiom-based moves.

5.2 Open value vocabulary

The **values_invoked** field is a free-form list. We deliberately avoid a fixed taxonomy at annotation time: Schwartz basic values [15] and moral foundations [10] were developed in predominantly Western survey populations and do not natively express Confucian-influenced values that are common in this show (*filial piety, face, harmony, collective duty*). Consolidation of the open vocabulary into a canonical taxonomy is a downstream analysis step rather than part of annotation itself (Section 9). This keeps the dataset’s labels reversible: if we later decide a different canonical taxonomy fits better, we re-cluster the same

Table 2
Mapping from schema fields to formal-argumentation frameworks.

Framework	Schema fields used
Dung AF	<code>relation_type=attack</code> , <code>related_to</code>
Bipolar AF	+ <code>relation_type=support</code>
Value-based AF	+ <code>values_invoked</code>
ASPIC+	+ <code>claim</code> , <code>premises</code> , <code>scheme</code>

open vocabulary without re-annotating.

5.3 Expert interventions

The schema’s *distinctive* layer concerns expert turns. When `is_expert_intervention` is true, `expert_norm_type` takes one of four values:

- **hidden premise** — the expert names an unstated assumption on which one or both teams’ arguments depend;
- **value reframe** — the expert introduces a different evaluative dimension;
- **fallacy identification** — the expert flags a reasoning error or rhetorical manipulation;
- **perspective shift** — the expert introduces a completely different viewpoint or stakeholder perspective.

This layer is the contribution most directly relevant to ongoing debates in the community around *quality norms* of argumentation [21]. It treats expert panellists as embodied normative judges whose moves can be encoded and, eventually, learned.

5.4 Bilingual annotation

The pipeline emits both the English and the Chinese rendering of `claim` and `premises` in a single JSON object; controlled-vocabulary fields keep their canonical English form. This preserves source-language content for downstream analysis, enables cross-lingual judge agreement (Section 7, Step 3), and positions the corpus as genuinely multicultural rather than as a translation.

5.5 From annotations to an argument graph

The `related_to` and `relation_type` fields define a directed graph over utterances that maps directly onto the standard formal frameworks (Table 2): attack edges become Dung-style attacks [8], support edges become bipolar supports [4], `values_invoked` underwrites a value-based AF [2], and the `claim/premises/scheme` fields supply ASPIC+ content [13].

6 Pilot Annotation

We piloted the pipeline on Season 6, Episode 4 of *I Can I BB* (YouTube id iDyC07rhqPQ; 101 min; topic “美术馆着火了，救名画还是救猫？” — *If a museum is on fire, save the painting or save the cat?*). Audio was extracted with `yt-dlp`, transcribed and diarised with WhisperX `large-v3` + `pyannote`

Table 3

An example annotation from the pilot. During the expert round of S6E4 (*If a museum is on fire, save the painting or save the cat?*), the expert recounts a case — a journalist deciding whether to publish a story that could help many people but harm a specific interviewee — in order to name an assumption shared by both debate teams: that a distant aggregate benefit is commensurable with an individual’s life. The annotation (produced by Gemma 4 on utterance 81) captures this as a **narrative** speech act with a **hidden premise** normative type. The **premises** list contains a second entry, and **relation_type**, **related_to** and **vote_delta** are omitted for space.

Utterance 81 — Gemma 4, expert panellist	
role	expert
speech_act	narrative
current_phase	expert_round
claim	The tension between saving many people (distant cries) and saving one person (near cries) is a false dichotomy when it involves sacrificing a real person for an imagined benefit.
claim_zh	当涉及为了想象中的利益而牺牲一个真实的人时，救远方的哭声与救近处的哭声之间的紧张关系是一个伪命题。
premises	A journalist friend once asked if they should publish a story to help many people by exposing social darkness, even if it would lead to the death of the person being interviewed.
premises_zh	一位记者朋友曾问过我，是否应该为了揭露社会阴暗面以帮助更多人而发布报道，即使这意味着被采访者会被残害而死。
scheme	argument from example
values_invoked	[individualism, ethics]
is_expert_intervention	true
expert_norm_type	hidden premise
confidence	0.85

(18 distinct speakers detected), and annotated by two backends: Gemma 4 31B (gemma4:31b) served locally through Ollama (temperature 0.1, batch size 8, context window 32,768) and Claude Sonnet 4 (claude-sonnet-4-20250514) via the Anthropic API. Both backends used the bilingual annotation mode, producing claim and premise fields in Chinese and English.

Volume and diarisation. The pilot produced 156 utterances (after same-speaker merging, threshold 2s). Gemma 4 generated 140 annotations (16 utterances dropped due to batch-level JSON failures); Claude Sonnet generated 156 annotations, covering every utterance. Both annotation sets include bilingual claim and premise fields (Chinese coverage: 136/140 for Gemma, 156/156 for Claude).

Scheme distribution. Table 4 reports the LLM-assigned scheme counts. Two observations are immediate: (i) *argument from values* is the largest *specific* category, consistent with the show’s high density of value-laden moves;

(ii) **other** is by far the largest category overall (68/140, 49% for Gemma; 62/156, 40% for Claude). The **other** class collects culturally specific moves (humour, personal anecdote, idiom reuse) which might explain the difficulty to assign those arguments to the Walton typology; we discuss this in Section 9. A related signal comes from the six Claude annotations that fall outside the inventory. Five carry labels that are not argumentation schemes at all, but values belonging to other fields of the schema — **narrative**, for instance, is a **speech_act** value. Only the sixth, argument from experience, is a genuine scheme proposal. The pattern suggests that a single prompt covering the whole schema invites confusion between fields; constraining the output per field is a straightforward remedy.

Table 4

Argumentation-scheme counts on the S6E4 pilot for Gemma 4 ($N = 140$) and Claude Sonnet ($N = 156$). The **other** bucket dominates in both backends. Rows sum to N . *Unparsed* counts annotations with an empty **scheme** string; *out-of-inventory* counts valid labels outside the seven-scheme inventory of Section 5.1 (**narrative**×3, **meta-comment**, **fallacy identification**, **argument from experience**), all emitted by Claude.

Scheme	Gemma	Claude
other	68	62
argument from values	27	28
argument from consequences	11	20
argument from example	14	10
reductio ad absurdum	9	4
argument from analogy	6	12
appeal to emotion	3	5
argument from authority	1	9
unparsed / empty	1	0
out-of-inventory	0	6
Total	140	156

Value vocabulary. The pilot returned 80 distinct value labels from Gemma and over 170 from Claude. In both cases the distribution is heavily skewed: most labels occur only once or twice, and — writing counts as Gemma/Claude — only empathy (12/6), compassion (7/9), humanity (9/1) and life (5/4) exceed four occurrences in either backend. The backends also differ non-trivially in which label they prefer for what appears to be the same concept — humanity is used nine times by Gemma and once by Claude. Many labels are near-synonyms (*compassion/empathy/humanity*; *cultural heritage/cultural preservation/historical continuity*). This argues for an explicit consolidation step downstream, but it also confirms the design choice of leaving the value field open at annotation time: the long tail contains items (*anti-elitism*, *aesthetic of destruction*, *intellectual honesty*, *narrative value*) that no pre-existing taxonomy would have offered as targets.

Expert interventions. Both backends marked a small subset of utterances as expert interventions, predominantly of type *hidden premise* and *value reframe*. The evaluation (Section 8) reveals that `expert_norm_type` is the weakest field, suggesting that classifying the *type* of normative move is harder for current LLMs than detecting its presence.

7 A six-step validation protocol

Before the corpus can be released, we need quantitative evidence that the LLM-generated annotations are of defensible quality, and we need to characterise where they fail. Full human annotation at the scale of even a 30-episode pilot is premature; LLM annotations alone, equally, are insufficient as a release-grade resource. We therefore adopt the following six-step validation protocol, designed to combine cheap, broad LLM-as-judge coverage [22] with targeted human evaluation where it matters most. Steps 1–4 have been executed on the pilot episode and are reported in Section 8; Steps 5–6 are in progress.

Step 1 — Annotation sets. Two annotation sets are produced over the same 156 utterances of episode `iDyC07rhqPQ`, differing only in annotator backend: the Gemma 4 31B set (140 annotations) and the Claude Sonnet set (156 annotations, produced with Anthropic’s API at $\sim$ USD 0.80). All subsequent steps treat the two sets symmetrically.

Step 2 — Judge self-consistency. Before measuring inter-judge agreement, measure each judge’s agreement with itself. On a $\sim$ 15-annotation subsample drawn from each annotation set, run each judge *twice*: once with the rubric items in the documented order and once with them reordered. Report intra-judge Krippendorff’s α per field. If a judge disagrees substantially with itself, the inter-judge numbers in Step 4 are uninterpretable and the protocol pauses until the judging prompt is stabilised.

Step 3 — 2×2 evaluation matrix. Two judges, both *disjoint* from the annotator set: **Qwen 3 32B** (`qwen3:32b`, local, Chinese-native) and **GPT-5 mini** (`gpt-5-mini`, OpenAI API). Neither shares training lineage with the candidates, removing self-preference bias [5]. Each judge scores every annotation on a 4-point ordinal rubric (0 = wrong; 3 = fully accurate) across nine fields (`claim` through `expert_norm_type`; N/A when inapplicable). Qwen scores the Chinese rendering; GPT-5 the English rendering, so disagreement additionally diagnoses cross-lingual translation fidelity. The result is a {Gemma, Claude} $\times$ {Qwen, GPT-5} matrix with all four cells equally interpretable.

Step 4 — Reporting. For each backend we report (a) per-field accuracy as the average rubric score from each judge, (b) inter-judge Krippendorff’s α per field, treating N/A as missing data and the remaining levels as ordinal, and (c) a *disputed set*: the annotations on which the two judges differ by ≥ 2 rubric points on at least one field. The disputed set is the input to Step 5.

Step 5 — Targeted human evaluation. Two trained Chinese-fluent human annotators (PhD students with prior argumentation training) are reviewing (a) $\sim$ 10 *randomly sampled* annotations, single-annotated and balanced across

the two backends, giving an unbiased quality estimate; and (b) the ~ 15 – 20 *disputed* annotations from Step 4, double-annotated by both, giving a failure-mode diagnostic plus an inter-annotator κ on the hardest cases.

Step 6 — Backend comparison. The 2×2 matrix from Step 3, calibrated against the human numbers from Step 5, yields a direct Gemma-vs-Claude comparison.

8 Preliminary Evaluation Results

We ran Steps 1–4 of the protocol on the pilot episode. Table 5 reports the per-field mean rubric scores (0–3 scale) from the 2×2 matrix; Table 6 reports the self-consistency α for each judge–annotation pair.

Table 5
Per-field mean rubric scores (0–3) across the 2×2 matrix. Higher is better.

Field	Gemma 4		Claude Sonnet	
	Qwen	GPT-5	Qwen	GPT-5
claim	2.44	2.61	2.71	2.86
premises	2.37	2.10	2.58	2.80
scheme	2.14	2.30	2.20	2.40
relation_type	2.59	1.84	2.59	2.38
role	2.14	2.12	2.01	2.11
speech_act	2.49	2.67	2.46	2.64
current_phase	1.83	2.06	1.48	2.01
values_invoked	2.37	2.33	2.60	2.65
expert_norm_type	0.00	0.33	0.69	0.11

Table 6
Judge self-consistency (Krippendorff’s α , ordinal, rubric-order shuffled, $n = 15$ subsample). Values above 0.67 are conventionally considered adequate; N/A indicates too few non-missing data points.

Field	On Gemma		On Claude	
	Qwen	GPT-5	Qwen	GPT-5
claim	0.63	0.69	0.76	0.89
premises	1.00	−0.07	0.70	−0.47
scheme	−0.61	0.32	0.58	0.57
relation_type	1.00	0.37	0.03	0.06
role	0.40	0.34	0.70	0.76
speech_act	0.69	0.38	0.66	−0.59
values_invoked	0.91	0.67	0.09	0.57
current_phase	0.72	0.79	1.00	0.19
expert_norm_type	N/A	N/A	N/A	N/A

Mean scores. Core fields (`claim`, `premises`, `scheme`, `speech_act`, `values_invoked`) fall in the 2.0–2.9 range (“mostly correct” to “fully accurate”). Claude Sonnet edges ahead on `claim` (2.71–2.86 vs. 2.44–2.61) and `premises` (2.58–2.80 vs. 2.10–2.37). The weakest field is `expert_norm_type` (0.00–0.69), confirming that classifying normative-move *types* remains hard for current LLMs.

Self-consistency. `claim` is the most robust ($\alpha = 0.63$ – 0.89); several fields show high variance: `scheme` (-0.61 – 0.58), `relation_type` (0.03 – 1.00), `values_invoked` (0.09 – 0.91). GPT-5 mini is the less self-consistent judge overall (mean $\alpha = 0.34$ against 0.58 for Qwen3, with negative values in three of sixteen cells against one), possibly because its temperature is locked at 1.0; at $n = 15$ this difference should be read as indicative only. These instabilities confirm the protocol’s design: self-consistency identifies the fields where judge scores need caution, and Step 5 human evaluation targets them. `expert_norm_type` returns N/A in all cells due to sparsity in the 15-annotation subsample — an issue that resolves at corpus scale.

9 Discussion and Limitations

Pilot scale and ASR fidelity. A single episode does not support empirical claims about argumentative behaviour; with $n = 15$ subsamples, individual α values are noisy and negative values likely reflect small-sample variance rather than systematic failures. Whisper `large-v3` degrades on overlapping speech and proper nouns; manual correction is part of the human-evaluation tier.

What is in other? We categorised Gemma’s $N = 68$ `other` annotations using Gemma 4 as a downstream classifier into five recurring patterns: humour and self-deprecation ($n = 10$), personal anecdote used as evidence ($n = 6$), audience-directed rhetorical questions ($n = 3$), classical-idiom (成语) reuse ($n = 1$), and a residual *other* ($n = 48$). The Claude bucket ($N = 62$) shows a similar distribution: humour 6 against Gemma’s 10, personal anecdote 7 against 6, rhetorical questions 1 against 3, and classical idiom 1 against 1. In both backends, however, the recurring patterns we can name account for a minority of the bucket — 20 of 68 annotations for Gemma, 15 of 62 for Claude — leaving a residual *other* of 48 and 47 respectively, close to a third of all annotations in each set, with no pattern we can currently characterise.

The Claude bucket ($N = 62$) shows a similar distribution: humour 6 against Gemma’s 10, personal anecdote 7 against 6, rhetorical questions 1 against 3, classical idiom 1 against 1, and residual *other* 47 against 48. A manual inspection of the residual reveals that it is almost entirely *non-argumentative* production content: sponsor reads, host procedural announcements, vote-count transitions, and the show’s closing sequence on YouTube. This decomposition suggests two distinct remedies for the next iteration. The 24–29% of `other` that carries non-Walton argumentative moves (humour, anecdote, idiom, rhetorical question) is what the announced scheme-inventory extension (Section 5.1) is intended to absorb. The remaining 71–76% is a different problem: it consists of

utterances that were sent to the annotator despite carrying no argument, and calls for a lightweight filter that decides, before schemes are assigned, whether an utterance warrants annotation at all.

Open value vocabulary. The open vocabulary captures cultural specificity but produces near-synonyms; downstream clustering onto a hybrid Schwartz/moral-foundations taxonomy [15,10] augmented with culturally specific values from the long tail is planned.

Judge bias mitigations. Judges are disjoint from the annotator set, removing self-preference; bilingual emit (Section 5.4) lets each annotation be judged in its source language; Step 5 anchors judge-level numbers against human κ .

Ethics and data rights. The episodes are made publicly available by iQIYI and can be accessed without payment. We will not redistribute the source recordings: the envisaged artefacts comprise the diarised transcript with utterance-level timestamps, the derived annotations, and the open-source processing pipeline. No audio or video is included, and re-running the pipeline requires each user to obtain the source video themselves. Audio is processed locally and is not retained beyond transcription, and speakers are identified only by structural role (debater-pro, debater-con, expert, host, audience) rather than by name. For the reproductions and extractions necessary to conduct the analysis, we rely on the text-and-data-mining exception for scientific research established by Article 3 of Directive (EU) 2019/790, which allows research organisations to carry out text and data mining on works to which they have lawful access.

10 Future Work and Conclusion

The next steps are: (a) complete the human evaluation (Step 5, currently underway), recording the calibration-anchor κ ; (b) export to AIF [7] for interoperability with OVA/AIFdb; (c) release a 30-episode batch using the current schema, in which `speech_act` carries communicative function and `current_phase` captures debate structure, with labels designed to map onto IAT illocutionary forces; and (d) publish a downstream demonstration converting the graph to a value-based AF with preferred extensions under different value orderings.

We have presented the construction methodology and a proof-of-concept pilot for what will, to our knowledge, be the first corpus combining multimodal Chinese debate, joint argument and value mining, expert normative interventions, and observable persuasion outcomes in a single multilingual resource.

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Axiomatization of Parity Modal Logic in Image-Finite Kripke Frames

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Abstract

Parity modal logic (PML) is obtained by extending basic modal logic with formulas of the form $O\varphi$ and $E\varphi$, which reads “there is an odd/even number of successors satisfying φ ”, respectively. We axiomatize PML with respect to image-finite Kripke frames, and give the soundness and completeness proof.

Keywords: Axiomatization, parity modal logic, logic and counting

1 Introduction

Basic modal logic has the power of talking about some basic numerical information, e.g. whether there is no successor satisfying φ , or there is at least 1 successor satisfying φ . In graded modal logic [4,8,12], the language has stronger expressive power, namely it can talk about whether there are more than n successors satisfying φ . Modal logic with cardinality comparison modalities [11,6] can express numerical information like “there are more successors satisfying φ than successors satisfying ψ ”. Modal logic extended with the majority modality can express “the majority of the successors satisfy φ ”, which has been studied in [9,10,5]. Axioms for modal logics with numerically weighted modalities has been studied in [7] where modalities are labelled with weights in the interval $[0, 1]$. For a survey on results on axiomatic completeness and complexity of formal languages for reasoning about finite numbers, see [13].

In the present paper, we introduce parity modal logic PML, which is obtained by extending basic modal logic with formulas of the form $O\varphi$ and $E\varphi$, which reads “there is an odd/even number of successors satisfying φ ”, respectively. We investigate the axiomatization of PML with respect to image-finite Kripke frames, where each world has finitely many successors, and give the soundness and completeness proof.

The special feature of this language is that the modalities O and E are neither monotone nor antitone, which means that even if $O\varphi$ is true at a world,

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$O(\varphi \vee \psi)$ and $O(\varphi \wedge \psi)$ are not necessarily true there, and similar for E . This is different from graded modal logic where the $\Diamond_n$ modalities (which means that at least n successors satisfy a certain formula) and the $\Box_n$ modalities (which means that at most $n - 1$ successors does not satisfy a certain formula) are monotone, majority operator $\mathbb{M}$ which is monotone, and cardinality comparison operator $\# \varphi \lesssim \# \psi$ which is monotone in φ and antitone in ψ . The non-monotonic features of the parity modalities make it interesting to study their performance in axiomatization.

The paper is organized as follows: Section 2 gives all the necessary preliminaries on PML. Section 3 gives the Hilbert system of PML with respect to image-finite Kripke frames and the soundness and completeness proofs. Section 4 gives the conclusions as well as future directions.

2 Preliminaries

This section gives the preliminaries of PML. In the paper, we assume familiarity with basic modal logic [2]. We use the abbreviation “iff” for “if and only if”, “w.r.t.” for “with respect to” and “w.l.o.g.” for “without loss of generality”.

2.1 Syntax

Given a countable set $\mathbf{Prop}$ of *propositional variables*, we define the *formulas* and *numerical terms* of PML as follows:

$$\begin{aligned} \text{formulas: } & p \mid \perp \mid \top \mid \neg\varphi \mid \varphi \wedge \psi \mid \Diamond\varphi \mid O\varphi \\ \text{numerical terms: } & \# \varphi \end{aligned}$$

where $p \in \mathbf{Prop}$. The abbreviations $\vee, \rightarrow, \leftrightarrow$ are as usual and $\Box\varphi$ is defined as $\neg\Diamond\neg\varphi$, $E\varphi$ is defined as $\neg O\varphi$.²

Intuitively, $O\varphi$ means that the current world has an odd number of successors satisfying φ , and $E\varphi$ means that the current world has an even number of successors satisfying φ , and $\# \varphi$ means the number of successors satisfying φ .

The next definition of counting depth is important in the completeness proof:

Definition 2.1 [Counting depth] The *counting depth* of formulas is defined recursively as follows:

- $d(p) = d(\perp) = d(\top) = 0$.
- $d(\neg\varphi) = d(\varphi)$.
- $d(\varphi \wedge \psi) = \max\{d(\varphi), d(\psi)\}$.
- $d(\Diamond\varphi) = d(O\varphi) = d(\varphi) + 1$.

Intuitively, the counting depth of φ is just the maximal number of O and $\Diamond$ symbols in a single branch of the syntax tree of φ .

² Notice that here we work in image-finite Kripke frames, therefore the number of successors of a world is either odd or even.

2.2 Semantics

Let $W \neq \emptyset$ be the *domain* of (possible) worlds, and $\mathcal{P}(W)$ be the *powerset* of W .

PML-formulas are interpreted on *Kripke frames* $\mathbb{F} = (W, R)$ where $W \neq \emptyset$ is the domain and $R \subseteq W \times W$ is the *accessibility relation*. A *Kripke model* is a tuple $\mathbb{M} = (\mathbb{F}, V)$ where $V : \mathbf{Prop} \rightarrow \mathcal{P}(W)$ is a *valuation* on W . The set of *successors* of s is denoted as $R_s = \{t : Rst\}$ and the set of worlds in $\mathbb{M}$ where φ is true is denoted as $\llbracket \varphi \rrbracket^{\mathbb{M}}$. A Kripke frame is:

- *finite*, if W is a finite set;
- *image-finite*, if R_s is a finite set for every $s \in W$.

Throughout the paper, we work with image-finite Kripke frames where each world has finitely many successors.

Definition 2.2 [Satisfaction relation] Given a Kripke model $\mathbb{M} = (W, R, V)$ and $s \in W$, the *satisfaction relation* is defined as follows:

- $\mathbb{M}, s \Vdash p$ iff $s \in V(p)$;
- $\mathbb{M}, s \Vdash \perp$: never;
- $\mathbb{M}, s \Vdash \top$: always;
- $\mathbb{M}, s \Vdash \neg\varphi$ iff $\mathbb{M}, s \Vdash \varphi$ does not hold;
- $\mathbb{M}, s \Vdash \varphi \wedge \psi$ iff both $\mathbb{M}, s \Vdash \varphi$ and $\mathbb{M}, s \Vdash \psi$ hold;
- For numerical terms,

$$\llbracket \# \varphi \rrbracket^{\mathbb{M}, s} = |R_s \cap \llbracket \varphi \rrbracket^{\mathbb{M}}|,$$

i.e. $\llbracket \# \varphi \rrbracket^{\mathbb{M}, s}$ is the cardinality of the set of successors of s where φ is true.

- For $\Diamond\varphi$,

$$\mathbb{M}, s \Vdash \Diamond\varphi \text{ iff } \llbracket \# \varphi \rrbracket^{\mathbb{M}, s} > 0.$$

- For $\mathbf{O}\varphi$,

$$\mathbb{M}, s \Vdash \mathbf{O}\varphi \text{ iff } \llbracket \# \varphi \rrbracket^{\mathbb{M}, s} \text{ is an odd number.}$$

The following facts hold:

Fact 2.3 The semantic conditions for $\Diamond, \Box, \mathbf{E}$ can be given as follows:

- $\mathbb{M}, s \Vdash \Diamond\varphi$ iff there exists $t \in W$ such that Rst and $\mathbb{M}, t \Vdash \varphi$.
- $\mathbb{M}, s \Vdash \Box\varphi$ iff for all $t \in W$ such that Rst , we have $\mathbb{M}, t \Vdash \varphi$.
- $\mathbb{M}, s \Vdash \mathbf{E}\varphi$ iff $\llbracket \# \varphi \rrbracket^{\mathbb{M}, s}$ is an even number.

3 The axiomatization of PML

In this section, we give a Hilbert system for PML which is sound and complete w.r.t. image-finite Kripke frames.

3.1 The Hilbert system PML^{fin}

The Hilbert system PML^{fin} w.r.t. all image-finite Kripke frames is given as follows:

- Axioms:
 - (i) Classical propositional tautologies;
 - (ii) $\Box(\varphi \rightarrow \psi) \rightarrow (\Box\varphi \rightarrow \Box\psi)$;
 - (iii) $\Box(\varphi \leftrightarrow \psi) \rightarrow (\mathbf{O}\varphi \leftrightarrow \mathbf{O}\psi)$;
 - (iv) $\mathbf{E}\varphi \leftrightarrow ((\mathbf{E}(\varphi \wedge \psi) \wedge \mathbf{E}(\varphi \wedge \neg\psi)) \vee (\mathbf{O}(\varphi \wedge \psi) \wedge \mathbf{O}(\varphi \wedge \neg\psi)))$;
 - (v) $\mathbf{O}\varphi \rightarrow \Diamond\varphi$.
- Rules:
 - (i) *Modus ponens*: from $\varphi \rightarrow \psi$ and φ one can get ψ ;
 - (ii) *Necessitation*: from φ one can get $\Box\varphi$.

Remark 3.1 The intuitive meaning of Axioms (iii)-(v) are as follows:

- Axiom (iii) means that if φ and ψ are satisfied at the same successors, then the parities of the number of successors satisfying φ and ψ are the same.
- Axiom (iv) means that the number of successors satisfying φ is even iff those successors can be partitioned into two parts, both of even sizes or both of odd sizes.
- Axioms (v) means that an odd natural number of successors satisfying φ implies the existence of successors satisfying φ .

The following standard definitions are necessary for later proofs:

Definition 3.2 Given any Hilbert system $\mathbf{H}$, any PML-formula φ , any Kripke frame $\mathbb{F}$ and any class $\mathcal{K}$ of Kripke frames,

- $\mathbf{H} \vdash \varphi$ means that φ is *provable* in $\mathbf{H}$;
- $\mathbb{F} \Vdash \varphi$ means that φ is *valid* on $\mathbb{F}$, i.e. $\llbracket \varphi \rrbracket^{\mathbb{F}, V} = W$ for any valuation V on $\mathbb{F}$;
- $\mathcal{K} \Vdash \varphi$ means that φ is *valid* on $\mathcal{K}$, i.e. $\mathbb{F} \Vdash \varphi$ for any Kripke frame $\mathbb{F} \in \mathcal{K}$;
- $\mathbf{H}$ is *sound* w.r.t. $\mathcal{K}$, if $\mathbf{H} \vdash \varphi$ implies $\mathcal{K} \Vdash \varphi$ for all formulas φ ;
- $\mathbf{H}$ is *complete* w.r.t. $\mathcal{K}$, if $\mathcal{K} \Vdash \varphi$ implies $\mathbf{H} \vdash \varphi$ for all formulas φ .

When $\mathbf{H}$ is obvious from the context, $\mathbf{H}$ can be omitted and we use $\vdash \varphi$ rather than $\mathbf{H} \vdash \varphi$.

Based on previous definitions, we prove the soundness of PML^{fin} w.r.t. image-finite Kripke frames:

Theorem 3.3 (Soundness of PML^{fin}) *For any PML-formula φ , if $\text{PML}^{\text{fin}} \vdash \varphi$ then $\mathbb{F} \Vdash \varphi$ for all image-finite Kripke frame $\mathbb{F}$, i.e. PML^{fin} is sound w.r.t. image-finite Kripke frames.*

Proof. We show the soundness of Axioms (iii)-(v), the rest are standard as in basic modal logic. For the soundness of the axioms, consider any image-finite Kripke frame $\mathbb{F} = (W, R)$, any valuation V on $\mathbb{F}$, and any $s \in W$.

Axiom (iii): If $\mathbb{F}, V, s \Vdash \Box(\varphi \leftrightarrow \psi)$, then $\llbracket \varphi \rrbracket^{\mathbb{F}, V} \cap R_s = \llbracket \psi \rrbracket^{\mathbb{F}, V} \cap R_s$, so $\llbracket \# \varphi \rrbracket^{\mathbb{F}, V, s} = \llbracket \# \psi \rrbracket^{\mathbb{F}, V, s}$, therefore they have the same parity, so $\mathbb{F}, V, s \Vdash \mathbf{O}\varphi \leftrightarrow \mathbf{O}\psi$.

Axiom (iv): $\mathbb{F}, V, s \Vdash \mathbf{E}\varphi$ iff $\llbracket \# \varphi \rrbracket^{\mathbb{F}, V, s}$ is an even number. Since $\llbracket \# \varphi \rrbracket^{\mathbb{F}, V, s} = \llbracket \#(\varphi \wedge \psi) \rrbracket^{\mathbb{F}, V, s} + \llbracket \#(\varphi \wedge \neg\psi) \rrbracket^{\mathbb{F}, V, s}$, we have that $\mathbb{F}, V, s \Vdash \mathbf{E}\varphi$ iff $\llbracket \#(\varphi \wedge \psi) \rrbracket^{\mathbb{F}, V, s} +$

$\llbracket \#(\varphi \wedge \neg\psi) \rrbracket^{\mathbb{F}, V, s}$ is an even number iff both $\llbracket \#(\varphi \wedge \psi) \rrbracket^{\mathbb{F}, V, s}$ and $\llbracket \#(\varphi \wedge \neg\psi) \rrbracket^{\mathbb{F}, V, s}$ are odd or both of them are even iff $\mathbb{F}, V, s \Vdash (\mathbf{E}(\varphi \wedge \psi) \wedge \mathbf{E}(\varphi \wedge \neg\psi)) \vee (\mathbf{O}(\varphi \wedge \psi) \wedge \mathbf{O}(\varphi \wedge \neg\psi))$.

Axiom (v): If $\mathbb{F}, V, s \Vdash \mathbf{O}\varphi$, then $\llbracket \# \varphi \rrbracket^{\mathbb{F}, V, s}$ is an odd natural number, therefore > 0 . so $\mathbb{F}, V, s \Vdash \Diamond\varphi$.

□

The next lemma is about the parities of partitions of the set of successors in terms of pairwise incompatible formulas.

Lemma 3.4 *The following formulas are provable in PML^{fin} :*

- (i) $\Box \neg(\varphi \wedge \psi) \wedge \mathbf{O}\varphi \wedge \mathbf{O}\psi \rightarrow \mathbf{E}(\varphi \vee \psi)$.
- (ii) $\Box \neg(\varphi \wedge \psi) \wedge \mathbf{O}\varphi \wedge \mathbf{E}\psi \rightarrow \mathbf{O}(\varphi \vee \psi)$.
- (iii) $\Box \neg(\varphi \wedge \psi) \wedge \mathbf{E}\varphi \wedge \mathbf{E}\psi \rightarrow \mathbf{E}(\varphi \vee \psi)$.
- (iv) $\bigwedge_{1 \leq i \neq j \leq n} \Box \neg(\varphi_i \wedge \varphi_j) \wedge \bigwedge_{i \in I} \mathbf{O}\varphi_i \wedge \bigwedge_{i \notin I} \text{and } 1 \leq i \leq n \mathbf{E}\varphi_i \rightarrow \mathbf{P} \bigvee_{i=1}^n \varphi_i$,
where $\mathbf{P}$ is $\mathbf{O}$ if $I \subseteq \{1, \dots, n\}$ is of an odd size, and $\mathbf{E}$ otherwise.
- (v) $\Box \neg(\varphi \wedge \psi) \wedge \mathbf{O}(\varphi \vee \psi) \rightarrow ((\mathbf{O}\varphi \wedge \mathbf{E}\psi) \vee (\mathbf{E}\varphi \wedge \mathbf{O}\psi))$.
- (vi) $\Box \neg(\varphi \wedge \psi) \wedge \mathbf{E}(\varphi \vee \psi) \rightarrow ((\mathbf{O}\varphi \wedge \mathbf{O}\psi) \vee (\mathbf{E}\varphi \wedge \mathbf{E}\psi))$.
- (vii) $\bigwedge_{1 \leq i \neq j \leq n} \Box \neg(\varphi_i \wedge \varphi_j) \wedge \mathbf{O} \bigvee_{i=1}^n \varphi_i \rightarrow \bigvee_{\text{odd } I \subseteq \{1, \dots, n\}} (\bigwedge_{i \in I} \mathbf{O}\varphi_i \wedge \bigwedge_{j \notin I} \mathbf{E}\varphi_j)$.
- (viii) $\bigwedge_{1 \leq i \neq j \leq n} \Box \neg(\varphi_i \wedge \varphi_j) \wedge \mathbf{E} \bigvee_{i=1}^n \varphi_i \rightarrow \bigvee_{\text{even } I \subseteq \{1, \dots, n\}} (\bigwedge_{i \in I} \mathbf{O}\varphi_i \wedge \bigwedge_{j \notin I} \mathbf{E}\varphi_j)$.

Proof.

- (i) From Axiom (iv) we have $\vdash \mathbf{O}((\varphi \vee \psi) \wedge \varphi) \wedge \mathbf{O}((\varphi \vee \psi) \wedge \neg\varphi) \rightarrow \mathbf{E}(\varphi \vee \psi)$. Obviously $\vdash \Box(((\varphi \vee \psi) \wedge \varphi) \leftrightarrow \varphi)$, therefore from Axiom (iii) we get $\vdash \mathbf{O}((\varphi \vee \psi) \wedge \varphi) \leftrightarrow \mathbf{O}\varphi$. Now we have $\vdash \mathbf{O}\varphi \wedge \mathbf{O}((\varphi \vee \psi) \wedge \neg\varphi) \rightarrow \mathbf{E}(\varphi \vee \psi)$. Obviously $\vdash \Box \neg(\varphi \wedge \psi) \rightarrow \Box(((\varphi \vee \psi) \wedge \neg\varphi) \leftrightarrow \psi)$. By Axiom (iii), we get $\vdash \Box \neg(\varphi \wedge \psi) \rightarrow (\mathbf{O}((\varphi \vee \psi) \wedge \neg\varphi) \leftrightarrow \mathbf{O}\psi)$. Now from this and $\vdash \mathbf{O}\varphi \wedge \mathbf{O}((\varphi \vee \psi) \wedge \neg\varphi) \rightarrow \mathbf{E}(\varphi \vee \psi)$, we get $\vdash \Box \neg(\varphi \wedge \psi) \wedge \mathbf{O}\varphi \wedge \mathbf{O}\psi \rightarrow \mathbf{E}(\varphi \vee \psi)$.
- (ii) Similar to item (i).
- (iii) Similar to item (i).
- (iv) By induction on n . The case $n = 2$ is obtained from the first three items. For the case $n = k + 1$, apply the induction hypothesis on k and 2 (notice that $\bigwedge_{i=1}^{n-1} \Box \neg(\varphi_i \wedge \varphi_n)$ is provably equivalent to $\Box \neg((\varphi_1 \vee \dots \vee \varphi_{n-1}) \wedge \varphi_n)$).
- (v) From Axiom (iv) we have $\vdash \mathbf{O}(\varphi \vee \psi) \rightarrow ((\mathbf{E}((\varphi \vee \psi) \wedge \varphi) \wedge \mathbf{O}((\varphi \vee \psi) \wedge \neg\varphi)) \vee (\mathbf{O}((\varphi \vee \psi) \wedge \varphi) \wedge \mathbf{E}((\varphi \vee \psi) \wedge \neg\varphi)))$. Similar to item (i), we have:
 - $\vdash \mathbf{O}((\varphi \vee \psi) \wedge \varphi) \leftrightarrow \mathbf{O}\varphi$;
 - $\vdash \mathbf{E}((\varphi \vee \psi) \wedge \varphi) \leftrightarrow \mathbf{E}\varphi$;
 - $\vdash \Box \neg(\varphi \wedge \psi) \rightarrow (\mathbf{O}((\varphi \vee \psi) \wedge \neg\varphi) \leftrightarrow \mathbf{O}\psi)$;
 - $\vdash \Box \neg(\varphi \wedge \psi) \rightarrow (\mathbf{E}((\varphi \vee \psi) \wedge \neg\varphi) \leftrightarrow \mathbf{E}\psi)$.

Combining them together, we have $\vdash \Box \neg(\varphi \wedge \psi) \wedge \mathbf{O}(\varphi \vee \psi) \rightarrow ((\mathbf{O}\varphi \wedge \mathbf{E}\psi) \vee (\mathbf{E}\varphi \wedge \mathbf{O}\psi))$.

- (vi) Similar to item (v).
- (vii) By induction on n . The case $n = 2$ is obtained from items (v) and (vi). For the case $n = k + 1$, apply the induction hypothesis on k and 2 (notice that we need to use induction hypothesis for both items (vii) and (viii)).
- (viii) Similar to item (vii).

□

3.2 Completeness

Now we show the completeness of PML^{fin} w.r.t. image-finite Kripke frames.

We first define maximally consistent sets (MCSs):

Definition 3.5 [(Maximally) consistent set] Given a Hilbert system $\mathbf{H}$ and two sets Γ, Σ of PML-formulas,

- Γ is *H-consistent*, if there are no formulas $\gamma_1, \dots, \gamma_n \in \Gamma$ such that $\mathbf{H} \vdash \gamma_1 \wedge \dots \wedge \gamma_n \rightarrow \perp$. When Γ is a singleton $\{\varphi\}$, we also say that φ is *H-consistent*.
- Γ is *maximally H-consistent* w.r.t. Σ , if $\Gamma \subseteq \Sigma$ is *H-consistent* and there is no proper superset $\Delta \supsetneq \Gamma$ such that Δ is *H-consistent* and $\Delta \subseteq \Sigma$.

When $\mathbf{H}$ is obvious from the context, $\mathbf{H}$ can be omitted and we say the formula set is (maximally) consistent (w.r.t. Σ).

Before the completeness proof, we first prove some lemmas which are useful in the main proof. We say that φ and ψ are provably equivalent if $\mathbf{H} \vdash \varphi \rightarrow \psi$ and $\mathbf{H} \vdash \psi \rightarrow \varphi$.

Lemma 3.6 Fix a finite set $\mathbf{P} = \{p_1, \dots, p_m\}$ of propositional variables and a counting depth threshold $n \in \mathbb{N}$, there are finitely many PML-formulas built up from propositional variables in $\mathbf{P}$ and of counting depth $\leq n$, modulo provability equivalence.

Proof. We prove by induction on n .

- When $n = 0$, it is easy to see that there are finitely different depth-0 PML-formulas built up from $\mathbf{P}$, modulo provability equivalence.
- When $n = k + 1$, suppose we have proved the claim for $n \leq k$, then for each PML-formula of counting depth $k + 1$, it is a Boolean combination of PML-formulas of counting depth $\leq k$ and formulas of the form $\mathbf{O}\varphi$ or $\Diamond\varphi$, where $d(\varphi) = k$. Since there are finitely many PML-formulas of counting depth $\leq k$ modulo provability equivalence, we have that there are finitely many possibilities of PML-formulas of counting depth $\leq k + 1$, modulo provability equivalence.

□

Now we fix a PML^{fin} -consistent formula $\neg\varphi$, assume that the propositional variables occurring in $\neg\varphi$ are among $\mathbf{P} = \{p_1, \dots, p_m\}$ and $d(\neg\varphi) = n$.

Definition 3.7 Define S_k to be the set of all PML-formulas ψ such that $d(\psi) \leq k$ and all propositional variables occurring in ψ are among $p_1, \dots, p_m$.

Lemma 3.8 (Lindenbaum Lemma) *Given any formula $\theta \in S_k$ which is PML^{fin}-consistent, $\{\theta\}$ can be extended to a MCS w.r.t. S_k .*

Proof. By the standard argument of Lindenbaum Lemma. $\square$

Given an MCS Γ w.r.t. S_k , since S_k is finite modulo provability equivalence and Γ is closed under provability equivalence for formulas of depth $\leq k$ (i.e. $\psi \in \Gamma$ and $\vdash \psi \leftrightarrow \theta$ and $d(\psi), d(\theta) \leq k$ imply that $\theta \in \Gamma$), we take one formula δ from each equivalence class of S_k (for the provability equivalence) and form a finite set Δ , and define $\text{Conj}(\Gamma) := \bigwedge \{\delta \in \Delta \mid \delta \in \Gamma\}$. Further define $T_k := \{\text{Conj}(\Gamma) \mid \Gamma \text{ is an MCS w.r.t. } S_k\}$. It is clear that T_k is finite for each k .

Lemma 3.9 *For each formula $\theta \in S_k$, we have $\vdash \theta \leftrightarrow \bigvee_{i=1}^{\ell} \text{Conj}(\Gamma_i)$ for some $\text{Conj}(\Gamma_i) \in T_k$, $i = 1, \dots, \ell$. In addition, for all $\text{Conj}(\Gamma) \in T_k$ not among $\text{Conj}(\Gamma_i)$ for $1 \leq i \leq \ell$, we have that $\vdash \neg(\theta \wedge \text{Conj}(\Gamma))$.*

Proof. We consider all MCSs $\{\Gamma_i\}_{i \in I}$ w.r.t. S_k such that $\theta \in \Gamma_i$. It is easy to see that there are finitely many Γ_i 's. We will show that $\vdash \theta \leftrightarrow \bigvee_{i \in I} \text{Conj}(\Gamma_i)$.

- We first show that $\vdash (\bigvee_{i \in I} \text{Conj}(\Gamma_i)) \rightarrow \theta$. Suppose otherwise, $(\bigvee_{i \in I} \text{Conj}(\Gamma_i)) \wedge \neg\theta$ is consistent, which implies that I is non-empty. Therefore $\bigvee_{i \in I} (\text{Conj}(\Gamma_i) \wedge \neg\theta)$ is consistent. Now $\text{Conj}(\Gamma_i) \wedge \neg\theta$ derives θ since $\theta \in \Gamma_i$, and it is trivial that $\text{Conj}(\Gamma_i) \wedge \neg\theta$ derives $\neg\theta$. Therefore all $\text{Conj}(\Gamma_i) \wedge \neg\theta$ can derive $\theta \wedge \neg\theta$, so $\vdash ((\bigvee_{i \in I} \text{Conj}(\Gamma_i)) \wedge \neg\theta) \rightarrow (\theta \wedge \neg\theta)$, a contradiction to the consistency of $(\bigvee_{i \in I} \text{Conj}(\Gamma_i)) \wedge \neg\theta$.
- For the other direction, if $\not\vdash \theta \rightarrow \bigvee_{i \in I} \text{Conj}(\Gamma_i)$, then $\theta \wedge \bigwedge_{i \in I} \neg\text{Conj}(\Gamma_i) \in S_k$ and it is consistent, by Lemma 3.8, it can be extended to an MCS Γ w.r.t. S_k . Then Γ must contain all $\neg\text{Conj}(\Gamma_i)$'s, so Γ is different from all Γ_i 's, but Γ contains θ , so Γ should be one of the Γ_i 's, a contradiction.

Therefore $\vdash \theta \leftrightarrow \bigvee_{i \in I} \text{Conj}(\Gamma_i)$.

To see that $\vdash \neg(\theta \wedge \text{Conj}(\Gamma))$, it suffices to see that $\theta \notin \Gamma$, i.e. $\neg\theta \in \Gamma$. $\square$

Lemma 3.10 *For any MCS Γ w.r.t. S_k (for $k \leq n$), Γ is uniquely determined by its propositional literal (i.e. propositional variables or their negations) members and members of the form $\Diamond\psi$ and members of the form $\text{O}\theta$, where ψ, θ are of the form $\text{Conj}(\Delta)$ with Δ being an MCS w.r.t. S_{k-1} .*

Proof. Indeed, it suffices to show that whenever we have determined the Γ -membership for formulas of the form $\Diamond\psi$ and $\text{O}\theta$ where ψ and θ are as described as in the lemma, then we can determine the Γ -membership for all formulas of the form $\Diamond\delta$ and $\text{O}\delta$, where $\delta \in S_{k-1}$.

By Lemma 3.9, for each $\delta \in S_{k-1}$, $\vdash \delta \leftrightarrow \bigvee_{i=1}^{\ell} \text{Conj}(\Gamma_i)$ for some $\text{Conj}(\Gamma_i) \in T_{k-1}$, $i = 1, \dots, \ell$. Therefore $\vdash \Diamond\delta \leftrightarrow \bigvee_{i=1}^{\ell} \Diamond\text{Conj}(\Gamma_i)$, and we can determine whether $\Diamond\delta \in \Gamma$ based on whether there exists a $\Diamond\text{Conj}(\Gamma_i)$ belonging to Γ .

Similarly, by Lemma 3.9, for each $\delta \in S_{k-1}$, $\vdash \delta \leftrightarrow \bigvee_{i=1}^{\ell} \text{Conj}(\Gamma_i)$ for some $\text{Conj}(\Gamma_i) \in T_{k-1}$, $i = 1, \dots, \ell$. Since $\vdash \text{Conj}(\Gamma_i) \wedge \text{Conj}(\Gamma_j) \rightarrow \perp$ for all $i \neq j$, by Lemma 3.4.(iv), we can uniquely determine the parity of δ , i.e. whether $O\delta \in \Gamma$ or $E\delta \in \Gamma$. $\square$

Now we can start defining the canonical model layer-by-layer:

Definition 3.11 [Canonical Model] We define the tree-shaped canonical model $\mathbb{M} = (W, R, V)$ step by step as follows:

- First of all, we put the MCS w.r.t. S_n containing $\neg\varphi$ that we obtained from Lemma 3.8 as the root node, denoted by Γ_n , and put it in the layer n .
- Suppose we have already defined layers $\geq k+1$, and then for layer k , consider a node $(\Gamma_n, \Gamma_{n-1, i_{n-1}}, \dots, \Gamma_{k+1, i_{k+1}})$ in layer $k+1$, it is essentially an MCS Σ w.r.t. S_{k+1} . Then by Lemma 3.10, it is determined by its propositional literal members and members of the form $\Diamond\psi$ and members of the form $O\theta$, where ψ, θ are of the form $\text{Conj}(\Delta)$ with Δ being an MCS w.r.t. S_k . Now for each MCS Δ w.r.t. S_k ,
 - if $\Diamond\text{Conj}(\Delta) \in \Sigma$ and $O\text{Conj}(\Delta) \in \Sigma$, then we add 1 copy of Δ as a successor of $(\Gamma_n, \Gamma_{n-1, i_{n-1}}, \dots, \Gamma_{k+1, i_{k+1}})$, and denote it as $(\Gamma_n, \Gamma_{n-1, i_{n-1}}, \dots, \Gamma_{k+1, i_{k+1}}, \Gamma_{k,1})$, and put it in the layer k ;
 - if $\Diamond\text{Conj}(\Delta) \in \Sigma$ and $O\text{Conj}(\Delta) \notin \Sigma$, then we add 2 copies of Δ as successors of $(\Gamma_n, \Gamma_{n-1, i_{n-1}}, \dots, \Gamma_{k+1, i_{k+1}})$, and denote them as $(\Gamma_n, \Gamma_{n-1, i_{n-1}}, \dots, \Gamma_{k+1, i_{k+1}}, \Gamma_{k,1})$, $(\Gamma_n, \Gamma_{n-1, i_{n-1}}, \dots, \Gamma_{k+1, i_{k+1}}, \Gamma_{k,2})$, respectively, and put them in the layer k ;
 - if $\Diamond\text{Conj}(\Delta) \notin \Sigma$, then $O\text{Conj}(\Delta) \notin \Sigma$, then we do not add any copy of Δ as successors of $(\Gamma_n, \Gamma_{n-1, i_{n-1}}, \dots, \Gamma_{k+1, i_{k+1}})$;
 - the situation $\Diamond\text{Conj}(\Delta) \notin \Sigma$ and $O\text{Conj}(\Delta) \in \Sigma$ is impossible due to axiom (v).
- Then we take the union of all the layers defined above together, and get the domain W .
- The relation R is defined as follows: sRt iff s is of the form $(\Gamma_n, \Gamma_{n-1, i_{n-1}}, \dots, \Gamma_{k+1, i_{k+1}})$ and t is of the form $(\Gamma_n, \Gamma_{n-1, i_{n-1}}, \dots, \Gamma_{k, i_k})$, for some $0 \leq k \leq n-1$.
- The valuation V is defined as $V(p) = \{s \in W \mid p \in s\}$.

Lemma 3.12 (Truth Lemma) For any α with $d(\alpha) = k \leq n$, for all s in layers $\geq k$, we have $\mathbb{M}, s \Vdash \alpha$ iff $\alpha \in s$.

Proof. We prove by induction.

- For atomic and Boolean cases, trivial.
- For the case of $\Diamond\theta$, suppose that $d(\theta) = k$, then by induction hypothesis, $\mathbb{M}, s \Vdash \theta$ iff $\theta \in s$, for all s in layer $\geq k$.

Now suppose that the s in consideration is at layer $k_0 \geq k+1$. By Lemma 3.9, $\vdash \theta \leftrightarrow \bigvee_{i=1}^{\ell} \text{Conj}(\Gamma_i)$ for some $\text{Conj}(\Gamma_i) \in T_k$, $i = 1, \dots, \ell$, so $\theta \leftrightarrow \bigvee_{i=1}^{\ell} \text{Conj}(\Gamma_i)$ belongs to every MCS. It is also easy to see that

$\Diamond\theta \leftrightarrow \bigvee_{i=1}^{\ell} \Diamond \text{Conj}(\Gamma_i)$ belongs to every MCS.

- ($\Rightarrow$) If $\mathbb{M}, s \Vdash \Diamond\theta$, then there is a t at layer $k_0 - 1 \geq k$ such that $\mathbb{M}, t \Vdash \theta$. By induction hypothesis, $\theta \in t$, so from $\theta \leftrightarrow \bigvee_{i=1}^{\ell} \text{Conj}(\Gamma_i)$ belonging to every MCS we have that there is a $\text{Conj}(\Gamma_i) \in t$, so by the definition of the canonical model, it must be the case that t is a copy of Γ_i , therefore $\Diamond \text{Conj}(\Gamma_i) \in s$, so from $\Diamond\theta \leftrightarrow \bigvee_{i=1}^{\ell} \Diamond \text{Conj}(\Gamma_i)$ belonging to every MCS we have that $\Diamond\theta \in s$.
- ($\Leftarrow$) If $\Diamond\theta \in s$, then from $\Diamond\theta \leftrightarrow \bigvee_{i=1}^{\ell} \Diamond \text{Conj}(\Gamma_i)$ belonging to every MCS we have that there is a $\Diamond \text{Conj}(\Gamma_i) \in s$, so by the definition of the canonical model, there are 2 or 1 copies of Γ_i added as successor(s) of s . W.l.o.g. we call one of this copy t , then t contains $\text{Conj}(\Gamma_i)$, by induction hypothesis, $\mathbb{M}, t \Vdash \text{Conj}(\Gamma_i)$, by soundness and $\vdash \theta \leftrightarrow \bigvee_{i=1}^{\ell} \text{Conj}(\Gamma_i)$ we get $\mathbb{M}, t \Vdash \theta$, so $\mathbb{M}, s \Vdash \Diamond\theta$.

- For the case of $\text{O}\theta$, suppose that $d(\theta) = k$, then by induction hypothesis, $\mathbb{M}, s \Vdash \theta$ iff $\theta \in s$, for all s in layer $\geq k$.

Now suppose that the s in consideration is at layer $k_0 \geq k + 1$. By Lemma 3.9, $\vdash \theta \leftrightarrow \bigvee_{i=1}^{\ell} \text{Conj}(\Gamma_i)$ for some $\text{Conj}(\Gamma_i) \in T_k$ (w.l.o.g. we assume that the Γ_i 's are pairwise different), so $\theta \leftrightarrow \bigvee_{i=1}^{\ell} \text{Conj}(\Gamma_i)$ belongs to every MCS. It is also easy to see that $\Box \neg(\text{Conj}(\Gamma_i) \wedge \text{Conj}(\Gamma_j))$ is provable for all $i \neq j$.

Denote all $\text{Conj}(\Gamma_i)$'s ($1 \leq i \leq \ell$) as $\alpha_1, \dots, \alpha_{\ell}$, by Lemma 3.4.(iii), we have:

- $\vdash \bigwedge_{1 \leq i \neq j \leq n} \Box \neg(\alpha_i \wedge \alpha_j) \wedge \text{O} \bigvee_{i=1}^n \alpha_i \rightarrow \bigvee_{\text{odd } I \subseteq \{1, \dots, n\}} (\bigwedge_{i \in I} \text{O}\alpha_i \wedge \bigwedge_{j \notin I} \text{E}\alpha_j).$
- $\vdash \bigwedge_{1 \leq i \neq j \leq n} \Box \neg(\alpha_i \wedge \alpha_j) \wedge \text{E} \bigvee_{i=1}^n \alpha_i \rightarrow \bigvee_{\text{even } I \subseteq \{1, \dots, n\}} (\bigwedge_{i \in I} \text{O}\alpha_i \wedge \bigwedge_{j \notin I} \text{E}\alpha_j).$

Since $\vdash \Box \neg(\alpha_i \wedge \alpha_j)$ for all $i \neq j$, we have:

- $\vdash \text{O} \bigvee_{i=1}^n \alpha_i \rightarrow \bigvee_{\text{odd } I \subseteq \{1, \dots, n\}} (\bigwedge_{i \in I} \text{O}\alpha_i \wedge \bigwedge_{j \notin I} \text{E}\alpha_j).$
- $\vdash \text{E} \bigvee_{i=1}^n \alpha_i \rightarrow \bigvee_{\text{even } I \subseteq \{1, \dots, n\}} (\bigwedge_{i \in I} \text{O}\alpha_i \wedge \bigwedge_{j \notin I} \text{E}\alpha_j).$

Now we are ready to show that $\mathbb{M}, s \Vdash \theta$ iff $\theta \in s$:

- ($\Rightarrow$) If $\mathbb{M}, s \Vdash \text{O}\theta$ and $\text{O}\theta \notin s$, then $\neg \text{O}\theta \in s$, i.e. $\text{E}\theta \in s$. By $\vdash \theta \leftrightarrow \bigvee_{i=1}^m \alpha_i$, we have that $\vdash \text{E}\theta \leftrightarrow \text{E} \bigvee_{i=1}^m \alpha_i$. By

$$\vdash \text{E} \bigvee_{i=1}^n \alpha_i \rightarrow \bigvee_{\text{even } I \subseteq \{1, \dots, n\}} (\bigwedge_{i \in I} \text{O}\alpha_i \wedge \bigwedge_{j \notin I} \text{E}\alpha_j),$$

we have that there is an $I \subseteq \{1, \dots, n\}$ such that $\text{O}\alpha_i \in s$ for $i \in I$, $\text{E}\alpha_j \in s$ for $j \notin I$ and $|I|$ is even. Then by the definition of the canonical model, we have 1 copy of Γ_i as successor of s for all $i \in I$, and 2 copies of Γ_j as successors of s for all $j \notin I$. Therefore we have $|I| + 2(\ell - |I|)$ -many successors of s containing θ . It is also easy to see that for all $\text{Conj}(\Gamma) \in T_k$ such that Γ is not among $\Gamma_1, \dots, \Gamma_{\ell}$, we have that $\vdash \text{Conj}(\Gamma) \rightarrow \neg\theta$, so all other successors of s do not contain θ . therefore, there is an even number of successors of s containing θ . By induction hypothesis, these successors are exactly the successors satisfying θ , so $\mathbb{M}, s \Vdash \text{E}\theta$, a contradiction.

($\Leftarrow$) If $\mathbb{M}, s \not\models O\theta$ and $O\theta \in s$, then we can use

$$\vdash O \bigvee_{i=1}^n \alpha_i \rightarrow \bigvee_{\text{odd } I \subseteq \{1, \dots, n\}} \left(\bigwedge_{i \in I} O\alpha_i \wedge \bigwedge_{j \notin I} E\alpha_j \right)$$

to derive a similar contradiction. $\square$

Now we reach the main theorem:

Theorem 3.13 *The Hilbert system PML^{fin} is sound and complete w.r.t. image-finite Kripke frames. Moreover, it is sound and complete w.r.t. finite Kripke frames.*

Proof. We have already proved soundness. For completeness, suppose $\not\models \varphi$ with $d(\varphi) = n$, then by Lemma 3.8, we can extend $\neg\varphi$ to an MCS Γ_n w.r.t. S_n . Then we can construct the canonical model $\mathbb{M} = (W, R, V)$, by Lemma 3.12, $\mathbb{M}, \Gamma_n \not\models \varphi$. The canonical frame is image-finite and of finite depth, hence is finite. $\square$

4 Conclusions

In the present paper, we axiomatize parity modal logic PML, and show the soundness and completeness of PML^{fin} w.r.t. image-finite Kripke frames. For future work, we mention the following:

- Parity modal logic is defined over image-finite Kripke frames. For the interpretation of this logic over all Kripke frames, we mention two possibilities: one is to treat infinite cardinalities as even (since they can be divided into two equal cardinality parts), while the other is to treat infinite cardinalities as neither odd nor even. In the first reading, we can still treat $E\varphi$ as $\neg O\varphi$, while in the second reading, we cannot treat $\bar{E}$ as the negation of O anymore.
- Parity modal logic talks about the number of successors satisfying a formula modulo 2. In the same spirit, we can consider the modal logic extended with modalities talking about the number of successors satisfying a formula modulo n for $n \geq 2$. This kind of language has been studied in [1,3], and it is interesting to investigate its axiomatization.
- There are also other kinds of extensions of basic modal logic talking about the numerical information about the number of successors satisfying a formula or two formulas, such as graded modal logic which can talk about “at least n successors satisfies φ ”, modal logic with cardinality comparison modality which can talk about “the number of φ -successors $\geq$ the number of ψ -successors”. It is worth investigating how to axiomatize the language combining parity modal logic and these languages.

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Collective Argumentation with Labellings: A Dynamic Impossibility Perspective

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Abstract

We study aggregation of three-valued argument labellings. Independence, neutrality, monotonicity, and uniqueness force dictatorship. Without uniqueness, regular revision gives another route to dictatorship on a seven-argument framework. A single fixed reset operator suffices, and we define it on all sets and inputs. Its action on non-labelling sets rules out every non-singleton oligarchy. The contraction result under uniqueness is a static corollary. Different component dictators can meet the stated static axioms. Within this class, however, commutation with all global revision operators, or all global contraction operators, requires one common dictator. The results separate the roles of static axioms and global updates.

Keywords: abstract argumentation, labellings, aggregation, social choice, belief revision

1 Introduction

Abstract argumentation frameworks represent conflict by means of arguments and attacks between them [11]. In the labelling approach [4], each argument receives one of three statuses: **in**, **out**, or **undec**. This three-valued representation is expressive, but it creates a sharp aggregation problem. In a multi-agent setting, each agent may submit a labelling, and the group should return a collective labelling. Pairwise support rules often fail to do so: the group may select two labels for the same argument, or no label at all.

The difficulty is not merely technical. It is a structural tension between local support and global labelling. A rule that decides each labelled pair separately is attractive because it is transparent and responsive to individual support. Yet a labelling is not an arbitrary set of labelled pairs: each argument must receive

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exactly one label. This uniqueness requirement severely restricts support-based aggregation. As in judgment aggregation [14,12,13], collective rationality and familiar social-choice axioms can force dictatorial behavior.

Consider the following simple profile. There are three agents and four arguments A, B, C, D , with A and B attacking each other, both A and B attacking C , and C attacking D . The agents report:

	A	B	C	D
1	in	out	out	in
2	out	in	out	in
3	undec	undec	undec	undec

Strict majority accepts (C, out) and (D, in) , but it accepts no label for A or B . The outcome is therefore not a collective labelling. This illustrates why aggregation of labellings cannot be reduced to independent pairwise voting without further constraints.

This paper studies the impossibility pressure from a dynamic perspective. Static aggregation asks how a profile should be combined at one moment. Dynamic aggregation asks whether aggregation behaves coherently when agents or the group revise their labellings. Suppose new information requires accepting or retracting a labelled pair. One may first aggregate the current individual labellings and then update the collective outcome; or one may first update each individual labelling and then aggregate. Dynamic rationality requires these two routes to agree.

Our contribution is threefold. First, we present a support-coalition model and a static dictatorship benchmark under independence, neutrality, monotonicity, and uniqueness. Second, without uniqueness, we prove on a seven-argument framework that groundedness, independence, neutrality, monotonicity, and commutation with regular revision characterize dictatorship. A fixed reset operator first forces a filter and then excludes non-singleton oligarchies; contraction under uniqueness is already a static corollary. Third, we distinguish static component modularity from dynamic synchronization: within the component-dictator class, either full global update class forces a common dictator. Complete-labelling preservation and local necessary conditions clarify the scope.

2 Labellings and Aggregation

An abstract argumentation framework is a pair $AF = \langle Arg, \rightarrow \rangle$, where Arg is a set of arguments and $\rightarrow \subseteq Arg \times Arg$ is an attack relation. We write $A \rightarrow B$ when A attacks B . Let $\Lambda = \{\text{in}, \text{out}, \text{undec}\}$ be the set of labels.

Definition 2.1 Let $AF = \langle Arg, \rightarrow \rangle$ be an argumentation framework. A labelling of AF is a total function $\mathcal{L} : Arg \rightarrow \Lambda$. We identify $\mathcal{L}$ with its graph, so $(A, l) \in \mathcal{L}$ means $\mathcal{L}(A) = l$.

The standard complete-labelling semantics of Caminada [4] can be stated as follows. An argument labelled **in** must have all its attackers labelled **out**; an

argument labelled **out** must have some attacker labelled **in**; and an argument labelled **undec** must have no attacker labelled **in** while not having all attackers labelled **out**. A labelling satisfying these three clauses for every argument is complete. The dynamic results below only require labellings as total three-valued assignments, but complete labellings motivate the rationality constraints that aggregation is expected to preserve.

Fix an argumentation framework AF and a nonempty finite set $N = \{1, \dots, n\}$ of agents. A profile is an n -tuple $\mathcal{L} = (\mathcal{L}_1, \dots, \mathcal{L}_n)$ of individual labellings. Let $\mathcal{P}_{AF} = \text{Arg} \times \Lambda$ be the set of labelled pairs.

Definition 2.2 An aggregation rule is a function $F : (\mathcal{L}_{AF})^n \rightarrow 2^{\mathcal{P}_{AF}}$, where $\mathcal{L}_{AF}$ is the set of all labellings of AF . The outcome $F(\mathcal{L})$ is a collective labelling when, for each argument A , exactly one label $l \in \Lambda$ satisfies $(A, l) \in F(\mathcal{L})$.

Definition 2.3 For a labelled pair (A, l) , its support coalition in profile $\mathcal{L}$ is

$$N_{(A,l)}^{\mathcal{L}} = \{i \in N \mid \mathcal{L}_i(A) = l\}.$$

We use the following standard axioms. F is independent if acceptance of (A, l) depends only on $N_{(A,l)}^{\mathcal{L}}$. It is neutral if equally supported labelled pairs are treated alike. It is monotonic if increasing support for an accepted labelled pair cannot make it rejected. It is grounded if it never accepts an unsupported labelled pair, that is,

$$F(\mathcal{L}) \subseteq \mathcal{L}_1 \cup \dots \cup \mathcal{L}_n.$$

It is unanimous if it accepts every labelled pair supported by all agents. It satisfies uniqueness if every outcome $F(\mathcal{L})$ assigns exactly one label to every argument. It preserves complete labellings if $F(\mathcal{L})$ is complete whenever all individual labellings in $\mathcal{L}$ are complete.

Definition 2.4 Whenever a rule is represented by a family $\mathcal{W} \subseteq 2^N$ such that

$$(A, l) \in F(\mathcal{L}) \iff N_{(A,l)}^{\mathcal{L}} \in \mathcal{W},$$

the members of $\mathcal{W}$ are winning coalitions. For a nonempty coalition $O \subseteq N$, the oligarchic rule F_O accepts exactly those labelled pairs supported by all members of O :

$$F_O(\mathcal{L}) = \{(A, l) \in \mathcal{P}_{AF} \mid O \subseteq N_{(A,l)}^{\mathcal{L}}\}.$$

A dictatorship is the special case $O = \{i\}$.

3 The Static Benchmark

The static benchmark explains why uniqueness-preserving aggregation is already narrow. Independence supplies locality: a labelled pair is decided from its own support coalition. Neutrality supplies uniformity: the same support coalition is treated alike across labelled pairs. Together they yield a single winning-coalition representation.

Theorem 3.1 *Let AF contain at least two arguments. If F is independent and neutral, then there is a family $\mathcal{W} \subseteq 2^N$ such that, for every profile $\mathcal{L}$ and every labelled pair (A, l) ,*

$$(A, l) \in F(\mathcal{L}) \iff N_{(A, l)}^{\mathcal{L}} \in \mathcal{W}.$$

Proof. For a coalition $C \subseteq N$, put $C \in \mathcal{W}$ iff some labelled pair with support coalition C is accepted at some profile. Independence makes this independent of the chosen profile for a fixed labelled pair. Neutrality transfers the decision between different labelled pairs when they have the same support coalition. If the two pairs concern the same argument but different labels, use a second argument as an intermediate pair. Hence acceptance depends only on the support coalition. $\square$

The two algebraic forms we need are filters and ultrafilters. A family $\mathcal{W} \subseteq 2^N$ is a proper filter when $N \in \mathcal{W}$, $\emptyset \notin \mathcal{W}$, it is upward closed, and it is closed under finite intersection. Since N is finite, every proper filter is generated by the intersection $O = \bigcap \mathcal{W}$, so $\mathcal{W} = \{C \subseteq N \mid O \subseteq C\}$. This is exactly oligarchy by O . If, in addition, for every $C \subseteq N$, exactly one of C and $N \setminus C$ is winning, the filter is a finite ultrafilter and O is a singleton; this is dictatorship.

Lemma 3.2 *Let F be represented by a winning-coalition family $\mathcal{W}$.*

- (i) *F is oligarchic iff $\mathcal{W}$ is a proper filter, i.e., $\emptyset \notin \mathcal{W}$, $N \in \mathcal{W}$, $\mathcal{W}$ is upward closed, and $\mathcal{W}$ is closed under intersection.*
- (ii) *F is dictatorial iff $\emptyset \notin \mathcal{W}$, $\mathcal{W}$ is closed under intersection, and $\mathcal{W}$ is maximal: for every $C \subseteq N$, exactly one of C and $N \setminus C$ is in $\mathcal{W}$.*

Proof. For (1), if F is oligarchic for O , then the winning coalitions are exactly the supersets of O , which form a proper filter. Conversely, if $\mathcal{W}$ is a proper filter, finiteness of N implies that $O = \bigcap \mathcal{W}$ is itself winning and nonempty. Upward closure then gives $\mathcal{W} = \{C \subseteq N \mid O \subseteq C\}$, so F is oligarchic.

For (2), a dictatorship by i has winning coalitions exactly those containing i , and these satisfy the stated conditions. Conversely, assume the conditions. Maximality and closure under intersection imply upward closure: if $C \in \mathcal{W}$ and $C \subseteq D$, then $D \notin \mathcal{W}$ would give $N \setminus D \in \mathcal{W}$, so $C \cap (N \setminus D) = \emptyset$ would be winning, contradiction. Since N is finite, choose a minimal winning coalition D . If D had two distinct members and $i \in D$, then maximality gives either $\{i\} \in \mathcal{W}$ or $N \setminus \{i\} \in \mathcal{W}$. The first case contradicts minimality unless $D = \{i\}$; the second makes $D \setminus \{i\}$ winning, again contradicting minimality. Thus some singleton is winning, and upward closure makes the rule dictatorial. $\square$

Proposition 3.3 *On every argumentation framework with at least two arguments, an aggregation rule is independent, neutral, monotonic, and uniqueness-preserving iff it is dictatorial.*

Proof. By Theorem 3.1, F is represented by a family $\mathcal{W}$ of winning coalitions. Uniqueness implies that for every partition of N into three coalitions, corresponding to support for **in**, **out**, and **undec** on a fixed argument, exactly one

part is winning. Hence $\emptyset \notin \mathcal{W}$, $N \in \mathcal{W}$, and for every $C \subseteq N$, exactly one of C and $N \setminus C$ is winning. Monotonicity gives upward closure.

It remains to see closure under intersection. Let $C, D \in \mathcal{W}$. If $C \cap D \notin \mathcal{W}$, then the partition $(C \cap D, C \setminus D, N \setminus C)$, together with maximality, forces $C \setminus D$ to be winning. Similarly $D \setminus C$ is winning. But then a partition containing $C \setminus D$ and $D \setminus C$ has two winning parts, contradicting uniqueness. Thus $C \cap D \in \mathcal{W}$. The family is a finite ultrafilter, hence principal at some agent i . The rule is the dictatorship of i . The converse is immediate. $\square$

4 Component Modularity

Before turning to dynamic change, it is useful to isolate a modular qualification to the static benchmark. The static proof collapses all labelled pairs into one winning-coalition family. This collapse has two parts: locality and uniformity. Independence says that each labelled pair is decided from its own support coalition. Neutrality says that the same support coalition is treated alike across all labelled pairs. When an argumentation framework is disconnected, the second requirement can be weakened: different connected components may use different support tests.

Let $\mathcal{G}(AF)$ be the set of connected components of the underlying undirected graph of AF . For a component $K \in \mathcal{G}(AF)$, put

$$AF_K = \langle K, \rightarrow \cap (K \times K) \rangle \quad \text{and} \quad \mathcal{P}_K = K \times \Lambda.$$

If $\mathcal{L}$ is a labelling, $\mathcal{L}|_K$ denotes its restriction to K .

Definition 4.1 An aggregation rule satisfies component separability if, for every component K , there is a rule

$$F_K : (\mathcal{L}_{AF_K})^n \rightarrow 2^{\mathcal{P}_K}$$

such that

$$F(\mathcal{L}) \cap \mathcal{P}_K = F_K(\mathcal{L}_1|_K, \dots, \mathcal{L}_n|_K)$$

for every profile $\mathcal{L}$.

Component separability says that the collective outcome on a component depends only on the individual labellings of that component. It is implied by full independence, but it is weaker: it allows dependencies among labelled pairs inside the same component.

Definition 4.2 F satisfies component-wise independence if, for every component K , every labelled pair $(A, l) \in \mathcal{P}_K$, and all restricted profiles on K , equality of the support coalition for (A, l) implies equality of the collective decision on (A, l) by the component rule F_K .

Definition 4.3 F satisfies component-wise neutrality if, for every component K , every profile $\mathcal{L}$, and all $(A, l), (B, m) \in \mathcal{P}_K$,

$$N_{(A, l)}^{\mathcal{L}} = N_{(B, m)}^{\mathcal{L}} \implies (A, l) \in F(\mathcal{L}) \iff (B, m) \in F(\mathcal{L}).$$

It satisfies cross-component neutrality if the same implication holds for labelled pairs lying in distinct components.

Thus global neutrality is the conjunction of component-wise and cross-component neutrality. Component-wise neutrality gives uniformity inside a component; cross-component neutrality synchronizes the components.

Proposition 4.4 *If F satisfies independence, then it satisfies component separability. Moreover, F satisfies global neutrality iff it satisfies both component-wise neutrality and cross-component neutrality.*

Proof. Suppose two full profiles have the same restriction to a component K . Then every labelled pair in $\mathcal{P}_K$ has the same support coalition in the two profiles. By independence, each such pair receives the same collective decision in both profiles. Hence the restriction of $F(\mathcal{L})$ to $\mathcal{P}_K$ depends only on $\mathcal{L}|_K$, yielding a component rule F_K .

The neutrality claim is immediate in one direction. Conversely, any two labelled pairs either lie in the same component or in two distinct components. In the first case component-wise neutrality applies; in the second case cross-component neutrality applies. $\square$

Proposition 4.5 *Component separability, component-wise neutrality, monotonicity, groundedness, unanimity, and uniqueness do not imply component-local dictatorship.*

Proof. Let $N = \{1, 2\}$ and let $AF = \langle \{A, B\}, \{A \rightarrow B\} \rangle$. There is only one connected component, so component separability is automatic. Define a selector

$$s(\mathcal{L}) = \begin{cases} 1, & \text{if } \mathcal{L}_1(A) = \text{in}, \\ 2, & \text{otherwise,} \end{cases}$$

and put $F(\mathcal{L}) = \mathcal{L}_{s(\mathcal{L})}$. The outcome is always one submitted labelling, so the rule is grounded, unanimous, and unique. It is component-wise neutral because, within any fixed profile, equally supported labelled pairs are either both supported by the selected agent or both not supported by her. It is also monotonic: if an accepted pair gains support, then with two agents its support becomes N , so it remains accepted regardless of which agent is selected.

The rule is not dictatorial. When agent 1 labels A as **in**, the rule follows agent 1; otherwise it follows agent 2. It also violates independence. In the two profiles

	A	B		A	B
1	in	in	1	out	in
2	out	out	2	in	out

the support coalition for (B, in) is $\{1\}$ in both cases. The first profile selects agent 1 and accepts the pair; the second selects agent 2 and rejects it. $\square$

The proposition marks an important boundary: component separability is a factorization condition, not a support-determinacy condition. To recover

local dictatorship one needs component-wise independence or an explicit local support representation.

Definition 4.6 F has a local support representation on K if there is a family $\mathcal{W}_K \subseteq 2^N$ such that, for every $(A, l) \in \mathcal{P}_K$,

$$(A, l) \in F(\mathcal{L}) \iff N_{(A, l)}^{\mathcal{L}} \in \mathcal{W}_K.$$

It has a component-modular support representation if it has a local support representation on every component.

Proposition 4.7 *Let K contain at least two arguments. If F satisfies component separability, component-wise independence, and component-wise neutrality on K , then F has a local support representation on K .*

Proof. The proof is the local version of Theorem 3.1. Define $C \in \mathcal{W}_K$ iff some labelled pair in $\mathcal{P}_K$ with support coalition C is accepted at some restricted profile. Component-wise independence makes the definition independent of the restricted profile for a fixed labelled pair. Component-wise neutrality transfers the decision between different labelled pairs inside K . If two labels of the same argument must be compared, use a second argument in K as an intermediate pair. $\square$

Theorem 4.8 *Suppose F has a component-modular support representation, is monotonic, and satisfies uniqueness. Then F is component-wise dictatorial: for every component K , there is an agent $d_K \in N$ such that, for all $(A, l) \in \mathcal{P}_K$,*

$$(A, l) \in F(\mathcal{L}) \iff d_K \in N_{(A, l)}^{\mathcal{L}}.$$

Moreover, F is a global dictatorship iff the assignment $K \mapsto d_K$ is constant.

Proof. Fix K . The local winning-coalition family $\mathcal{W}_K$ is subject to the same ultrafilter argument as in Proposition 3.3. Uniqueness applied to the three labels of any argument in K gives $\emptyset \notin \mathcal{W}_K$, $N \in \mathcal{W}_K$, and maximality with respect to complements. Monotonicity gives upward closure, and the same three-partition argument gives closure under intersection. Hence $\mathcal{W}_K$ is a finite ultrafilter and is principal at some d_K . If all d_K 's are the same agent, the rule is globally dictatorial. If two components have different dictators, no single agent determines all labelled pairs. $\square$

Proposition 4.9 *Let AF have at least two connected components and let $d : \mathcal{G}(AF) \rightarrow N$ assign an agent to each component. Define*

$$F^d(\mathcal{L}) = \bigcup_{K \in \mathcal{G}(AF)} \{(A, l) \in \mathcal{P}_K \mid \mathcal{L}_{d(K)}(A) = l\}.$$

Then F^d satisfies component separability, component-wise independence, component-wise neutrality, monotonicity, groundedness, unanimity, uniqueness, and preservation of complete labellings. If d is not constant, F^d is not a global dictatorship.

Proof. Inside each component, F^d is simply the dictatorship of $d(K)$. Thus it satisfies the stated support and monotonicity properties locally, and its outcome on each component is one of the submitted component labellings. Groundedness, unanimity, and uniqueness follow immediately. If all individual labellings are complete, each selected restriction $\mathcal{L}_{d(K)}|_K$ is complete on AF_K ; since there are no attacks between components, their union is complete on AF . If d is not constant, two components follow different agents, so no single dictator controls the whole outcome. $\square$

Proposition 4.10 *If a component-dictator rule F^d satisfies cross-component neutrality, then d is constant. Hence cross-component neutrality restores global dictatorship.*

Proof. Suppose $d(K) \neq d(H)$. Choose a coalition $C \subseteq N$ containing $d(K)$ but not $d(H)$, and construct a profile in which some pair in $\mathcal{P}_K$ and some pair in $\mathcal{P}_H$ both have support C . The pair in K is accepted, while the pair in H is rejected. This violates cross-component neutrality. $\square$

Proposition 4.11 *Assume every component of AF has at least two arguments. Under component separability, component-wise independence, component-wise neutrality, monotonicity, and uniqueness, the admissible rules are exactly the component-dictator rules F^d .*

Proof. Proposition 4.7 gives a local support representation on every component. Theorem 4.8 then forces a local dictator $d(K)$ on each component, so the rule is F^d . Conversely, Proposition 4.9 shows that every such F^d satisfies the listed axioms. $\square$

The restriction to nontrivial components avoids a small degeneracy: with only one argument, component-wise neutrality cannot compare different labels with the same nonempty support coalition in a single profile, because the three label-support coalitions are disjoint. The modular lesson is limited but important. Complete-labelling preservation can coexist with the absence of a global dictator if disconnected components use different local dictators. What global neutrality rules out is not component separability itself, but the lack of synchronization among components. These are static conclusions. The local revision result below gives a necessary condition wherever a local support representation is available; it does not establish that arbitrary component rules can be combined while retaining dynamic rationality. Proposition 9.3 will show that global updates themselves synchronize component dictators.

5 Complete-Labelling Preservation

The uniqueness benchmark in Proposition 3.3 is deliberately abstract: it requires every outcome to be a labelling. In argumentation, a more semantic collective-rationality condition is preservation of complete labellings. The next result shows that completeness can generate the same ultrafilter conditions on a suitably rich framework. Its role is diagnostic: the dictatorship pressure is not an artefact of imposing uniqueness in isolation.

Consider a disconnected framework with two components. The left component has arguments A, A', B, B', C , mutual attacks $A \leftrightarrow A'$ and $B \leftrightarrow B'$, and attacks $A \rightarrow C$ and $B \rightarrow C$. The right component has arguments F, G, H, I , mutual attacks $F \leftrightarrow G$, attacks $F \rightarrow H$ and $G \rightarrow H$, and an attack $H \rightarrow I$. The first component will be used to force closure under intersection; the second will be used to force maximality.

Theorem 5.1 *On the framework just described, every aggregation rule satisfying groundedness, unanimity, independence, neutrality, and preservation of complete labellings is dictatorial.*

Proof. By Theorem 3.1, independence and neutrality represent F by a single family $\mathcal{W}$ of winning coalitions. Groundedness gives $\emptyset \notin \mathcal{W}$. It is enough to show closure under intersection and maximality; then $\mathcal{W}$ is a finite ultrafilter.

Let $\mathcal{C}, \mathcal{D} \in \mathcal{W}$. We show $\mathcal{C} \cap \mathcal{D} \in \mathcal{W}$. In the left component, let agents in $\mathcal{C}$ label A' as **in** and A as **out**, and agents outside $\mathcal{C}$ label A as **in** and A' as **out**. Similarly, let agents in $\mathcal{D}$ label B' as **in** and B as **out**, and agents outside $\mathcal{D}$ label B as **in** and B' as **out**. Finally, label the argument C as **in** exactly for agents in $\mathcal{C} \cap \mathcal{D}$, and as **out** otherwise. These assignments can be extended to complete labellings of the full framework.

The support coalitions for (A, out) , (B, out) , and (C, in) are $\mathcal{C}$, $\mathcal{D}$, and $\mathcal{C} \cap \mathcal{D}$, respectively. Since $\mathcal{C}$ and $\mathcal{D}$ are winning, the collective outcome labels both attackers A and B of the argument C as **out**. Preservation of complete labellings then requires C to be labelled **in**. Hence $\mathcal{C} \cap \mathcal{D} \in \mathcal{W}$.

For maximality, fix $\mathcal{E} \subseteq N$. In the right component, let agents in $\mathcal{E}$ use the complete labelling with F **in**, G **out**, H **out**, and I **in**; let agents outside $\mathcal{E}$ use the complete labelling with G **in**, F **out**, H **out**, and I **in**. By unanimity, (H, out) is accepted. In any complete labelling, H can be **out** only if at least one of its attackers is **in**. Therefore the collective outcome must accept either (F, in) or (G, in) . Their support coalitions are $\mathcal{E}$ and $N \setminus \mathcal{E}$. Thus at least one of these coalitions is winning. They cannot both be winning, because then the outcome would contain incompatible labels for the mutual attackers F and G , violating completeness. Hence exactly one of $\mathcal{E}$ and $N \setminus \mathcal{E}$ is winning.

So $\mathcal{W}$ is a finite ultrafilter and is principal at some agent. The corresponding rule is dictatorial. $\square$

This theorem also clarifies the modular point. The proof uses the same winning-coalition family on both disconnected components. If cross-component neutrality is relaxed, the two components may use different local ultrafilters. Completeness alone does not synchronize them; global neutrality does.

6 Dynamic Rationality

The static theorem concerns one profile at one time. A dynamic setting is different. Agents may receive new information and revise their labellings; the collective agent may also be expected to revise its collective outcome. This creates an order-independence question: should one first aggregate and then update, or first update each individual and then aggregate?

The distinction matters technically. Without dynamic rationality, many non-dictatorial rules can be used period by period: quota rules that accept a labelled pair supported by at least q agents are grounded, neutral, independent, monotonic, and non-dictatorial for many choices of q . What they typically fail is not one-shot admissibility but commutation with the update operation. Revision may change the support of several logically connected labelled pairs at once; dynamic rationality links the original profile with the profile generated by belief change, while independence constrains only a single support pattern at a single profile.

Definition 6.1 Let $|$ be an operator that maps a labelled-pair set $X \subseteq \mathcal{P}_{AF}$ and a pair $p \in \mathcal{P}_{AF}$ to a labelled-pair set $X|p$. An aggregation rule F is dynamically rational with respect to $|$ if, for every profile $\mathcal{L}$ and every $p \in \mathcal{P}_{AF}$,

$$F(\mathcal{L}_1|p, \dots, \mathcal{L}_n|p) = F(\mathcal{L}_1, \dots, \mathcal{L}_n)|p.$$

Thus aggregation commutes with the update operation. Equivalently, the following square should commute:

$$\begin{array}{ccc} (\mathcal{L}_1, \dots, \mathcal{L}_n) & \xrightarrow{|p \text{ agentwise}} & (\mathcal{L}_1|p, \dots, \mathcal{L}_n|p) \\ F \downarrow & & \downarrow F \\ F(\mathcal{L}_1, \dots, \mathcal{L}_n) & \xrightarrow{|p} & F(\mathcal{L}_1, \dots, \mathcal{L}_n)|p. \end{array}$$

The requirement relates different profiles generated by an update; it is therefore stronger than any one-shot support condition.

Definition 6.2 A revision operator $\dashv$ maps $X \subseteq \mathcal{P}_{AF}$ and $p \in \mathcal{P}_{AF}$ to $X \dashv p$. It is regular if it satisfies:

$$p \in X \dashv p \quad \text{and} \quad X = X \dashv p \text{ iff } p \in X.$$

It is labelling-preserving if $\mathcal{L} \dashv p \in \mathcal{L}_{AF}$ for every labelling $\mathcal{L}$ and every $p \in \mathcal{P}_{AF}$.

Revision incorporates a labelled pair while keeping labellings as labellings. Regularity records success and conservativeness: the input pair is accepted, and nothing changes if it was already accepted.

Definition 6.3 A contraction operator $\vdash$ maps $X \subseteq \mathcal{P}_{AF}$ and $p \in \mathcal{P}_{AF}$ to $X \vdash p$. It is regular if it satisfies:

$$p \notin X \vdash p \quad \text{and} \quad X = X \vdash p \text{ iff } p \notin X.$$

It is labelling-preserving if $\mathcal{L} \vdash p \in \mathcal{L}_{AF}$ for every labelling $\mathcal{L}$ and every $p \in \mathcal{P}_{AF}$.

These definitions adapt the basic AGM ideas of success and conservativeness [1] to labelled-pair sets without imposing distance, priority, or minimal change. Commutation with the full class is therefore a strong robustness requirement. The operators may change several arguments, including arguments in different components. Their action on non-labelling collective outcomes is

also substantive: the reset extension below is used to exclude non-singleton oligarchies, not merely to obtain intersection closure. We make no claim that these postulates describe every intended application or a minimal sufficient operator class.

For explicit witnesses, put

$$M_{(A,l)}(X) = (X \setminus (\{A\} \times \Lambda)) \cup \{(A,l)\},$$

and fix the cycle $\tau(\text{in}) = \text{out}$, $\tau(\text{out}) = \text{undec}$, $\tau(\text{undec}) = \text{in}$. Given a fixed pair p_0 and a fixed total labelling T containing p_0 , define, for every $X \subseteq \mathcal{P}_{AF}$ and $p = (A, l)$,

$$R_{p_0, T}(X, p) = \begin{cases} X, & p \in X, \\ X \cup \{p\}, & p \notin X, X \notin \mathcal{L}_{AF}, \\ T, & p \notin X, X \in \mathcal{L}_{AF}, p = p_0, \\ M_p(X), & p \notin X, X \in \mathcal{L}_{AF}, p \neq p_0. \end{cases} \quad (1)$$

For a fixed total labelling T omitting p_0 , define

$$C_{p_0, T}(X, p) = \begin{cases} X, & p \notin X, \\ X \setminus \{p\}, & p \in X, X \notin \mathcal{L}_{AF}, \\ T, & p \in X, X \in \mathcal{L}_{AF}, p = p_0, \\ M_{(A, \tau(l))}(X), & p \in X, X \in \mathcal{L}_{AF}, p \neq p_0. \end{cases} \quad (2)$$

The cases are disjoint and exhaustive. Each operator has success and leaves X unchanged exactly in the conservative case. On total labellings, reset or label replacement again yields a total labelling. Hence these are regular labelling-preserving operators on their entire domains. Their parameters are fixed before choosing any profile or test coalitions.

Why Majority Fails Dynamically Let $N = \{1, 2, 3\}$ and let majority accept a labelled pair whenever at least two agents support it. Majority is grounded, neutral, independent, monotonic, and non-dictatorial. Yet it need not commute with revision.

Consider the framework with arguments A, A', B, B', C, C', D , mutual attacks between each X and X' for $X \in \{A, B, C\}$, and attacks $A' \rightarrow D$, $B' \rightarrow D$, $C' \rightarrow D$. Let the three agents report:

	A	A'	B	B'	C	C'	D
L^1	in	out	undec	undec	in	out	undec
L^2	in	out	in	out	undec	undec	undec
L^3	undec	undec	in	out	in	out	undec

The majority outcome contains (A, in) and (B, in) . Use the fixed operator R_{p_0, L^1} from (1), with $p_0 = (A, \text{in})$. If we aggregate first, (A, in) is already accepted, so the collective outcome is unchanged and (B, in) remains accepted.

If we revise first, L^3 becomes L^1 , so only agent 2 supports (B, in) . Majority no longer accepts it. Hence majority violates dynamic rationality.

The example reveals the general mechanism. Majority coalitions are upward closed, but intersections of majority coalitions need not be majority coalitions. Revision can transform a profile so that a pair supported before by a winning coalition is supported after revision only by the intersection of two winning coalitions. Dynamic rationality therefore forces intersection closure.

7 Revision Characterizes Dictatorship

Theorem 7.1 *On the seven-argument framework of the majority example, among rules satisfying groundedness, neutrality, independence, and monotonicity, the rules dynamically rational with respect to every regular labelling-preserving revision operator are exactly the dictatorships.*

Proof. Fix the framework of the majority example, and the single operator $R = R_{p_0, L^1}$ from (1), with $p_0 = (A, \text{in})$ and L^1 from that example. These choices do not depend on the profile or on any coalitions. By Theorem 3.1, independence and neutrality give a winning-coalition family $\mathcal{W}$.

Groundedness gives $\emptyset \notin \mathcal{W}$. To see that $N \in \mathcal{W}$, take a profile unanimously supporting p_0 . Revision by p_0 leaves every individual labelling unchanged, while success puts p_0 in the revised collective outcome. Commutation with R therefore makes p_0 accepted originally. Monotonicity gives upward closure.

For intersection closure, let $\mathcal{C}, \mathcal{D} \in \mathcal{W}$. Use the three labellings of the majority example: agents in $\mathcal{C} \setminus \mathcal{D}$ use L^1 , agents in $\mathcal{C} \cap \mathcal{D}$ use L^2 , and agents outside $\mathcal{C}$ use L^3 . Then p_0 has support $\mathcal{C}$, while (B, in) has support $N \setminus (\mathcal{C} \setminus \mathcal{D})$, a superset of $\mathcal{D}$. Both pairs are accepted. Aggregate-then-revise leaves the outcome unchanged because it contains p_0 , whether or not it is a total labelling. Revising individuals first leaves L^1, L^2 unchanged and resets L^3 to L^1 . The support for (B, in) becomes exactly $\mathcal{C} \cap \mathcal{D}$. Commutation forces this coalition to be winning.

Thus $\mathcal{W}$ is a proper filter over finite N . Lemma 3.2 yields $F = F_O$ for a nonempty coalition O . Suppose $|O| \geq 2$. Let every member of O label A as out, and let two members disagree on B ; complete all labellings arbitrarily. This is permitted because the domain consists of all total labellings. The outcome

$$X = F_O(\mathcal{L}) = \bigcap_{i \in O} \mathcal{L}_i$$

contains (A, out) , omits p_0 , and has no label for B ; hence it is not a total labelling. Equation (1) gives

$$R(X, p_0) = X \cup \{p_0\}, \quad F_O(R(\mathcal{L}_1, p_0), \dots, R(\mathcal{L}_n, p_0)) = L^1.$$

Indeed, all members of O are reset to L^1 . The first set contains (A, out) , whereas L^1 does not. This contradicts commutation with the same fixed R . Therefore $|O| = 1$, so F is dictatorial.

Conversely, the dictatorship $F(\mathcal{L}) = \mathcal{L}_i$ satisfies all four static axioms. For any labelling-preserving update $|$,

$$F(\mathcal{L}_1|p, \dots, \mathcal{L}_n|p) = \mathcal{L}_i|p = F(\mathcal{L})|p.$$

It therefore commutes with every regular labelling-preserving revision operator. $\square$

Remark 7.2 The necessity proof uses only R_{p_0, L^1} , fully defined for every set and every input by (1). Even the step $N \in \mathcal{W}$ uses its distinguished input p_0 . Thus commutation with this one fixed operator, together with the four static axioms, already characterizes dictatorship on this framework. No witness is selected anew for a coalition or profile. This does not establish minimality among operator classes or extend the result to restricted domains of complete labellings.

The filter is an intermediate conclusion, not a classification of dynamically rational rules. Static monotonicity gives upward closure; revision adds intersection closure and, through its specified non-labelling extension, excludes every non-singleton kernel. Uniqueness is not assumed in this route to dictatorship.

8 Contraction under Uniqueness: A Static Corollary

Corollary 8.1 *On the three-argument framework in which every argument attacks the other two, every independent, neutral, monotonic, uniqueness-preserving aggregation rule is dictatorial. Consequently, it commutes with every regular labelling-preserving contraction operator.*

Proof. The framework has at least two arguments, so Proposition 3.3 gives dictatorship. A dictatorship commutes with every labelling-preserving update by the identity in the proof of Theorem 7.1. $\square$

Groundedness, contraction, and dynamic rationality are unnecessary for this dictatorship conclusion. In particular, zero update witnesses suffice. This is a static corollary, not an additional impossibility theorem caused by contraction. The revision theorem instead dispenses with uniqueness and uses dynamic rationality.

9 Local Conditions and Global Synchronization

The global revision theorem uses one winning-coalition family. Where only a local support representation is available, its filter argument gives the following necessary condition. We do not infer that arbitrary combinations of such local rules commute with global operators. The subsequent local contraction result follows entirely from static assumptions.

Theorem 9.1 (Local revision: a necessary condition) *Let K be a component with at least two arguments. Suppose F has a local support representation $\mathcal{W}_K$ on K , is grounded and monotonic, and is dynamically rational with respect to every regular labelling-preserving revision operator. Then the behavior of F on K is oligarchic: there is a nonempty coalition $O_K \subseteq N$ such that, for every $(A, l) \in \mathcal{P}_K$,*

$$(A, l) \in F(\mathcal{L}) \iff O_K \subseteq N_{(A, l)}^{\mathcal{L}}.$$

Proof. It suffices to show that $\mathcal{W}_K$ is a proper filter. Groundedness gives $\emptyset \notin \mathcal{W}_K$. The argument used in Theorem 7.1 shows that $N \in \mathcal{W}_K$: if all agents support a pair in K , revision by that pair leaves all individual labellings unchanged and success forces the pair into the revised collective outcome. Monotonicity gives upward closure.

For intersection closure, let $\mathcal{C}, \mathcal{D} \in \mathcal{W}_K$. Choose two distinct arguments $A, B \in K$. Construct three component labellings L^1, L^2, L^3 such that L^1 labels A as **in** and B as **out**, L^2 labels both A and B as **in**, and L^3 labels A as **out** and B as **in**; label the remaining arguments arbitrarily. Use L^1 for agents in $\mathcal{C} \setminus \mathcal{D}$, L^2 for agents in $\mathcal{C} \cap \mathcal{D}$, and L^3 for agents outside $\mathcal{C}$. Then (A, in) is supported by $\mathcal{C}$, while (B, in) is supported by a superset of $\mathcal{D}$, so both are accepted.

Extend L^1, L^2, L^3 to full labellings by a common arbitrary assignment outside K . If $\mathcal{C} \cap \mathcal{D}$ were not winning on K , use the global operator $R_{(A, \text{in}), L^1}$ of (1). Its reset target is the fixed full extension of L^1 . After revising individuals first, the support for (B, in) on K is exactly $\mathcal{C} \cap \mathcal{D}$, so the pair is rejected. But aggregate-then-revise leaves the collective outcome unchanged because (A, in) was already accepted. This contradicts dynamic rationality. Hence $\mathcal{W}_K$ is closed under intersection and is a finite proper filter. Its kernel $O_K = \bigcap \mathcal{W}_K$ gives the local oligarchy. $\square$

Corollary 9.2 (Local static consequence) *Let K be a component with at least two arguments. Suppose F has a local support representation $\mathcal{W}_K$, is monotonic, and satisfies uniqueness on K . Then its behavior on K is dictatorial. In particular, adding groundedness and dynamic rationality for contraction does not strengthen this conclusion.*

Proof. All three-part partitions of N can be realized as the label supports of an argument in K . Uniqueness therefore says exactly one part wins. Applying this to $(N, \emptyset, \emptyset)$ excludes $\emptyset$ and makes N winning; applying it to $(E, N \setminus E, \emptyset)$ gives complement maximality. Monotonicity gives upward closure. If winning $\mathcal{C}, \mathcal{D}$ had a losing intersection, the partitions $(\mathcal{C} \cap \mathcal{D}, \mathcal{C} \setminus \mathcal{D}, N \setminus \mathcal{C})$ and $(\mathcal{C} \cap \mathcal{D}, \mathcal{D} \setminus \mathcal{C}, N \setminus \mathcal{D})$ would force both $\mathcal{C} \setminus \mathcal{D}$ and $\mathcal{D} \setminus \mathcal{C}$ to win. They are disjoint and can be parts of one three-part partition, a contradiction. Thus $\mathcal{W}_K$ is a finite ultrafilter and is principal at one agent. No update operator is used. $\square$

The static possibility of different dictators in Proposition 4.9 does not imply dynamic compatibility. The present operator classes allow an update triggered

in one component to change another. Within the component-dictator class this forces synchronization, without a cross-component neutrality assumption.

Proposition 9.3 *Let AF have at least two connected components and let F^d be the component-dictator rule of Proposition 4.9. The following are equivalent:*

- (i) d is constant.
- (ii) F^d commutes with every regular labelling-preserving revision operator.
- (iii) F^d commutes with every regular labelling-preserving contraction operator.

Proof. If d is constant, F^d is a global dictatorship, so it commutes with every labelling-preserving update.

For each converse, suppose $d(K) = i \neq j = d(H)$. Choose $A \in K$, $B \in H$, and the following labels; fix all other arguments to **out** for every agent, and set both A, B to **out** for all agents other than i, j :

	A	B
$\mathcal{L}_i$	in	out
$\mathcal{L}_j$	out	in
$X = F^d(\mathcal{L})$	in	in

Let T label A as **out**, B as **in**, and all other arguments as **out**. These are total labellings; completeness is not required.

For revision use $R_{p_+, T}$ of (1), with $p_+ = (B, \mathbf{in})$. Since $p_+ \in X$, aggregate-then-revise leaves X unchanged and retains $(A, \mathbf{in})$. Revising individuals first resets $\mathcal{L}_i$ to T and leaves $\mathcal{L}_j$ unchanged. Component K follows i , so the resulting aggregate labels A as **out**. Commutation fails.

For contraction use $C_{p_-, T}$ of (2), with $p_- = (B, \mathbf{out})$. Since $p_- \notin X$, aggregate-then-contract again retains $(A, \mathbf{in})$. Contracting individuals first resets $\mathcal{L}_i$ to T , while leaving $\mathcal{L}_j$ unchanged, and the aggregate labels A as **out**. Commutation again fails. Both operators are fully defined on all sets and inputs, with parameters fixed independently of the profile. The construction works unchanged when each component has at least two arguments: the additional arguments receive the fixed label **out**. $\square$

Proposition 9.3 is a characterization within the class F^d , not a characterization of arbitrary aggregation rules. Theorem 9.1 states only a local necessary condition. No sufficiency claim for independently chosen local oligarchies or dictators follows. A separate theory of component-local updates would have to restrict both where an update acts and which component information it reads; we do not replace the global operator classes by such a restriction here.

10 Related Work

This paper belongs to the study of collective argumentation, but it focuses on labellings rather than on aggregating the attack relation itself. Work by Coste-Marquis et al. [9], Tohme et al. [15], and Chen and Endriss [7] studies aggregation or preservation problems for argumentation frameworks and attacks. Caminada and Pigozzi [5] study aggregation of argument labellings

directly; Booth et al. [3] develop interval aggregation methods. Our approach is complementary: it uses support-coalition axioms to identify impossibility pressures, and it emphasizes dynamic order-independence.

The static part is closest in spirit to judgment aggregation [14,12,13], where consistency-preserving rules are likewise rare and the rarity motivates rather than undermines the benchmark. Unlike standard judgment aggregation, however, our framework works with three labels rather than binary acceptance and rejection, and the main theorems concern dynamic order-independence rather than static collective rationality alone.

The dynamic part is inspired by AGM belief revision [1] and by work on changing argumentation systems [2,6,10]. The closest dynamic aggregation reference is Chen and Ju [8]. The present paper studies a different object of change from much of the dynamic argumentation literature: the argumentation framework is fixed, while the agents' labellings and the collective labelled-pair outcome are revised or contracted. Consequently, dynamic rationality constrains winning coalitions of labelling aggregators rather than operators for adding or deleting arguments and attacks.

Commutation with every regular labelling-preserving operator is demanding; the class permits large resets and constrains non-labelling outcomes only weakly. Remark 7.2 proves that one fully defined revision operator suffices for dictatorship, not that its class is minimal. Contraction under uniqueness needs no update assumption. The dynamic contribution is revision without uniqueness and global synchronization within the component-dictator class. Informative weaker update classes remain to be identified.

11 Conclusion

There are two routes to dictatorship. Static independence, neutrality, monotonicity, and uniqueness suffice. On the seven-argument framework, groundedness, independence, neutrality, monotonicity, and commutation with regular revision suffice without uniqueness. The fixed reset witness first forces a filter and then excludes non-singleton oligarchies through its non-labelling extension. Global and local contraction under uniqueness are static corollaries.

Different component dictators can satisfy the stated static properties, including complete-labelling preservation. Within that class, commutation with all global revision operators, or all global contraction operators, is equivalent to a common dictator. Local necessary conditions do not justify arbitrary combinations of component rules. Future work should examine weaker update classes, restricted labelling domains, precisely defined component-local updates, weaker independence or neutrality, and iterated updates.

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LLM-Driven Argumentation for Explainable Text Classification

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Abstract

We propose a neuro-symbolic workflow for explainable text classification. A large language model performs *argumentation mining* by extracting reusable argumentative cues and organizing them into a weighted bipolar argumentation framework, while *argumentative reasoning* under a gradual semantics yields an inspectable conclusion. Rather than producing the final prediction directly, the large language model is used to help construct the argumentation framework, which then serves as an interpretable and revisable classifier. We illustrate the approach on fake-job and hate-speech detection and outline an evaluation agenda centered on argument quality, detector grounding, and end-to-end classification behaviour.

Keywords: computational argumentation, large language models, neuro-symbolic AI, explainable AI, text classification.

1 Introduction

Text classification is commonly approached as a mapping from unstructured language to class labels. Deep learning models, ranging from BERT-style encoders to GPT-style large language models, have produced strong predictive performance when trained to learn this mapping from data [2]. However, the reasoning behind their predictions is often difficult to inspect because the final label is produced by an opaque statistical process. This is a limitation when decisions should be transparent, contestable, or open to revision.

Computational argumentation offers a natural way to address this limitation because it makes explicit which considerations support and attack a conclusion.

In practice, however, argumentation-based approaches often depend on manually constructed knowledge bases, rule sets, or other domain artifacts, which can limit scalability and require substantial expert effort. Large language models offer a complementary strength because they can process large amounts of text and help identify recurring argumentative cues, but their own reasoning remains imprecise and opaque [14]. The integration of both approaches can combine the strengths of each while mitigating their weaknesses. Formal argumentation frameworks give a structured way to reason, while LLMs reduce reliance on hand-crafted knowledge and can process large corpora of raw text.

We propose a neuro-symbolic workflow that combines both approaches. A large language model analyses a corpus to discover recurring reasons for and against a class label and encode them in a weighted bipolar argumentation framework, while the argumentation framework performs the final reasoning. Under this approach, the large language model is not the predictor that is later explained. Instead, it is used to help construct an explicit argumentative structure, which subsequently serves as the classifier. This makes the reasoning process inspectable and, in principle, revisable through changes to arguments, relations, and weights.

This paper presents a position and a research prototype within the broader agenda on argumentative human-AI decision-making [13]. More specifically, we operationalize two synergistic tasks at the intersection of LLMs and computational argumentation: *argumentation mining*, which extracts reusable argumentative cues from text, and *argumentative reasoning*, which evaluates the resulting structures under a formal semantics. In the current prototype, these tasks define a two-stage workflow. First, argumentation mining identifies recurring cues in corpora and organizes them into arguments with weighted support and attack relations. Second, argumentative reasoning determines which arguments are detected for a new instance and evaluates them under a gradual semantics to produce a transparent classification. The paper motivates this workflow with two representative domains, reports preliminary prototype results, and demonstrates a reasoning instantiation using weighted bipolar argumentation.

2 Domains and Dataset Selection

We motivate our approach with two domains: fake job posting detection and hate speech detection. Both are binary text classification tasks, but they illustrate different reasons why an argumentative formulation may be useful. Fake job detection involves recurring and relatively observable warning signals, while hate speech detection depends more strongly on context-sensitive counter-considerations.

2.1 Fake Job Posting Detection

Fake job posting detection considers the problem of distinguishing fraudulent job advertisements from genuine ones. We use the Real / Fake Job Posting

Prediction dataset¹, which contains labelled examples of real and fake job postings. This domain is suitable because fraudulent postings often contain recurring warning signals that are comparatively easy to observe in text, such as exaggerated earning claims, vague low-barrier recruitment language, urgency, and sparse organizational detail. These cues can be modelled as arguments that support a fraud conclusion, while cues such as a clearly identified company, a verified recruiter profile, or concrete job responsibilities can be modelled as attacking arguments. Existing approaches to fake job detection mainly rely on black-box models that do not make such considerations explicit [3,1].

Example fragments from the dataset illustrate this pattern. One fraudulent posting states: “URGENT Full & Part Time Workers Needed. No Experience Required ... Earn Easily \$400 to \$500 Extra Per Day”. Another says: “Offering CASH Positions, Immediate Start ... You can earn \$300 to \$400 extra a day. No experience required”. Other fragments repeat the same combination of urgency, minimal qualification demands, and unusually easy earnings, suggesting that these recurring cues can be encoded as supporting arguments for a fraud conclusion.

2.2 Hate Speech Detection

Hate speech detection considers the problem of identifying hateful or offensive content in text. We use the HateXplain dataset, which contains labelled examples of hateful and non-hateful posts [9]. This domain is suitable for a different reason than fake job detection. Here, classification often depends not only on explicit lexical cues but also on context, quotation, and disapproving or self-referential use. This makes the need for both support and attack relations especially clear. Explicit slurs, dehumanizing language, or targeted hostile generalizations may support a hate-speech label, while quotation for discussion, disapproving framing, or educational context may attack it.

Examples from HateXplain illustrate this contrast. The post “pakistan is created for muslims so no muslim refugee in india anymore” expresses explicit exclusion of a target group with little mitigating context, so it should activate support for a hate-speech classification. By contrast, the examples “my grandson a homosexual and i love him i love my big gay grandson” and “you might just think i am a dumb slut but i will have you know i am actually an intellectual slut” show that identity terms or insults can appear in affectionate or self-referential contexts rather than as attacks on a target group. A useful classifier should therefore not only detect potentially hateful expressions, but also represent how context, speaker stance, and target orientation redirect their interpretation.

3 Argumentation Framework

Arguments for and against a class label are represented in a formal argumentation framework, for which several prominent formalisms are available. Basic

¹ <https://www.kaggle.com/datasets/shivamb/real-or-fake-fake-jobposting-prediction>

abstract argumentation provides a clean and influential starting point [6], while structured approaches offer deeper rule-level interpretability with premises and inference rules [5]. However, both are likely too rigid for the present setting. In both domains, the relevant evidence is cumulative and graded rather than purely binary. Several weak warning signals may jointly support a label, while a contextual factor may partially undermine that support without fully defeating it.

For these reasons, we use weighted bipolar argumentation as a practical starting point. It combines explicit support and attack relations with graded strengths [4]. The weighted nature of the framework is especially useful here because it allows different evidential strengths to be represented without forcing an all-or-nothing treatment of noisy cues. A request for sensitive financial information may count more strongly toward a fraud label than a vague job description alone. Similarly, a dehumanizing slur may count more strongly toward a hate-speech label than a mild insult, while quotation or disapproving framing may partially mitigate that support.

4 Neuro-Symbolic Pipeline

We frame the proposed workflow in terms of the two tasks introduced earlier: *argumentation mining* and *argumentative reasoning*. Together, they define how the system moves from raw text to an inspectable class decision.

Argumentation mining is the offline discovery stage. Starting from a dataset of text instances with binary class labels in a domain where classification depends on recurring reasons or counter-reasons, a large language model analyses a sample of instances to identify candidate argumentative cues for or against the label. The goal is not to predict labels directly, but to extract reusable observations in the form of candidate arguments and attack relations that can later be operationalized in a formal argumentation framework. In the fake-job domain these may include cues such as urgency, high earnings, and low required experience. In the hate-speech domain they may include explicit slurs, dehumanizing language, quotation, disapproving framing, or self-referential use.

These candidate cues are then organized into a global weighted bipolar argumentation framework in which the target class is represented as a conclusion node and the mined cues are represented as arguments connected by weighted support and attack relations. In the current prototype, an LLM is used to formulate candidate arguments and to propose relations and corresponding weights, based on the prevalence and discriminative value of cues in the corpus. Recent work suggests that LLMs can effectively assign weights to relations in a bipolar argumentation framework [8,7], which makes this a practical starting point. Another option is to learn or calibrate the weights by testing inference behaviour on labelled data [11]; we leave this to future work.

Argumentative reasoning is the inference stage. For a new text instance, a classifier detects which mined arguments are activated and with what confidence. In the current prototype, this is implemented with an LLM-based detector, but the aim is to ground argument activation in deterministic clas-

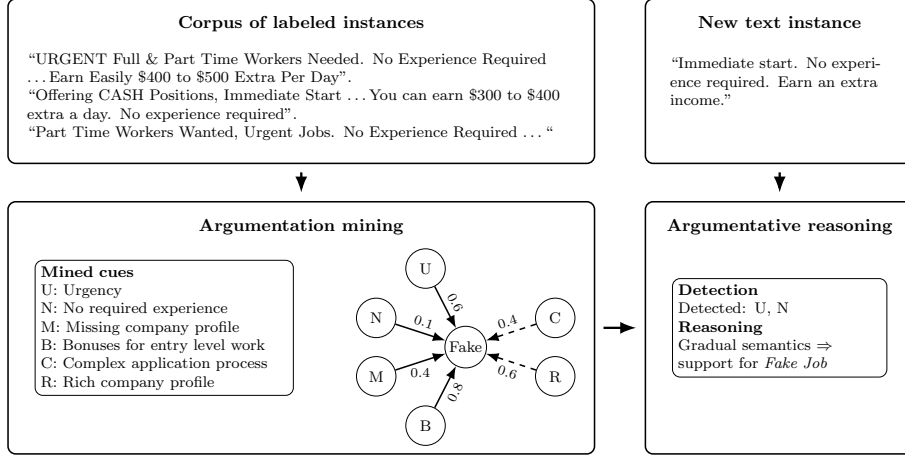


Fig. 1. Neuro-symbolic workflow illustrated with a fake-job example. The left panel shows argumentation mining from text and the right panel shows argumentative reasoning on a new instance using the mined framework. Support relations are shown with solid arrows and attack relations with dashed arrows.

sifiers wherever possible. The activated framework is then evaluated under a gradual semantics, yielding a final strength for the conclusion node. The class label is derived from this evaluated argumentative structure rather than from a direct model score, which makes it possible to inspect which arguments fire, how they interact, and why the final classification is reached. Figure 1 summarizes this workflow.

5 Evaluation Agenda and Challenges

This paper presents preliminary work in progress rather than a completed end-to-end empirical study. The present contribution is therefore primarily a research direction and prototype design for connecting large language models with formal argumentation in explainable text classification. A full evaluation should compare the resulting argumentation-based classifier against strong deep learning-based baselines while also examining whether the explanations it produces are informative and well grounded.

The current prototype provides an initial feasibility signal in the fake-job domain. The mined cues and argumentation framework in Figure 1 were obtained from a single pass of argumentation mining on a sample of the corpus using a large language model.² The LLM was prompted with a description of the task, instructions for the argumentation mining process, and examples of the desired output. In preliminary experiments, a first argumentation-based implementation achieved non-trivial classification performance on a held-out sample of 100 fake-job instances, with accuracy 0.70, precision 0.69, and re-

² We used Gemini 2.5 Flash-Lite as a comparatively strong lightweight LLM.

call 0.72. We do not present these numbers as a full comparative evaluation. Rather, they indicate that the proposed workflow can already be instantiated end-to-end and can serve as a basis for more systematic refinement and assessment.

We distinguish three levels of evaluation. The first concerns end-to-end classification behaviour on held-out instances, using measures such as precision, recall, and F1. This performance should be evaluated against a strong deep learning-based baseline classifier. The goal is not necessarily to surpass that baseline. Reaching similar classification performance would already be a positive result if it is achieved together with greater explainability, inspectability, and revisability.

The second level concerns the quality of the mined arguments: are they grounded in the data and reusable across instances? This level depends on human judgment, since the central question is whether the mined arguments capture meaningful and inspectable considerations rather than merely plausible-sounding cues. For example, the mined cue labelled *Complex application process* (Figure 1) may sound plausible, but it can be difficult to ground consistently in the text because it is not always clear what should count as a complex application process.

The third level concerns analysis of the argumentation framework in operation: which arguments fire, how they interact through support and attack relations, and whether the resulting explanation is plausible. This level, too, requires human judgment, because the value of the framework lies not only in predictive behaviour but also in whether its explicit argumentative structure yields an understandable explanation.

These evaluation questions are closely tied to several main technical challenges. One challenge is the usability of mined arguments and whether they can be reliably detected at inference time. A second challenge is structural consistency, since independently proposed relations may create redundant, spurious, or contradictory links. Third, the mined arguments may be related only to the main claim rather than to intermediate claims or to each other, thus only creating shallow argumentative structures with no interaction between arguments. Finally, even with well-motivated arguments, the final behaviour depends strongly on how weights are initialized and calibrated and on the resulting interaction of support and attack relations, which may be difficult to tune.

6 Related Research

Previous work has explored combining classifiers to identify premises in text with an argumentative structure to draw conclusions. For example, Mumford et al. [10] use Abstract Dialectical Frameworks in conjunction with a learned classifier to predict the outcome of legal cases. Similarly, Schraagen et al. [12] use a classifier to identify premises and then apply the ASPIC framework to draw conclusions on online crime reports. These approaches are similar in spirit to the present proposal, but differ in that both the classifier and the

argumentative structure are hand-crafted rather than being mined from text.

More recently, work on Argumentative LLMs [7] shows that weighted bipolar argumentation can be used to externalize model reasoning into an inspectable and revisable structure. Additionally, work on ArgRAG [15] indicates that a weighted bipolar argumentation framework can be mined from text and used to produce an inspectable conclusion.

7 Conclusion

This paper proposes a neuro-symbolic approach to explainable text classification in which a large language model mines reusable arguments and attack relations from text and a weighted bipolar argumentation framework performs the final reasoning. The paper motivates this approach with two representative domains, fake job posting detection and hate speech detection, which together show why both recurring cues and contextual counter-arguments matter. We also argue that weighted bipolar argumentation is a practical formalism for this setting as it can represent both positive and negative evidence with graded strength. The proposed workflow suggests a concrete way to connect LLM-based pattern discovery with formal, inspectable symbolic reasoning in text classification. Preliminary prototype results in the fake-job domain suggest that the approach is already feasible, even though a full comparative evaluation remains future work.

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Safety, Oscillation, Liveness, and Fairness in Quantitative Argumentation Dialogues

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Abstract

We introduce the notions of *safety*, *liveness*, and *fairness*, as commonly used in temporal reasoning and distributed systems, to quantitative (bipolar) argumentation dialogues where repeated inferences are drawn from argumentation graphs with weighted nodes. Between inferences, these graphs are updated. Safety and liveness capture whether arguments' final strengths always and eventually (respectively) attain a specific threshold of credibility. Fairness notions assess how the safety of arguments is distributed within a sequence of argumentation graphs. Additionally, we introduce the notion of *oscillation* to capture the stability of topic arguments with respect to the threshold of credibility. We formally show how these notions are related and why providing general guarantees for these properties is analytically challenging.

Keywords: formal argumentation, safety, oscillation, liveness, fairness.

1 Introduction

Quantitative (bipolar) argumentation is a formal argumentation variant in which inferences are drawn from graphs whose nodes (*arguments*) with weights (*initial strengths*) are connected by *support* and *attack* relations. Argumentation *semantics* then draw inferences from a *quantitative bipolar argumentation graph* (QBAG) by updating arguments' initial strengths to *final strengths*, considering the QBAG's topology. QBAGs are of substantial interest to the formal argumentation community, notably because of their potential to complement machine learning approaches and help make them *explainable* [21,27] and *contestable* [26]. QBAG applications are emerging that facilitate the explainability and reliability of large language model-based inference [28,9,12]. In some applications, argumentation can be viewed as a dynamic process in which QBAGs are manipulated, and repeated inferences are drawn. Accordingly, it may be interesting to inquire about the dynamic properties of QBAG-based inference.

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This paper systematizes such inquiries by drawing inspiration from well-known notions of *safety*, *liveness*, and *fairness*, which are prominent in various temporal reasoning approaches, such as linear temporal logic [1,10], distributed processes [2], and Petri nets [16]⁴. In addition, we introduce the notion of *oscillation*, which aims to capture the stability of a topic argument. Collectively, we refer to these notions as *SOLF*.

Our notions of SOLF are introduced against the backdrop of a *threshold of credibility*. Intuitively, arguments whose final strengths *attain* the threshold are deemed credible, whereas arguments that fail to meet the threshold are not. Our SOLF notions aim to assess the credibility of a set of topic arguments over a dynamic process. This process is specified by a sequence of QBAGs (which we call a *chain* or *dialogue*). The *topic arguments* are a subset of arguments occurring in all of the sequence's graphs. Note that we assume the threshold of credibility is use-case dependent, analogous to the interpretation of final strengths in QBAGs in general.

Given this backdrop, our notions allow us to ask the following questions:

Safety: Do all of the topic arguments always attain the threshold? This notion of safety can, for instance, help us determine whether obviously credible arguments always retain reasonable strengths in the dialogue, or whether the dialogue derails. This interpretation of safety aligns with the canonical Linear Temporal Logic (LTL) reading, namely that “*bad things*” *do not happen during execution*, where “bad things” refers to the topic arguments falling below the threshold [1].

Oscillation: If the topic arguments are credible at the beginning, do they retain their credibility throughout the process? Oscillation describes the behavior of the topic arguments' final strengths throughout the dialogue, i.e., whether they remain stable or flip-flop across the threshold. Intuitively, this reflects that such arguments are subject to substantial debate.

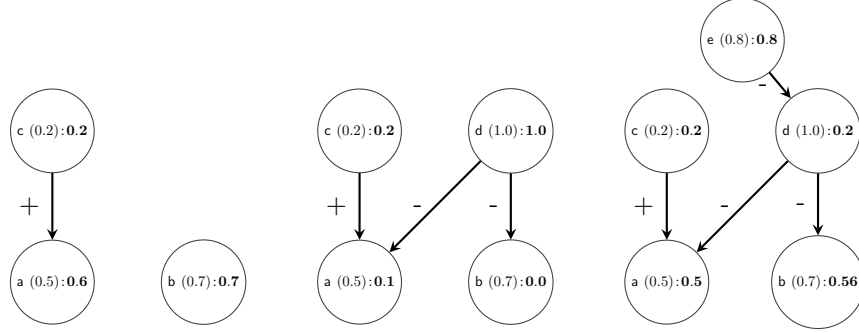
Liveness: Do all of the topic arguments eventually become credible, i.e., do they attain the threshold at the end of the dialogue? Again, this interpretation of liveness is aligned with the standard reading in LTL, namely “*good things*” *do happen (eventually)*, where “good things” refers to the topic arguments attaining the threshold [1].

Fairness: Do all topic arguments get to have an equal chance to attain credibility? We present several discrete and gradual fairness notions, which are based on safety.

We informally demonstrate our intuitions in the following example.

Example 1.1 Consider the QBAGs in Figure 1. Each argument is labelled $x(y) : z$, denoting the name, initial strength, and final strength, respectively. Intuitively, initial strengths represent the credibility of arguments per

⁴ We do not formally relate to these notions of *tracial control*, as we are interested in conceptually somewhat different questions, namely the dynamics of quantitative argument strengths.



1.1: Initial QBAG G . 1.2: Normal expansion G' . 1.3: Normal expansion G'' .

Fig. 1. From left to right, the initial QBAG expands by successively adding d (normal expansion G') and e (normal expansion G''), respectively. Such scenarios capture the essence of *dynamic argumentation*.

se, whereas arguments' final strengths reflect the justification reached through argument interactions. We denote $\sigma_G(x)$ as the final strength of x in the QBAG G and apply DFQuAD semantics, as defined in the preliminaries. Interactions are represented via attack and support relations, labelled $-$ and $+$, respectively. The QBAGs in Figure 1 portray a *normal expansion chain*: new arguments and relationships are added, but relationships between existing arguments (and arguments' initial strengths) remain unchanged.

Our SOLF notions are introduced with a *threshold of credibility*; credible arguments are assumed to have final strengths that attain this threshold. We understand safe argument sets to be credible throughout the process. For instance, $\{c\}$ is safe for the threshold $t = 0.1$ in Figure 1. Liveness stipulates that arguments always become credible eventually, i.e., at the end of the dialogue chain, their final strengths attain the threshold. $\{b\}$ is live w.r.t. the threshold of 0.1, since $\sigma_{G''}(b) \geq 0.1$. Extending this observation, $\{b\}$ also *oscillates* w.r.t. 0.1, since its final strength is below 0.1 in G' , but attains it again in G'' , thus crossing the threshold. Our notions of fairness then cover two dimensions: several binary classifications based on safety, and gradual notions based on the established measures of the Gini index and Shannon entropy.

In the example, $\{a, b\}$ is *cautiously fair*, which requires that if any of the topic arguments becomes credible (at the end), so does the whole set. Indeed, $\{a, b\}$ is cautiously fair because at least $\{a\}$ or $\{b\}$ is live. However, our gradual fairness notions provide more nuance: $\{a, b, c\}$ yields fairness scores of ≈ 0.98 and ≈ 0.46 for the *Shannon-based* and *Gini-based* notions (defined in Section 3), respectively. Intuitively, this occurs because $\{c\}$ has a better chance at justification than $\{a\}$ or $\{b\}$, indicating imperfect uncertainty (Shannon entropy) and inequality of justification distribution among arguments (Gini index).

In the following sections, we introduce relevant preliminaries (Section 2), to then formally present our SOLF notions, observing some of their properties, as well as how the notions are interrelated (Section 3). We also discuss our results in the context of related research (Section 4) and sketch why giving guarantees w.r.t. the SOLF notions may be challenging using an empirical demonstration of semantics behavior, thus highlighting directions for future work (Section 5).

2 Preliminaries

Let $\langle \mathbb{I}, \preceq \rangle$ be a partial order. Typically, $\mathbb{I}$ will be a real interval alongside the canonical order (e.g., $\mathbb{I} = [0, 1]$). A *quantitative bipolar argumentation graph* (QBAG) represents arguments with *initial strengths* in $\mathbb{I}$, and support and attack relations between these arguments. Intuitively, an initial strength may represent the initial confidence in the argument without any influence; greater initial strengths represent higher confidence.

Definition 2.1 [QBAG] A *Quantitative Bipolar Argumentation Graph* (QBAG) is a quadruple $\langle Args, \tau, Att, Supp \rangle$, where $Args$ is a non-empty finite set of elements called *arguments*, $Att, Supp \subseteq Args \times Args$ are the *attack* and the *support* relations, respectively, over the arguments, with $Att \cap Supp = \emptyset$, and $\tau : Args \rightarrow \mathbb{I}$ is a total function assigning an *initial strength* $\tau(x)$ to each argument $x \in Args$.

Figure 1 features some QBAGs that are explained in Example 1.1. Unless stated otherwise, let $G = \langle Args, \tau, Att, Supp \rangle$ be an arbitrary but fixed QBAG, with finite $Args$. Given an argument $x \in Args$, the attackers and supporters of x are denoted by

$$Att_G(x) := \{a \in Args : \langle a, x \rangle \in Att\},$$

$$Supp_G(x) := \{a \in Args : \langle a, x \rangle \in Supp\},$$

respectively. We drop the subscript G whenever it is clear from context. Given two arguments $x, y \in Args$, we say that x *reaches* y if and only if there is a path from x to y in the graph $\langle Args, Att \cup Supp \rangle$. An *acyclic* QBAG does not contain any loops, i.e., no argument $x \in Args$ can reach itself. For any $Args' \subseteq Args$, we define the *restriction of G to $Args'$* as

$$G \downarrow_{Args'} := \langle Args', \tau \cap (Args' \times \mathbb{I}), Att \cap (Args' \times Args'), Supp \cap (Args' \times Args') \rangle.$$

As we are interested in dynamic argumentation, we require notions that relate several QBAGs.

Definition 2.2 [Sub-QBAG] We say that $G' = \langle Args', \tau', Att', Supp' \rangle$ is a *sub-QBAG* of G , denoted by $G' \sqsubseteq G$, if and only if $Args' \subseteq Args$, $Att' \subseteq Att$, $Supp' \subseteq Supp$, and $\tau'(x) = \tau(x)$, for every $x \in Args'$. If $G' \sqsubseteq G$ but $G' \neq G$, we may denote this by $G' \sqsubset G$.

Our idea of argumentation dialogues is captured via QBAG chains. Intuitively, chains represent evolutions of a given QBAG along arbitrary dimensions, such as expansions and initial strength updates.

Definition 2.3 [QBAG Chains] A *QBAG chain* (short: *chain*) is a sequence of QBAGs $\mathcal{G} = (G_1, \dots, G_n)$.

Given a chain $\mathcal{G} = (G_1, \dots, G_n)$, when we write $G \in \mathcal{G}$, we mean that for some G_i , $1 \leq i \leq n$, it holds that $G_i = G$. We may call G_1 and G_n the *start point* and *end point* of $\mathcal{G}$, respectively.

The notion of QBAG chains capture a broad range of QBAG modifications: attack and support additions/removals, strength change updates and expansions. If a chain only encompasses expansions, i.e., the addition of new arguments to the previous QBAG in the chain, we call it a *QBAG expansion chain*.

Definition 2.4 [QBAG Expansion Chains] A *QBAG expansion chain* (or simply *expansion chain*) is a chain $(G_1, \dots, G_n)$, with $G_i \sqsubset G_{i+1}$ for $1 \leq i < n$. An expansion chain $\mathcal{G}$ is called:

- (i) *normal* if and only if for every $G_i = \langle \text{Args}, \tau, \text{Att}, \text{Supp} \rangle$ and $G_j = \langle \text{Args}', \tau', \text{Att}', \text{Supp}' \rangle$, with $G_i, G_j \in \mathcal{G}$ and $G_i \sqsubset G_j$ it holds that $\forall \langle x, y \rangle \in (\text{Att}' \cup \text{Supp}') \setminus (\text{Att} \cup \text{Supp})$, either x or y is in $\text{Args}' \setminus \text{Args}$;
- (ii) *weak* if and only if $\mathcal{G}$ is normal and for every $G_i = \langle \text{Args}, \tau, \text{Att}, \text{Supp} \rangle$ and $G_j = \langle \text{Args}', \tau', \text{Att}', \text{Supp}' \rangle$, with $G_i, G_j \in \mathcal{G}$ and $G_i \sqsubset G_j$ it holds that $\forall \langle x, y \rangle \in (\text{Att}' \cup \text{Supp}') \setminus (\text{Att} \cup \text{Supp}) : y \notin \text{Args}$.

Example 2.5 Referring to Figure 1, $\mathcal{G} = (G, G', G'')$ is a (*normal* but not *weak*) expansion chain, where G and G'' are the start and end points of $\mathcal{G}$, respectively.

Note that while the examples provided in this paper focus on expansion chains, our SOLF notions are applicable to arbitrary chains.

Definition 2.6 [Gradual semantics] A *gradual semantics* is a function $\sigma_G : \text{Args} \rightarrow \mathbb{I} \cup \{\perp\}$ mapping the arguments in a QBAG G to the partial order $\mathbb{I}$, along with the symbol $\perp$ for undefined [3].

Modular semantics are well-studied sub-class of gradual semantics, where the final strength of an argument is assigned by *aggregating* the (final) strengths of its attackers and supporters, and then computing the *influence* on the argument given the aggregation result and the argument's initial strength. More formally, an *aggregation function* has the form $f(A, S)$, given multi-sets of attack values A and support values S . The application of an influence function $g_s(f(A, S))$ then yields an update of an argument's strength value $s \in \mathbb{I}$. Modular semantics first *initialize* all arguments by assigning their initial strengths as the temporary strength value, and then update these values by applying aggregation and influence function to these values following the topological order of the QBAG (in the direction of the binary relations).

In our examples, we make use of DFQuAD semantics [22], given $\mathbb{I} = [0, 1]$.

Definition 2.7 [DFQuAD] Let $A = \text{Att}(x)$ and $S = \text{Supp}(x)$, for $x \in \text{Args}$.

For the aggregation function

$$f(A, S) = \prod_{b \in A} (1 - \sigma(b)) - \prod_{c \in S} (1 - \sigma(c))$$

and initial strength $\tau(a)$, DFQuAD is defined over acyclic graphs as per the equation given below:

$$g(\tau(a), f(A, S)) := \tau(a) - \tau(a) \cdot \max\{0, -f(A, S)\} + (1 - \tau(a)) \cdot \max\{0, f(A, S)\}.$$

Assumption. Notice that for cyclic QBAGs, modular semantics may not converge and thus yield undefined final strengths in some cases [20]. Because of this, we focus on acyclic QBAGs, which is a common assumption in the literature [17,7,18,6], i.e., we assume that final strengths are comparable.

3 Safety, Oscillation, Liveness, and Fairness Properties

In this section, we formally introduce our SOLF notions for QBAGs. The notions are validated w.r.t. an initial QBAG G , a chain of updates $\mathcal{G}$, a semantics σ , and a threshold $t \in \mathbb{I}$, which we call the *threshold of credibility*. Throughout this section, we fix a partial order $\mathbb{I}$, arbitrary QBAGs $G = \langle \text{Args}, \tau, \text{Att}, \text{Supp} \rangle$, $G' = \langle \text{Args}', \tau', \text{Att}', \text{Supp}' \rangle$, $G'' = \langle \text{Args}'', \tau'', \text{Att}'', \text{Supp}'' \rangle$, and a QBAG chain $\mathcal{G} = (G_1, \dots, G_n)$. We also fix a non-empty finite set of topic arguments $T \subseteq \bigcap_{1 \leq i \leq n} \text{Args}_{G_i}$, i.e., the arguments occurring in the entire chain, to analyze the SOLF notions.

Safety and Liveness

Intuitively, safe arguments remain credible throughout the whole span of the dialogue, i.e., their final strengths remain above the threshold of credibility when inferences are repeatedly drawn.

Definition 3.1 [Safety] We say that a topic set T is safe w.r.t. $\mathcal{G}$, σ , and t if and only if for every $G \in \mathcal{G}$, and $x \in T$, it holds that $\sigma_G(x) \geq t$.

When we discuss whether a topic set is safe w.r.t. some chain $\mathcal{G}$, semantics σ , and threshold t , we may leave the specification of any of $\mathcal{G}$, σ , or t implicit. For other SOLF notions, we proceed analogously; notably, in our examples, we generally assume DFQuAD semantics.

When assessing liveness, we are interested only in the credibility of the topic arguments at the end of the dialogue, regardless of their credibility throughout the process.

Definition 3.2 [Liveness] We say that a topic set T is live w.r.t. $\mathcal{G}$, σ , and t if and only if T is safe w.r.t. the (singleton) QBAG chain (G_n) , σ , and t .

We revisit our initial example to illustrate the safety and liveness notions.

Example 3.3 In Figure 1, the arguments $\{c\}$ and $\{b\}$ are safe and live, respectively, w.r.t. (G, G', G'') and the threshold $t = 0.1$. However, if the threshold of credibility is changed to $t = 0.3$, $\{c\}$ is neither safe nor live, whereas $\{b\}$ continues to be live.

Since safety concerns the credibility of topic arguments over the whole process, it is immediate that safety implies liveness.

Theorem 3.4 *If a topic set T is safe w.r.t. $\mathcal{G}$, σ , and t , then it is live w.r.t. $\mathcal{G}$, σ , and t .*

Proof. Let $T \subseteq \text{Args}$ be safe w.r.t. $\mathcal{G} = (G_1, \dots, G_n)$, σ , and t . This means $\forall G \in \mathcal{G}, \forall x \in T$ it holds that $\sigma_G(x) \geq t$, and thus also $\sigma_{G_n}(x) \geq t$. The previous statement is the condition for liveness, as G_n is the end point of the chain. $\square$

In general, the converse of Theorem 3.4 does not hold, as shown in Example 3.3. However, under certain restrictions on the expansion sets and the choice of semantics, a partial converse can be achieved. In the subsequent theorem, we propose two of such cases.

Theorem 3.5 *Suppose one of the following cases holds:*

- (i) *The chain $\mathcal{G}$ has length 1, or*
- (ii) *$\mathcal{G}$ is a weak expansion chain and σ is a modular semantics.*

Then safe and live arguments coincide w.r.t. $\mathcal{G}$, σ , and t .

Proof. Assume T is live w.r.t. $\mathcal{G}$, σ , and t , where the chain $\mathcal{G}$ is denoted by $(G_1, \dots, G_n)$. Now, we prove our claim case by case.

Case (i): If $\mathcal{G}$ has length 1, then $\mathcal{G} = (G_1)$, and hence $G_1 = G_n$. Since the start and end point of $\mathcal{G}$ coincide, by Definition 3.2, $\sigma_{G_1}(x) \geq t$, for every $x \in T$. Since $\mathcal{G}$ is a singleton, $\forall G \in \mathcal{G}, \sigma_G(x) \geq t$, which is the condition for safety.

Case (ii): By the strong directionality argument [14] (Principle 4, Lemma 6.1), weak expansions under any modular semantics σ do not alter the final strengths of arguments in G . Thus, $\forall G, G' \in \mathcal{G}$ and $x \in T$, $\sigma_G(x) = \sigma_{G'}(x)$. Accordingly, it also holds $\forall x \in T$ that $\sigma_{G_n}(x) \geq t$ if and only if $\forall G \in \mathcal{G} : \sigma_G(x) \geq t$. $\square$

Oscillation

Safety and liveness concern the behavior of topic arguments at or above the threshold of credibility, i.e., whether the arguments remain consistently credible, or eventually attain credibility at the end. However, these notions do not capture cases where arguments remain relatively stable, only to lose credibility at a few specific instances during (or at the end of) the dialogue. For instance, an argument may remain consistently above the threshold of credibility throughout the expansion chain, but drop below the threshold at the end point. Such an argument is then not live and hence not safe. However, it is still worthwhile to investigate the threshold-crossing behavior of such arguments.

To study these cases, we introduce the notion of *oscillation*. This notion aims to answer whether or not an argument's credibility remains stable over the whole expansion chain. The idea is formally introduced through fluctuations across the threshold of credibility. Intuitively, fluctuations quantify the “flip-

flopping” behavior exhibited by an argument. Formally, we say that $x \in T$ *moves across a threshold t* w.r.t. σ and QBAGs G, G' if and only if either $\sigma_G(x) < t \leq \sigma_{G'}(x)$, or $\sigma_{G'}(x) < t \leq \sigma_G(x)$ holds. This gives us a key prerequisite to define k -fluctuations as follows.

Definition 3.6 [k -Fluctuations] We say that $x \in T$ shows k -*fluctuations* w.r.t. $\mathcal{G} = (G_1, \dots, G_n)$, σ , and t iff there exist natural numbers $1 < i_1 < \dots < i_k < n$ such that x moves across the threshold t w.r.t. σ and every $G_{i_j}, G_{i_{j+1}} \in \mathcal{G}$. We say x shows *exactly k -fluctuations* if and only if x shows at least k -fluctuations and does not show $k + 1$ -fluctuations.

Intuitively, fluctuations are determined relative to the threshold of credibility, which must be (at least) met for threshold attainment. If an argument’s final strength is equal to the threshold and attains it at the next QBAG in the chain, no fluctuations are counted. Similarly, no fluctuations are counted if an argument’s final strength is above the threshold of credibility and becomes equal to the threshold at the next QBAG in the chain. We apply k -fluctuations to topic sets, which gives rise to the notion of *oscillation*.

Definition 3.7 [Oscillation] We say that a topic set T *oscillates* w.r.t. $\mathcal{G}$, σ , and t if and only if each $x \in T$ shows at least 1-fluctuation w.r.t. $\mathcal{G}$, σ , and t . We say that a topic set is (at least / exactly) n -*oscillating* if each of its arguments shows (at least / exactly) n -fluctuations.

Let us illustrate these definitions with the help of an example.

Example 3.8 In Figure 1, the chain $\mathcal{G} = (G, G', G'')$ has the topic set $\{a, b\}$ as 2-oscillating w.r.t. the threshold 0.2, which means that $\{a\}$ and $\{b\}$ cross the threshold at least twice. On the other hand, $\{c\}$ shows no fluctuation w.r.t. $t = 0.2$; accordingly, it is safe and not oscillating.

Intuitively, oscillating topic sets have the *potential* to be live and safe, but in practice, they may *actually* be live but *never* safe.

Theorem 3.9 *If a topic set T is safe w.r.t. $\mathcal{G}$, σ , and t then it does not oscillate w.r.t. $\mathcal{G}$, σ , and t .*

Proof. Suppose T is safe, then by definition $\sigma_G(x) \geq t, \forall G \in \mathcal{G}, \forall x \in T$. Now, if T oscillates, then each $x \in T$ must cross the threshold t at least once. This is impossible, as it would contradict the safety of T . $\square$

On the other hand, oscillating topic sets can be live, given an appropriate number of fluctuations. This can be observed in Example 3.8 and Figure 1, where $\{a, b\}$ is live and exactly 2-oscillating.

In certain cases, we can guarantee that no fluctuations are observed in the topic set. These include cases of weak expansions and evolutions that only append downstream arguments. We formally state this below as a corollary; the proof is a straight-forward application of Theorem 3.5 and Theorem 3.9.

Corollary 3.10 *Suppose that one of the following cases holds:*

- (i) *The chain $\mathcal{G}$ has length 1, or*

(ii) $\mathcal{G}$ is a weak expansion chain and σ is a modular semantics.

Then, for all $x \in \text{Args}$, x shows no fluctuations w.r.t. $\mathcal{G}, \sigma$, and $t \in \mathbb{I}$. $\square$

Discrete Fairness

Conceptually, our notions of fairness evaluate whether the topic arguments have an equal shot to achieve safety or liveness. Accordingly, these notions reflect the concept of *resource distribution* (of final strengths equal to or greater than the threshold) among arguments in the topic set. Intuitively, these notions condition the safety or liveness of all topic arguments on the existence of safety or liveness in one topic argument in terms of a necessary condition. We introduce three notions of discrete fairness, namely *ideally fair*, *live fair*, and *cautiously fair*. The notion of ideally fair stipulates that the safety of one topic argument implies the safety of the whole topic set. Live fair is the counterpart for liveness of ideally fair, i.e., if one topic argument is live, then the whole topic set is live. Lastly, cautiously fair is a weakening of the ideally fair condition. If any topic argument is safe, then the whole topic set is at least live.

Definition 3.11 [Discrete Fairness] We say a topic set T , w.r.t. $\mathcal{G}, \sigma$, and t , is

- (i) **Ideally Fair** if and only if some $\{x\} \subseteq T$ is safe implies T is safe.
- (ii) **Live Fair** if and only if some $\{x\} \subseteq T$ is live implies T is live.
- (iii) **Cautiously Fair** if and only if some $\{x\} \subseteq T$ is safe implies T is live.

In our running example of Figure 1, sets of arguments satisfy different fairness notions.

Example 3.12 Refer again to Figure 1, and consider $\{a, b, c\}$ to be the topic set. Example 3.3 highlights that $\{c\}$ is safe w.r.t. (G, G', G'') and threshold 0.2. The final strengths of all topic arguments final attain 0.2 at the end point G'' , thereby making the set $\{a, b, c\}$ both cautiously fair and live fair. However, ideal fairness is violated by $\{a, b, c\}$ as $\{c\}$ is safe, whereas $\{a\}$ and $\{b\}$ are live but not safe.

The following connections are present between the discrete notions of fairness.

Theorem 3.13 For any topic set T , the following holds w.r.t. $\mathcal{G}, \sigma$, and t :

- (i) *Live fairness implies cautious fairness;*
- (ii) *Ideal fairness implies cautious fairness;*
- (iii) *Ideal fairness implies live fairness in the presence of a safe argument.*

Proof. For the first case, assume T is live fair w.r.t. $\mathcal{G}, \sigma$, and t . If there is a safe $\{x\} \subseteq T$, then by Theorem 3.4, $\{x\}$ is live, and consequently T is live due to live fairness. This satisfies the constraint for being cautiously fair.

For the second case, suppose T is ideally fair, and $\{x\} \subseteq T$ is safe. Now, by ideal fairness of T , we get that T is safe. By Theorem 3.4, T is also live.

For the third case, suppose T is ideally fair with a safe topic argument x , which implies T is safe, and by Theorem 3.4, T is live, and therefore live

fairness is satisfied. $\square$

Additionally, given a safe topic argument, ideal fairness prohibits the oscillation of topic arguments, thereby implying an “uninteresting” behavior of the dialogue with respect to the concerned arguments.

Theorem 3.14 *W.r.t. $\mathcal{G}$, σ , and t , if T is ideally fair, then T does not oscillate whenever there is a safe $\{x\} \subseteq T$.*

Proof. Suppose T is ideally fair w.r.t. $\mathcal{G}, \sigma$, and t , with a safe $\{x\} \subseteq T$. This implies every $x \in T$ satisfies $\sigma_G(x) \geq t$ for every $G \in \mathcal{G}$, thus showing no fluctuations; accordingly, T does not oscillate. $\square$

Gradual Fairness

Conceptually, our notions of safety and liveness analyze whether the topic arguments are credible over the course of a dialogue: safety stipulates that the topic arguments attain the specified threshold, whereas liveness ensures threshold attainment at the end of the dialogue. Hence, we may ask *to what extent* all topic arguments have equal opportunities to attain credibility over the course of the dialogue, i.e., how balanced the distribution of threshold-attaining arguments is among the topic set across the QBAGs in a given chain.

To that end, we introduce gradual notions of fairness, aimed to measure the fairness of threshold-attainment among arguments in the topic set. Specifically, we consider two notions: Gini-based fairness and Shannon-based fairness. The former is based on the Gini index [11], which traditionally measures “inequalities” in the distribution of wealth, income, or consumption. In our setting, we measure how closely threshold attainment within a topic set is aligned with an *ideal scenario*: in this (typically counterfactual) scenario, the total number of threshold-attaining instances within the topic set is equally distributed among each topic argument.

Definition 3.15 [Fairness Line and Safety Curve] Let $\mathcal{G}$ be a chain, T be a non-empty finite set of topic arguments, and t be a threshold. For any argument $x \in T$, we define

$$S_{\mathcal{G}, \sigma, t}(x) := |\{G \in \mathcal{G} : \sigma_G(x) \geq t\}|, \quad (1)$$

which is the number of QBAGs in $\mathcal{G}$ for which x *attains* t . We drop the subscripts whenever the context is clear. Let $S_{\text{total}} = \sum_{y \in T} S(y)$ denote the sum of threshold attainments for all the arguments.

The *fairness line* $f(x)$ is the linear segment joining $(0, 0)$ to $(|T|, S_{\text{total}})$, defined piecewise as:

$$f(x) := \begin{cases} 0 & \text{if } x < 0, \\ \frac{S_{\text{total}}}{|T|} \cdot x & \text{if } 0 \leq x < |T|, \\ S_{\text{total}} & \text{if } x \geq |T|. \end{cases} \quad (2)$$

Let $n = |T|$, and let $z_0, \dots, z_{n-1}$ be the arguments in T arranged in increasing order of the number of times that each argument attains the threshold, i.e.,

if $0 \leq j < i < n$, then $S(z_j) \leq S(z_i)$. The *safety curve* is a piecewise linear spline defined as:

$$\zeta_{\mathcal{G},T,\sigma}(x) := \begin{cases} 0 & \text{if } x < 0, \\ S(z_0) \cdot x & \text{if } 0 \leq x < 1, \\ \left(\sum_{0 \leq j < i} S(z_j) \right) + S(z_i) \cdot (x - i) & \text{if } i \leq x < i + 1 \\ & \text{for } i \in \{1, \dots, n-1\}, n > 1, \\ S_{\text{total}} & \text{if } x \geq n. \end{cases} \quad (3)$$

Gini-based fairness measures fairness in terms of the area enclosed between the fairness line and the safety curve. Formally, it is introduced as follows.

Definition 3.16 [Gini-based Fairness] For the topic set T , define the unnormalized Gini-based fairness score using Equation 3 w.r.t. $\mathcal{G}$, σ , and t :

$$\mathcal{F}_{Gini}^u(\mathcal{G}, T, \sigma, t) := \int_0^{|T|} |\zeta_{\mathcal{G},\sigma,t}(x) - f(x)| dx. \quad (4)$$

Thus, the (normalized) Gini-based fairness score is defined as

$$\mathcal{F}_{Gini}(\mathcal{G}, T, \sigma, t) := \left(\frac{2}{1 + e^{-\mathcal{F}_{Gini}^u(\mathcal{G}, T, \sigma, t)}} - 1 \right). \quad (5)$$

We illustrate the fairness notion with the help of the following example.

Example 3.17 Consider the set of topic arguments $T = \{a, b, c\}$ in Figure 1. Here, a and b remain above the threshold $t = 0.2$ for the QBAGs G, G'' , w.r.t. the chain (G, G', G'') . The argument c , with $\{c\}$ being safe, attains t for the entire chain, i.e., in all three QBAGs, while a and b attain t in only two QBAGs. Thus, T can be ordered as (a, b, c) , corresponding to $\langle a, 2 \rangle \preceq \langle b, 2 \rangle \prec \langle c, 3 \rangle$ when arranged by the number of threshold-attaining instances over the whole chain.

Figure 2 plots the *fairness line* (dotted line) using Equation 2. This line represents the situation where the seven threshold-attainment instances are equally distributed among the topic arguments in T . The *fairness line* connects the points $(0, 0)$ and $(3, 7)$, as per Equation 2. The *safety curve* (solid line) in Figure 2 is plotted as per Equation 3, and the actual fluctuations shown by T . Intuitively, the fairness line shows the cumulative sum of fluctuations w.r.t. the topic arguments according to the aforementioned ordering. Fairness is now measured in terms of the area enclosed between the safety curve and the fairness line. In the example, the Gini-based fairness score is ≈ 0.46 .

As noted in Example 3.17, the last equation ensures monotonicity of the Gini-based fairness notion; as the safety curve deviates further from the fairness line, $\mathcal{F}_{Gini}$ approaches 1. The case of perfect equality yields a score of 0.

Theorem 3.18 *Fairness of fluctuations is a positive function having its range in the unit interval $[0, 1)$. The value 0 represents perfect equality, i.e., when the safety curve coincides with the fairness line.*

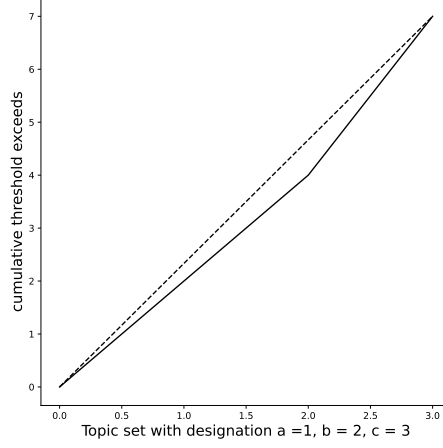


Fig. 2. Plot of the fairness line versus ζ . The solid line is due to the safety curve ζ , and the fairness line is represented using dashes. The area enclosed between the two curves measures the inequality in fluctuation.

Proof. Notice that $\mathcal{F}_{Gini}^u(\mathcal{G}, T, \sigma, t) \geq 0$ implies that $0 \leq \mathcal{F}_{Gini}(\mathcal{G}, T, \sigma, t) < 1$. In the case where $\mathcal{F}_{Gini}^u(\mathcal{G}, T, \sigma, t) = 0$, i.e., when the fairness line and ζ coincide, $\left(\frac{2}{1+e^{-\mathcal{F}_{Gini}^u(\mathcal{G}, T, \sigma, t)}}\right) = 1$ and by definition, $\mathcal{F}_{Gini}(\mathcal{G}, T, \sigma, t) = 0$. $\square$

Our second gradual fairness notion is based on Shannon entropy [24].

Definition 3.19 [Threshold-attainment Probability] The *threshold-attainment probability* for the set of topic arguments T w.r.t. $\mathcal{G}$, σ , and a threshold $t \in \mathbb{I}$ is defined as

$$p_{\mathcal{G}, t, \sigma}(x) := \frac{S_{\mathcal{G}, t, \sigma}(x)}{\sum_{y \in T} S_{\mathcal{G}, t, \sigma}(y)},$$

which is the ratio between the total threshold-attaining QBAGs for x in $\mathcal{G}$ and the total number of threshold-attaining QBAGs for all the arguments in the entire set of topic arguments T .

We drop the subscript whenever it is clear from context. Notice that p is *undefined* in the absence of threshold-attainment behavior, i.e., if no argument in T attains the threshold.

Given an arbitrary threshold-attainment behavior, we want to measure how *surprising* it is.

Definition 3.20 [Information Function] Formally, we define the *information function* I_p in terms of a topic argument x with probability $p(x)$ as

$$I_p(x) := \log_b \left(\frac{1}{p(x)} \right) = -\log_b(p(x)) = \frac{-\log(p(x))}{\log b},$$

where $b = |T|$.

Note that $I_p(\mathbf{x})$ is either non-negative or *undefined*. I_p is undefined whenever $p_{\mathcal{G},\sigma,t}(\mathbf{x}) = 0$ or $\log b = 0$; the former corresponds to the cases where $\mathbf{x}$ is consistently below the credibility threshold throughout the dialogue. When accounting for Shannon fairness, we leave out such cases, i.e, Shannon based fairness ignores consistently marginal arguments. On the other hand, the case where $\log b = 0$ occurs is when the topic set is a singleton, and in this case, we trivially assign the value 1.

Formally, Shannon-based fairness is defined as the expectation of the surprise factor.

Definition 3.21 [Shannon-based Fairness] For the topic arguments T w.r.t. the chain $\mathcal{G}$, semantics σ and threshold $t \in \mathbb{I}$, Shannon-based fluctuation fairness is defined as:

$$\mathcal{F}_{Shannon}(\mathcal{G}, T, \sigma, t) := \begin{cases} \sum_{\mathbf{x} \in T: p(\mathbf{x}) > 0} p(\mathbf{x}) \cdot I_p(\mathbf{x}) & \text{if } p \text{ defined, } b > 1, \\ 1 & \text{otherwise.} \end{cases} \quad (6)$$

which is expanded to

$$\mathcal{F}_{Shannon}(\mathcal{G}, T, \sigma, t) := \begin{cases} \frac{-1}{\log b} \cdot \sum_{\mathbf{x} \in T: p(\mathbf{x}) > 0} p(\mathbf{x}) \cdot \log(p(\mathbf{x})) & \text{if } p \text{ defined, } b > 1, \\ 1 & \text{otherwise.} \end{cases} \quad (7)$$

where $b = |T|$.

Example 3.22 Let us determine the Shannon-based fairness for the topic set $T = \{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ and threshold $t = 0.2$ in Figure 1. We first compute the values of $S(\mathbf{x})$ for each $\mathbf{x} \in T$, which are $\{\langle \mathbf{a}, 2 \rangle, \langle \mathbf{b}, 2 \rangle, \langle \mathbf{c}, 3 \rangle\}$, and the sum of these values is 7. The probability of fluctuation is given by $p(\mathbf{a}) = \frac{2}{7}$, $p(\mathbf{b}) = \frac{2}{7}$, and $p(\mathbf{c}) = \frac{3}{7}$.

Since $b = 3$,

$$\mathcal{F}_{Shannon}(\mathcal{G}, T, \sigma, t) = \frac{-1}{\log 3} \left(\frac{2}{7} \cdot \log \frac{2}{7} + \frac{2}{7} \cdot \log \frac{2}{7} + \frac{3}{7} \cdot \log \frac{3}{7} \right) \approx 0.98 .$$

The next result provides a case where Shannon-based fairness is maximized.

Theorem 3.23 If $\forall \mathbf{x}, \mathbf{y} \in T, S(\mathbf{x}) = S(\mathbf{y})$ w.r.t. $\mathcal{G}, \sigma$ and $t \in \mathbb{I}$, then $\mathcal{F}_{Shannon}(\mathcal{G}, T, \sigma) = 1$.

Proof. If $\forall \mathbf{x} \in T, S(\mathbf{x}) = 0$, $\mathcal{F}_{Shannon}(\mathcal{G}, T, \sigma, t) = 1$ by definition, because p is undefined.

Suppose that $\forall \mathbf{x}, \mathbf{y} \in T, S(\mathbf{x}) = S(\mathbf{y})$ and that $\forall \mathbf{x} \in T, S(\mathbf{x}) > 0$. This immediately implies that $\forall \mathbf{x}, \mathbf{y}, p(\mathbf{x}) = p(\mathbf{y})$. This, in turn, implies $p(\mathbf{x}) = \frac{1}{m}$,

where $m = |T|$, and consequently $b = |T| = m$. Notice that for each $x \in T$,

$$I_p(x) = -\log_b(p(x)) = -\log_m\left(\frac{1}{m}\right) = 1.$$

Thus, the result is given by the following chain of equalities:

$$\mathcal{F}_{Shannon}(\mathcal{G}, T, \sigma, t) = \sum_{x \in T} p(x) \cdot I_p(x) = \sum_{x \in T} \frac{1}{m} = 1.$$

□

Theorem 3.23 shows the interplay between binary and gradual notions of fairness. If a topic T is ideally fair in the presence of some strongly safe $\{x\} \subseteq T$, then $\mathcal{F}_{Shannon}(\mathcal{G}, T, \sigma, t) = 1$. Similarly, given the previous scenario, $\mathcal{F}_{Gini}(\mathcal{G}, T, \sigma, t) = 0$.

Theorem 3.24 *If T is ideally fair w.r.t. $\mathcal{G}, \sigma, t$, then $\mathcal{F}_{Gini}(\mathcal{G}, T, \sigma, t) = 0$ and $\mathcal{F}_{Shannon}(\mathcal{G}, T, \sigma, t) = 1$ whenever there exists $\{x\} \subseteq T$ such that $\{x\}$ is safe w.r.t. $\mathcal{G}, \sigma, t$.*

Proof. Notice that if T is ideally fair, then one of its arguments is safe, and all the arguments in T are safe, and all of them attain the threshold in all the QBAGs. Hence, $\forall x, y \in T$, we have $S(x) = S(y)$, and this value is the number of QBAGs in $\mathcal{G}$. This means p is a uniform distribution, and by Theorem 3.23, $\mathcal{F}_{Shannon}(\mathcal{G}, T, \sigma, t) = 1$.

The previous observation implies that ζ is a linear function satisfying $\zeta(0) = 0$ and $\zeta(|T|) = \sum_{x \in T} S(x)$. This, in turn, implies that ζ coincides with the fairness line, and by Theorem 3.18, $\mathcal{F}_{Gini}(\mathcal{G}, T, \sigma, t) = 0$. □

4 Discussion

Our work contributes to the line of research on *argumentation dynamics*, which, broadly speaking, studies argumentative inference, where the representation of the exchange of arguments undergoes changes. This naturally reflects the dynamic nature of dialectical reasoning. Argumentation dynamics are frequently studied for different variants of computational argumentation—see [8] for a survey of works on abstract argumentation dynamics.

For the specific case of QBAGs, dynamics are often applied in order to facilitate some notion of *explainability*, notably by assessing the impact (a change to) one argument has on the final strength of another one [27, 13]. In these works, dynamics are thus essentially used as an analytical tool. Conversely, our work introduces analytical tools for studying dynamics that occur in sequences of QBAGs, which we call *chains* or, intuitively, *dialogues*. It may thus be most closely related to [15], which studies which changes—also called *explanations*—made to a QBAG affect the relative strength of two topic arguments. The current work could be extended accordingly to define explanations indicating why SOLF notions are satisfied or violated.

5 Future Work

Our study of SOLF notions for quantitative (bipolar) argumentation dialogues suggests the following notable directions for future work.

Providing SOLF guarantees depending on argumentation semantics.

Our work analyzes only simple cases where some of our SOLF notions can be guaranteed, notably cases based on weak expansions, where changes do not affect the topic arguments. Indeed, we view the SOLF notions as analysis tools that can be applied to *specific instances* of QBAG chains. Guaranteeing safety, oscillation, liveness, and fairness more generally can be challenging. Figure 3 illustrates that even relatively simple scenarios can make it difficult to generally show that SOLF notions are satisfied or violated.

Consider the QBAG in Figure 3.1, with topic arguments a and b , and assume that the strength of f is variable. Consider a scenario where we have a chain of QBAGs that are all based on the one in the figure. Assume that the strength of f increases, either due to an initial strength change or the influence of additional supporters, e.g., from 0.1 to 0.5 and finally to 0.9. The effects of these changes on the final strengths of a and b are shown in Figure 3.2. Notice that, intuitively, the effect of strength changes to f on a and b is *non-monotonic*: it “switches” from positive to negative for a and inversely for b —this phenomenon is, e.g., discussed in [13].

This makes it difficult to assess whether a set of topic arguments, even when consisting of a single argument, is safe or live, and analogously, whether a set of topic arguments is fair. In our example, $\{a\}$ is safe and live w.r.t. the thresholds 0.1 and 0.175, respectively. However, $\{b\}$ fails to satisfy both safety and liveness w.r.t. the above-mentioned thresholds. Both a and b oscillate w.r.t. the threshold 0.175, but only b oscillates w.r.t. the other threshold of 0.1. $\{b\}$ ’s liveness failure prevents $\{a, b\}$ from satisfying the binary notions of fairness w.r.t. the thresholds 0.1 and 0.175.

Defining SOLF notions for partially ordered final strengths. In this work, we have made the crucial assumption that our assessments of safety, oscillation, liveness, and fairness pertain to topic arguments that are *comparable*. However, this may not always be the case in QBAGs, as arguments may have *undefined* final strengths, practically, because a modular semantics may not converge for cyclic graphs. Pragmatically, one may claim that undefined final strengths give rise to an edge case whose consideration introduces overhead and thus confusion. However, in some cases, dealing with arguments of incomparable strengths can be useful. This is useful, for instance, when moving from quantitative to labelling-based abstract argumentation, where labels may form a lattice of argument strengths (cf. [25]), as sketched in [15].

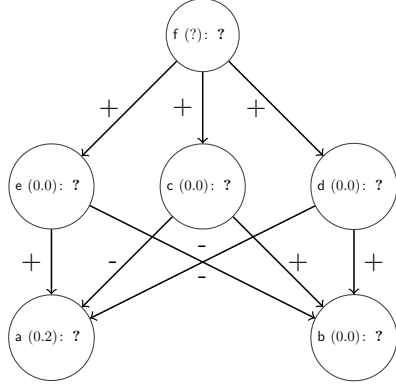
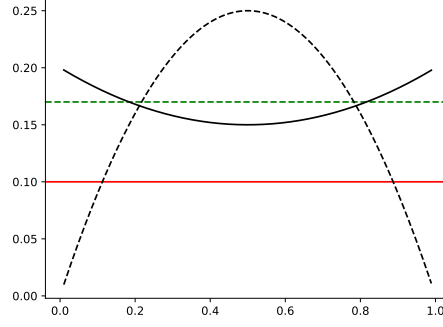
3.1: QBAG with variable initial strength of f .3.2: $\sigma(a)$ (solid) and $\sigma(b)$ (dashed), given $\tau(f)$.

Fig. 3. SOLF properties are difficult to guarantee, even in seemingly simple scenarios where we change the initial strength of a single argument that reaches two topic arguments. The horizontal lines in Figure 3.2. indicate different threshold values (of 0.1 and 0.175, respectively).

Introducing SOLF notions to other argumentation variants. The aforementioned bridge to abstract argumentation, via labelling-based approaches that give rise to notions akin to final strengths, can enable the introduction of our SOLF notions to other variants of computational argumentation. In the context of abstract argumentation, our notions could then be integrated with a wealth of established results, e.g., regarding *stability*, *robustness*, and other invariant notions [23,5,4,19]. Conversely, one could extend the scope of these existing notions to a quantitative setting.

Acknowledgements

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Some Observations about Nonmonotonic and Pseudo-Nonmonotonic Effects in Quantitative Bipolar Argumentation

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Abstract

In quantitative bipolar argumentation, *non-monotonic* effects describe that gradually increasing the initial strength of one argument may first decrease and then increase the final strength of another one, or vice versa. Analogously, *pseudo-nonmonotonic* effects describe non-monotonic effects on the difference between two arguments' final strengths. These effects may complicate dialectical reasoning, given an agent's objective to either maximize an argument's final strengths, or to achieve a desired ranking of several arguments. We make new observations about (pseudo-)nonmonotonic effects and notably show their occurrence in topologically simple graphs. Our findings highlight potential challenges with quantitative bipolar argumentation, in particular as a facilitator of interactive explainability.

Keywords: Computational Argumentation, Formal Argumentation, Quantitative Bipolar Argumentation Graphs, Non-monotonic Effects

1 Introduction

Quantitative Bipolar Argumentation (QBA) is a form of computational argumentation in which arguments and their relationships are represented as graphs—called Quantitative Bipolar Argumentation Graphs (QBAGs)—with nodes modeling arguments with *initial strengths*, connected via (disjoint) attack and support relations. Considering a graph's topology, initial strengths are then updated to *final strengths* by so-called *gradual semantics*. Due to its somewhat “continuous” nature that allows bridging sub-symbolic inference with knowledge representation and reasoning, QBA is considered a potential facilitator of *explainable* and *contestable* artificial intelligence [4,9,2]. However, QBA itself faces some explainability challenges, i.e., dialectical reasoning in QBAGs is not always straightforward.

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One challenge that has been observed in recent works is the occurrence of so-called *non-monotonic effects* [3,9]. Given a QBAG with a *topic* argument t and a *contributor* x , non-monotonic effects describe that by changing the Initial Strength (IS) of x monotonically, we can turn x 's marginal effect on the Final Strength (FS) of t from positive to negative, or vice versa. An example of a non-monotonic effect can be observed in Figure 1b. Non-monotonic effects make it more difficult to reason about the effect of changes to QBAGs with respect to a specific topic argument, as affecting the IS of a contributor monotonically may lead to first increasing and then decreasing the FS of the topic—accordingly, an agent may struggle to consistently argue for or against a topic.

The notion of non-monotonic effect can be extended to the FS-based ranking of two topic arguments t and t' . This yields the notion of *pseudo-nonmonotonic effect*: by changing the IS of x , we can switch the marginal effect of x on t from greater than its marginal effect on t' to smaller, or vice versa. Pseudo-nonmonotonic effects make it more difficult to reason about the effect of changes to QBAGs whenever we are interested in the comparison of several arguments' FS. Figure 1 gives examples of nonmonotonic and pseudo-nonmonotonic effects.

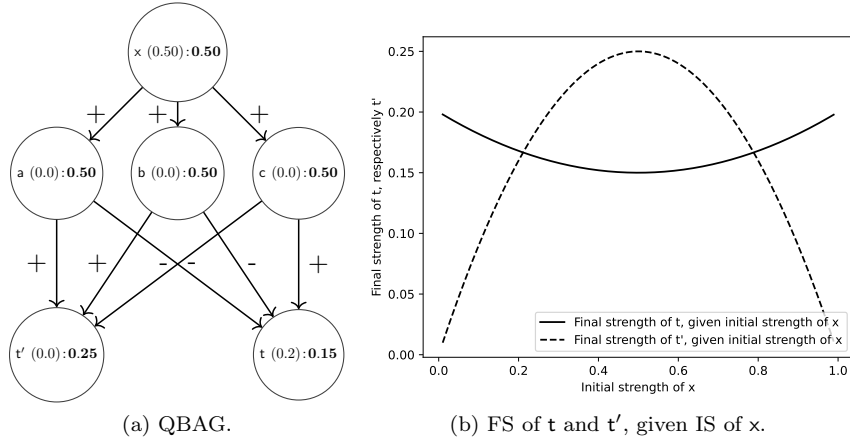


Fig. 1. QBAG with nonmonotonic and pseudo-nonmonotonic effects. A node labeled $y(i):f$ represents argument y with IS $\tau(y) = i$ and FS $\sigma(y) = f$. Edges labeled $+$ and $-$, respectively, represent supports and attacks. FS is computed using DFQuAD semantics.

Pseudo-nonmonotonic effects have not been studied explicitly so far. The study of nonmonotonic effects is limited to results amounting to the observation that the effects are absent whenever all paths between a contributor and a topic have the same *polarity*, i.e., when a contributor cannot (indirectly) attack and support a topic at the same time [11]. In this paper, we expand the study of these effects by observing that: (i) nonmonotonic effects exist even in attack-only QBAGs; (ii) pseudo-nonmonotonic effects can typically be constructed given nonmonotonic effects; (iii) pseudo-nonmonotonic effects occur in very simple multitree-like QBAGs. The findings highlight that dialectical reasoning

with QBAGs poses challenges in even simple scenarios, as indirectly supporting an argument at one point in a dialogue may amount to attacking the same argument as the dialogue further evolves (or vice versa).

2 Preliminaries

Let $\mathbb{I} \subseteq \mathbb{R}$ be a (real) interval, typically $[0, 1]$. A QBAG is a quadruple $G = (Args, \tau, Att, Supp)$, in which $Args$ is a finite set of elements called *arguments*, $\tau : Args \rightarrow \mathbb{I}$ is the *initial strength function*, and Att and $Supp$ are disjoint binary relations on $Args$, called *attack* and *support*, respectively. Next, we assume a QBAG $G = (Args, \tau, Att, Supp)$. We say G is: (i) *attack-only* or *support-only* iff $Supp = \emptyset$ or $Att = \emptyset$, respectively; (ii) *multitree-like* iff for all $x, y \in Args$ there exists at most one path from x to y in the graph $(Args, Att \cup Supp)$. Given a path from $x \in Args$ to $y \in Args$, we say its *polarity* is *even* iff the number of attack edges on the path is even (including zero), and *odd* otherwise. For $x \in Args$, $s \in \mathbb{I}$, we define $G_{\leftarrow \tau(x)=s} := (Args, \tau', Att, Supp)$, where $\forall y \in Args$ it holds that $\tau'(y) = \tau(y)$ if $y \neq x$ and $\tau'(y) = s$, otherwise.

A gradual semantics σ updates IS to FS, considering the topology of a given QBAG. Here, we are interested in so-called *modular semantics* that assign FS by first *aggregating* final strengths of an argument's attackers and supporters and then determine the *influence* of the aggregation result on the argument's IS. Accordingly, a modular semantics is a composition of an *aggregation function*, aggregating the final strength of the attackers $Att(x)$ and supporters $Supp(x)$ of the argument x at hand, and an *influence function*, calculating the influence of this aggregation, given $\tau(x)$. Given acyclic graphs, which we focus on in this work, modular semantics can update IS to FS by means of a single pass through the QBAG following the direction of attack and support relations. Our examples use *DFQuAD* semantics [8], defined as follows for $x \in Args$ (assuming acyclic QBAGs):

$$f(x) := \tau(x) - \tau(x) \times \max\{0, -g(Att(x), Supp(x))\} + (1 - \tau(x)) \times \max\{0, g(Att(x), Supp(x))\},$$

with $g(Att(x), Supp(x)) = \prod_{y \in Att(x)} (1 - \sigma(y)) - \prod_{y \in Supp(x)} (1 - \sigma(y))$. Finally, given $t, x \in Args$, we define the gradient-based contribution function $\text{Ctrb}_{t,G}^\partial : Args \rightarrow \mathbb{R}$ as $\text{Ctrb}_{t,G}^\partial(x) := \frac{\partial f_t}{\partial \tau(x)}(\tau(x_1), \dots, \tau(x_n))$, where $x_1, \dots, x_n$ are all arguments from which there is a path to t in $(Args, Att \cup Supp)$, and f is a composition of an aggregation function and an influence function.

3 Nonmonotonic and Pseudo-Nonmonotonic Effects

We proceed by formalizing the notions of nonmonotonic and pseudo-nonmonotonic effects as sketched in the introduction. Notice that while these notions depend on the behavior of a given gradual semantics, we leave the semantics implicit. Also, we provide counter-examples only for DFQuAD semantics. However, additional counter-examples for a range of other semantics

are provided in an executable notebook at <http://s.cs.umu.se/gwp785>⁴.

Definition 3.1 [Nonmonotonic and Pseudo-Nonmonotonic Effects] Given a QBAG $G = (Args, \tau, Att, Supp)$, we say that a:

Nonmonotonic effect occurs in G iff there exist arguments $x, t \in Args$ s.t. for some $s, s' \in \mathbb{I}$ it holds that $\text{Ctrb}_{t, G \leftarrow \tau(x)=s}^\partial(x) < 0$ and $\text{Ctrb}_{t, G \leftarrow \tau(x)=s'}^\partial(x) > 0$.

Pseudo-nonmonotonic effect occurs in G iff there exist arguments $x, t, t' \in Args$ s.t. for some $s, s' \in \mathbb{I}$ it holds that $\text{Ctrb}_{t, G \leftarrow \tau(x)=s}^\partial(x) < \text{Ctrb}_{t', G \leftarrow \tau(x)=s}^\partial(x)$ and $\text{Ctrb}_{t, G \leftarrow \tau(x)=s'}^\partial(x) > \text{Ctrb}_{t', G \leftarrow \tau(x)=s'}^\partial(x)$.

Notice that pseudo-nonmonotonic effects require merely a reversal in a contributor's marginal effect on two topics, and not a reversal in the arguments' final strength-based ordering.

Assuming a *modular* gradual semantics under which every argument without attackers and supporters retains its IS as its FS, non-monotonic effects trivially bring pseudo-nonmonotonic effects with them.

Proposition 3.2 *If a nonmonotonic effect occurs in a QBAG $G = (Args, \tau, Att, Supp)$ then there exists a QBAG $G' = (Args \cup \{t' \notin Args\}, \tau \cup (t', s^* \in \mathbb{I}), Att, Supp)$ in which a pseudo-nonmonotonic effect occurs, assuming a modular gradual semantics under which every argument without attackers and supporters retains its IS as its FS.*

Proof. Since a nonmonotonic effect occurs in G , there exist $t, x \in Args$ s.t. for some $s, s' \in \mathbb{I}$ it holds that $\text{Ctrb}_{t, G \leftarrow \tau(x)=s}^\partial(x) < 0$ and $\text{Ctrb}_{t, G \leftarrow \tau(x)=s'}^\partial(x) > 0$. Given the assumption about semantics behavior, the same must then hold given G' , i.e., $\text{Ctrb}_{t, G' \leftarrow \tau(x)=s}^\partial(x) < 0$ and $\text{Ctrb}_{t, G' \leftarrow \tau(x)=s'}^\partial(x) > 0$. Clearly, the gradient-based contribution of every argument in $Args$ to t' is 0 in both $G'_{\leftarrow \tau(x)=s}$ and $G'_{\leftarrow \tau(x)=s'}$. Consequently, it holds that $\text{Ctrb}_{t, G' \leftarrow \tau(x)=s}^\partial(x) < \text{Ctrb}_{t', G' \leftarrow \tau(x)=s}^\partial(x)$ and $\text{Ctrb}_{t, G' \leftarrow \tau(x)=s'}^\partial(x) > \text{Ctrb}_{t', G' \leftarrow \tau(x)=s'}^\partial(x)$, which closes the proof. $\square$

Let us informally claim that nonmonotonic effects do not occur in QBAGs in which each two paths between the same two arguments have the same polarity (given, e.g. DFQuAD semantics). This essentially follows from formal observations made in [11] (Proposition 7) and similarly in [9] (Proposition 4). Intuitively, changing the IS of a contributor in such QBAGs must always positively (or not at all) affect the FS of a topic if the polarity is positive, whereas the effect is always negative (or zero) if the polarity is negative. If a topic cannot reach each a contributor, the effect is zero. For the sake of conciseness, we abstain from a more formal observation of this phenomenon here.

Interestingly, nonmonotonic effects occur in attack-only QBAGs, as evidenced by Figure 2 for DFQuAD semantics.

⁴ In the notebook, we use QE [6], DFQuAD, squared DFQuAD [7], Euler-based [1], and Euler-based top [5] semantics. Only for Euler-based semantics, we did not find pseudo-nonmonotonic effects in multitree-like QBAGs; we leave it to future work to investigate whether this may give rise to a more general result.

Proposition 3.3 *Nonmonotonic effects occur in attack-only QBAGs, given DFQuAD semantics.*

Proof. Consider Figure 2; e.g., set $G_{\leftarrow\tau(x)=0.2}$, then $G_{\leftarrow\tau(x)=0.6}$. $\square$

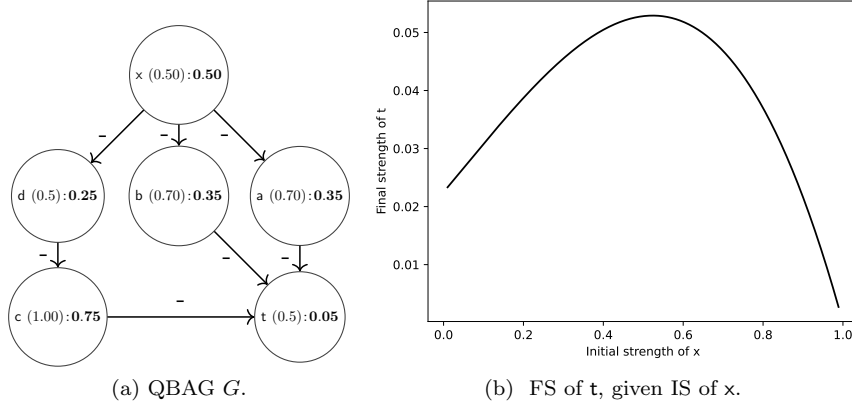


Fig. 2. Nonmonotonic effect in an attack-only QBAG, given DFQuAD semantics.

Accordingly, pseudo-nonmonotonic effects occur in attack-only QBAGs, too.

Corollary 3.4 *Pseudo-nonmonotonic effects occur in attack-only QBAGs, given DFQuAD semantics.*

Proof. The proof follows directly from Propositions 3.2 (we can construct pseudo-nonmonotonic effects from nonmonotonic effects without using supports) and 3.3 (nonmonotonic effects occur in attack-only QBAGs). The assumption about semantics behavior in Proposition 3.2 trivially holds for DFQuAD semantics. $\square$

Pseudo-nonmonotonic effects also occur in even simpler QBAGs, notably in graphs without any attacks.

Proposition 3.5 *Pseudo-nonmonotonic effects occur in support-only QBAGs, given DFQuAD semantics.*

Proof. Consider Figure 3; e.g., set $G_{\leftarrow\tau(x)=0}$, then $G_{\leftarrow\tau(x)=0.9}$. $\square$

Intuitively, *nonmonotonic* effects cannot occur in such graphs given DFQuAD, as increasing support in absence of attack can never decrease an argument's FS. Another class of somewhat “simple” QBAGs in which pseudo-nonmonotonic effects can occur are multitree-like QBAGs, in which two arguments are connected via at most one path.

Proposition 3.6 *Pseudo-nonmonotonic effects occur in multitree-like QBAGs, given DFQuAD semantics.*

Proof. Consider Figure 4; e.g., set $G_{\leftarrow\tau(x)=0.1}$, then $G_{\leftarrow\tau(x)=0.9}$. $\square$

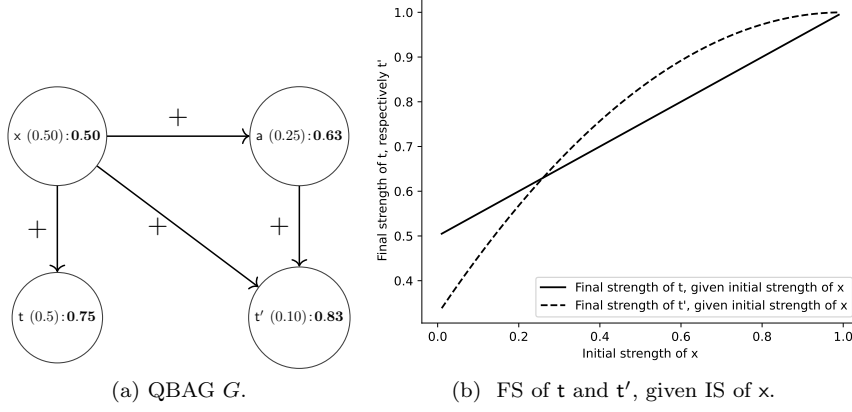


Fig. 3. Pseudo-nonmonotonic effect in a support-only QBAG, given DFQuAD.

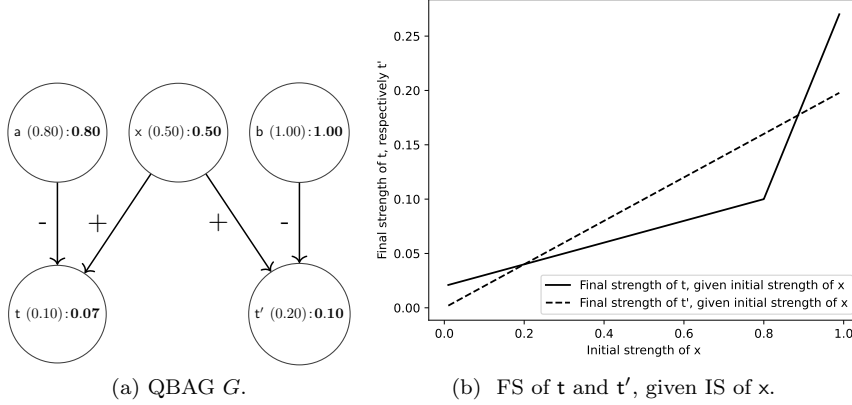


Fig. 4. Pseudo-nonmonotonic effect in a multitree-like QBAG, given DFQuAD.

4 Conclusion

In this paper, we have provided a brief introduction to nonmonotonic and pseudo-nonmonotonic effects in QBAGs. These notions are conceptually related to gradient-based attribution functions of arguments' final strengths, as studied in [10,3]. We have observed that nonmonotonic effects can occur in attack-only QBAGs, and that pseudo-nonmonotonic effects can even occur in support-only and simple multitree-like QBAGs. Our findings indicate that dialectical reasoning with QBAGs can be challenging, even in simple scenarios. Future work may study nonmonotonic and pseudo-nonmonotonic effects for additional semantics, identify further special cases for which the absence of these effects can be guaranteed, and design dialectical reasoning procedures that account for them.

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